

Ethiopia's National Forest Inventory



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Ministry of Environment,
Forest and Climate
Change

Ethiopia's National Forest Inventory Final Report



Food and Agriculture Organization
of the United Nations



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Forest and Climate Change

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Message from the State Minister



I am very glad to see that our Ministry, in a few short years of its establishment, managed to complete the first National Forest Inventory (NFI) and produce the NFI Report. I am also pleased to note that this was largely undertaken by Ethiopians with technical support from the Food and Agriculture Organization of the United Nations (FAO), which is yet another big achievement. The national, expert-driven process has helped us build our institutional capacity which will enable periodic forest resources monitoring of the forest cover over time.

As we all know, forests play a vital role in livelihoods of Ethiopians by ensuring food security, alleviating poverty, providing environmental services (water supply for domestic, agricultural and energy production), and opportunities for green growth and development, among others. While the contribution from the forestry sector to gross domestic product (GDP) remains largely underestimated in official statistics, our recent findings indicate that the sector's direct contributions to the national GDP and to other sectors are about 6% and over 7% respectively.

For Ethiopia with fragile mountainous ecosystem, the forests play an even greater role in protecting the country from erosion, floods, droughts and other adverse impacts of climate change, while potentially improving rainfall regimes across the country. Clearly, forestry is the most important and strategic sector that needs due attention by all citizens. Nevertheless, despite such dependence of our lives on forests, we have lost the resource to deforestation due to population pressure and increasing demand for agricultural and grazing lands, and unsustainable fuel wood consumption. Cognizant of this serious threat to the forest resources, the government is addressing deforestation and forest degradation mainly through its REDD+ program.

Forest resources information is crucial for various strategic decisions, sector policy formulation and for monitoring forest trends and for this reason, the national forest inventory was designed and implemented. I am happy that many years after the Woody Biomass Inventory, we have successfully completed the first NFI which is more comprehensive and holistic, designed to collate a wide range of information including data for our REDD+ program. The estimates and results from our current NFI will guide development planning in the sector while making data and information available for international reporting obligations. The current NFI is not an end; rather it is a beginning for future periodic monitoring and inventories.

I would like to commend the efforts and hard work of all those involved in the NFI, particularly the NFI field crew for getting through this challenging work in the field and the coordinating team.

I would also like to thank the Royal Norwegian government for its financial support and the World Bank and FAO, without the support of which this work would have been impossible. I congratulate all involved for the successful completion of this tiresome but vital effort.

Finally, I hope that the NFI results will be of help to all stakeholders, besides the Ministry and government agencies in the sector.

H.E. Kebede Yimam,

State Minister of Environment, Forest and Climate Change

Foreword

Understanding forest resources and their changes is key to national and international environmental and developmental policy processes and information on forest resources is required by many international agreements, including the United Nations Framework Convention on Climate Change (UNFCCC), the Convention on Biological Diversity (CBD), the United Nations Convention to Combat Desertification (UNCCD), the United Nations Forest Instrument (UNFI) and the Sustainable Development Goals (SDGs).

Consequently, the demand for reliable and up-to-date forest information at national level has grown considerably. The 16th and 19th Conference of the Parties (COP 16 and COP 19) decided that robust national forest monitoring systems (NFMS) should provide data and information that are transparent and consistent over time, which are suitable for measuring, reporting and verifying (MRV) anthropogenic forest-related emissions by sources and removals by sinks, forest carbon stocks, and forest-area changes for REDD+.

Ethiopia's national forest inventory (NFI), the first of its type, involved a detailed field survey of the country's forests to assess the extent, composition, biomass and condition of the entire national forest estate by visiting 539 Sampling Units (2,077 permanent forest inventory plots) and spanned from February 2014 to July 2016. These plots now represent a bank of baseline data. NFI analysis results were computed using general models, formulas and conversion factors and are presented by four biomes. The result of this survey will enable Ethiopia to fulfill national and international reporting obligations and to monitor the sustainable development of the forest resource in Ethiopia. Data from the NFI will be used to estimate carbon stocks through the calculation of forest biomass in forests and greenhouse gas emissions and reductions associated with land use change.

Data processing and analysis was completed to allow for the preparation of this publication in April 2018, the preparation of which was a highly collaborative process involving MEFC and FAO. The NFI results were subjected to a thorough assessment including a validation workshop before publication. This important publication is the result of hard work of all who were engaged in the consultation process. As you read the report, I hope you will gain insight into the breadth and depth of our activities.

This NFI report is structured in five Sections. Section 1 provides general background on Ethiopia's commitment to carbon-neutral development and its relevant strategy that leads to achieving its vision and the NFI is one of the main elements that enables decision making. Section 2 deals with methodology for undertaking the NFI and data analysis followed by results in Section 3. Conclusion and the way forward are given in Section 4. The contents of the different sections are summarized in the Executive Summary Section while details are provided in the appendices.

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Acronyms and Abbreviations

AEZ	Agro-Ecological Zones	LUCS	Land Use Cover Section
AGB	Above Ground Biomass	LULC	Land Use Land Cover
AGC	Above Ground Carbon	MEF	Ministry of Forests
BGB	Below Ground Biomass	MEFCC	Ministry of Environment, Forest and Climate Change
BGC	Below Ground Carbon	MoA	Ministry of Agriculture
C	Carbon	MP	Measurement point
CDM	Clean Development Mechanism	MRV	Measuring, Reporting and Verification
CI95	Confidence Interval with 95% Confidence level	NFI	National Forest Inventory
CRGE	Climate Resilient Green Economy	NFMA	National Forest Monitoring and Assessment
CSP	Circular Sub Plot	OF	Open Foris
DBH	Diameter at the Breast Height	PVM	Potential Vegetation Map
DEM	Digital Elevation Model	QA/QC	Quality Assessment/ Quality Control
FAO	Food and Agriculture Organization	REDD+	Reducing Emissions from Deforestation and Forest Degradation
FCPF	Forest Carbon Partnership Facility	R-PP	Readiness Preparation Proposal
FDT	Fallen Deadwood Transect	RRW	REDD Readiness Wheel
FRA	Global Forest Resources Assessment	RS	Remote Sensing
FREL	Forest Reference Emissions Level	RSP	Rectangular Sub Plot
GDP	Gross Domestic Product	SOC	Soil Organic Carbon
GHG	Greenhouse Gas	SRTM	Shuttle Radar Topography Mission
GIS	Geographic Information System	SU	Sampling Unit
GPS	Global Position System	TA	Technical Assistance
GTP	Growth and Transformation Plan	TCP	Technical Cooperation Programme
GWDDDB	Global Wood Density Database	TOF	Trees Outside of Forest
IPCC	Intergovernmental Panel on Climate Change	UNFCCC	United Nations Framework Convention on Climate Change
LSP	Litter Sub Plot	WBSIPP	Woody Biomass Inventory and Strategic Planning Project
LUCC	Land Use Cover Classes	WD	Wood Density

Executive Summary

In the first quarter of 2016, the Government of Ethiopia (GoE) announced that the number of people affected by severe drought caused by El Niño stood at 10.2 million in the period 2015/2016.

Ethiopia is seriously committed to addressing climate change. The Climate Resilient Green Economy (CRGE) strategy that was officially introduced during the 17th Conference of the Parties (COP 17) in Durban in 2011 is a clear manifestation of commitment to pursue a carbon-neutral development regime. The CRGE strategy has identified a number of low carbon-emitting initiatives across key economic sectors (forestry, agriculture, energy, livestock, transport and industry) that will help Ethiopia achieve its vision of building a climate-resilient middle-income economy by 2025. The successful implementation of these initiatives would help Ethiopia eliminate projected emissions of over 250 Mt CO₂e a year by 2030 compared to the actual Business as Usual (BAU) scenario. The strategy aims to maintain the annual Greenhouse Gas (GHG) level of the country at the 2010's level, i.e., 150 Mt CO₂e.

The forests in the country are under direct and indirect threats. Direct threats include small and large-scale agricultural land conversion, increased wood extraction for fuel and construction, and pressure caused by increased livestock grazing. The indirect threats comprise gaps in the application of forest policy and regulations; tenure/unclear forest user rights; lack of private investment in forestry development; population growth; and inadequate land use planning and participatory forest management (PFM). As a result, the forestry sector is the second largest contributor of GHG emissions in the country after agriculture. Under BAU, greater pressure on forests will increase the rate of deforestation and forest degradation and prevent the country from reaping its long-term environmental and economic benefits.

In this context the knowledge of the status of the forest and the Trees Outside of Forest (TOF) is a country priority. The CRGE strategy includes the adoption of the REDD+ programme as operational instrument for mitigate the CO₂ emissions in the forest sector.

Objectives

Ethiopia conducted the National Forest Inventory (NFI) field campaign between 2014 and 2016. The NFI started as a FAO - Technical Co-operation Project (TCP) in 2013 with the aim to pilot and test the National Forest Inventory methodology, and was then scaled up with the aid of Forest Carbon Partnership Facility (FCPF) and Bio-Carbon Fund for broader implementation of Reducing Emissions from Deforestation and Forest Degradation (REDD+) Readiness at the Ministry of Environment, Forest and Climate Change (MEFCC) in late 2014 and continued with the FAO technical support.

Specific objectives of the Ethiopian NFI were:

- The identification of a methodological approach to support the Measuring, Reporting and Verification (MRV) implementation by taking into account the national context through a participatory process;
- The design and set up of for a long-term monitoring system of the forests and trees outside of forest based on a network of permanent plots located across the country;
- The collection of variables that respond to the national data needs;
- The establishment of a database to store and manage the field information for analysis and dissemination.

The forest definition adopted for NFI is: *Forest as a land with of 10% tree canopy cover, a 0.5 ha area and a 5 m tree height in average.*

Methodology

The sampling design chosen for NFI is stratified systematic cluster sampling. Using available geospatial layers of Ethiopia and large scale ecological studies the whole country was classified into five strata that give a macro ecological characterization of five macro areas. The number of the sampling units varies within the strata depending on the estimated probability to find trees and forests. A total of 627 sampling units were established across the five strata. Every sampling unit has an area of 1 km² and is composed of 4 plots (with cumulative plot area of 2 ha).

A series of variables were chosen to give a complete characterization of Ethiopian forests, including biomass, carbon, biodiversity, health status and disturbances. The quality of the final results is ensured by rigorous Quality Assessment/Quality Control (QA/QC) procedures.

The methodology adopted can be replicated as all the sampling units are permanent and the data collection can be improved according to the findings on this first field campaign. All the collected NFI data has been stored in a database at the MRV laboratory of the MEFC.

In 2015 the stratification scheme was changed because Ethiopia decided to adopt a classification that better describes the vegetation characteristics of the country. The following biomes were adopted:

- ζ *Acacia-Commiphora*
- ζ *Combretum-Terminalia*
- ζ *Dry Afromontane*
- ζ *Moist Afromontane*

This change resulted in the adoption of more specific analysis methods. All the results are thus presented by biome, and not by original NFI strata. All results have been computed using general models, formulas and conversion factors described in this report.

Results

Out of the 627 planned sampling units, 539 accessible ones containing a total of 2,077 accessible sample plots were visited. About 49,829 trees and 2,029 stumps were recorded and analyzed, and a cumulative area of 1,029 ha was surveyed by the field crews in 27 months.

Approximately 40.3% of the total growing stock in terms of above-ground biomass (AGB) is located in natural regenerated forests, followed by cultivated land (28.5%) Other Wooded Land (22.9%), Other Land Natural (5.8%), Planted Forests (1.4%) and Other Land Built-up (1.1%). Hence, there is a considerable number of trees outside of forests (ToFs). ToFs are considered as an important part of the total growing stock, because woodlands dominate the landscape of Ethiopia. Although built-up land does not have a large share of the national area, it happens to contain a considerable high number of trees per hectare, which indicates the importance of trees also in the urban and peri-urban areas. Results by biome demonstrate that in Ethiopia (in both forest and non- forest land) trees in the Moist Afromontane biome contain 43.3% of the total volume.

Tree species count shows that the most frequent species in forest are *Crotonmacrostachyus* (4.9 trees/ha), *Syzygium guineense* (4.6 trees/ha) and *Juniperus procera* (3.0 trees/ha). In the Other Wooded Land *Acacia bussei* (5.7 trees/ha) is the most frequent tree species, followed by *Commiphora erythraea* (4.8 trees/ha) and *Boswellia rivae* (3.7 trees/ha).

Based on the NFI data, 65.8% of forest lands are somehow disturbed, and 58.8% of Other Wooded Lands are disturbed by human activity. These are rather high figures but typical to countries, like Ethiopia, with high population pressure on forest resources. In forest land, the main disturbance is assigned to the encroachment by agriculture. In addition, a relatively high frequency of *Croton macrostachyus*, may show that Ethiopia's forest ecosystems, particularly in the highlands, are degraded. This species is often an indication of forest degradation because it colonizes open areas. Further indication of degradation is the prevalence of *Acacia bussei* in Other Wooded Land. This species generally tends to dominate recently regenerated sites in the dryAfromontane areas.

The overall health condition of forests and trees in Ethiopia seems to be at a good level, approximately 94% of the trees are considered healthy and 3.2% of trees are unhealthy. The causative agents for unhealthy trees in descending order are: climate, human, fire, insect pests, diseases and animals.

Table 1.1 Summary table of the main result variables by FRA classes (the bole volume is shown here for all tree species, not only for the commercial species)

FRA – Major LUCC	N. of trees (N ha ⁻¹)	Bole volume (m3 ha ⁻¹)	Volume (m3 ha ⁻¹)	AGB (t ha ⁻¹)
Forest -Natural regenerated forest	121.5	93.1	131.7	125.1
Forest – Forest plantation	178.3	55.1	72.4	71.3
Other Wooded Land	61.2	4.7	7.4	7.6
Other land - Natural	10.3	5.4	7.1	6.7
Other land - Cultivated	25.4	11.3	17.2	16.4
Other land – Built-up	49.4	10.5	17.5	17.1
Other land – Quarry or Mining	35.1	6.1	7.2	7.3
Water - Inland water	25.2	5.1	7.2	7.2

Conclusions

The main conclusions from the NFI initiative are as follows:

- From the NFI results, there are indications of considerable degradation in Forests and OWL. Results show that population pressure is high given that 65.8% of forest lands and 58.8% of OWLs are considered as “disturbed” by human activity. In forest land, the main cause of disturbance is encroachment by agriculture. Results on the basal area in forest lands also indicate that the forests appear to be degraded, with basal area (BA) in forests being 13.3 m²ha⁻¹ and in High woodland only 3.8 m²ha⁻¹. When looking at the basal area results by biome (forest only plots), in *Acacia-Commiphora*, *Combretum-Terminalia*, Dry Afromontane and Moist Afromontane estimates are 6.5, 8.4, 12.4 and 20.7 m²ha⁻¹, respectively. Only Moist Afromontane forest has a basal area greater than 20 m²ha⁻¹, a further indication of degradation;
- All the established sampling units can be considered as permanent and can be revisited during the next NFI cycles. Re-measurements of the sampling units will allow for better understanding of the status of forests and especially forest dynamics in the country. A continuous NFI is recommended and entails a certain proportion of sampling units being re-measured every year (as for example in Finland and Poland);
- All collected data are stored in an Open Foris Collect database and managed by the MRV staff in the Directorate’s experts at MEFCC. This database can be used for further analysis and scientific research purposes. Despite results being reported by region, it should be noted that the first NFI was designed to give results primarily for the national level. An intensification of NFI sampling in the future can expand the use of NFI data to subnational levels and improve the accuracy of the estimates;
- The implemented NFI methodology can be adapted and improved, by adding new variables or stratification schemes. In 2016, Ethiopia already institutionalized MRV

1 Guillozet, K., Bliss, J.C. & Kelecha, T.S. Small-scale Forestry (2015) 14: 121.
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in a decentralized way, thus allowing all the administrative levels to be part of the process. MEFC have several options to improve NFI mechanisms, between them is a formalization of NFI as part of MRV that will regularly be carried out by (decentralized) institutions.

- The following actions are recommended in order for Ethiopia to enhance its NFI:
 - o Formal approval of the new national forest definition;
 - o Assign permanent teams to the NFI in order to create specialized and professional teams;
 - o Take into account the change in the national forest definition in order to harmonize the data analysis and reporting consistently;
 - o Decentralize field operations so that they are better coordinated technically in order to secure faster and more efficient field data collection phase;
 - o Establish clear protocols for data flow between the NFI levels (i.e. operators);
 - o Reinforce links between remote sensing analysis, inventory planning and field data collection;
 - o Use NFI field data as validation of RS results (forest cover/change detection analysis);
 - o Conduct further training and supervision to reduce the number of measurement errors and improve the QA/QC procedures;
 - o Include a botanist in each team;
 - o Reinforce links with the academia by allowing the research teams to access the NFI data and to improve the quality of the variables collected. This will yield an increased number of topical publications, thesis, and books that can emphasize the importance, complexity and uniqueness of the Ethiopian forests.

1 Introduction

Ethiopia's vision and strategy for CRGE² is articulated in its vision document endorsed by the parliament at the end of 2011. This initiative, comprising a strategy for climate resilient development and another for a green economy, seeks to improve resilience to climate change, ensure abatement, enhance avoidance of future emissions, as well as foster both economic development and less carbon dependent growth. The strategy focuses on adapting to climate change to minimize the potential risks and to maximize the potential benefits.

The country's commitments on developing its green economy include the protection and re-establishment of forests and their economic and ecosystem services, including the reduction of emissions from deforestation and forest degradation. In this context, Ethiopia has chosen the REDD+ initiative as a way to mitigate climate change through the forestry sector by implementing activities that should be measured, reported and verified through an appropriate national forest monitoring system.

The Government of Ethiopia, in partnership with the World Bank's FCPF and the BioCarbon Fund, identified the Food and Agriculture Organization (FAO) of the United Nations as a qualified partner to provide the necessary Technical Assistance (TA) to address the lack of monitoring capacities and experiences in applying international standards through capacity transfer, capacity building and awareness on the various components of its national forest MRV system.

Ethiopia decided in this context to implement its first NFI with the aim of constructing a countrywide scenario of the forest, for the generation of reliable information on forest resources to be used as a baseline for policy formulation, institutional capacity building, planning, conservation and utilization of natural resources on a sustainable basis.

The implementation of the NFI started in 2013 as a FAO-TCP project to pilot the National Forest Inventory and was then scaled up with the aid of FCPF and BioCarbon Fund for broader implementation of REDD+ Readiness at MEFCC through a Technical Agreement. FAO was tasked with supporting the implementation of a national forest monitoring and MRV system for REDD+ readiness in Ethiopia. The government was supported by FAO in addressing the lack of monitoring capacities and experience in applying international standards in the framework of REDD+ through capacity building and awareness raising on reference/baseline scenario construction and MRV systems as well as the establishment of a system for monitoring forest area change.

The field campaign and the results of the NFI are described in this report. This was the first NFI campaign in Ethiopia and it was carried out during the period from March 2013 to November 2016.

2 CRGE, 2011.

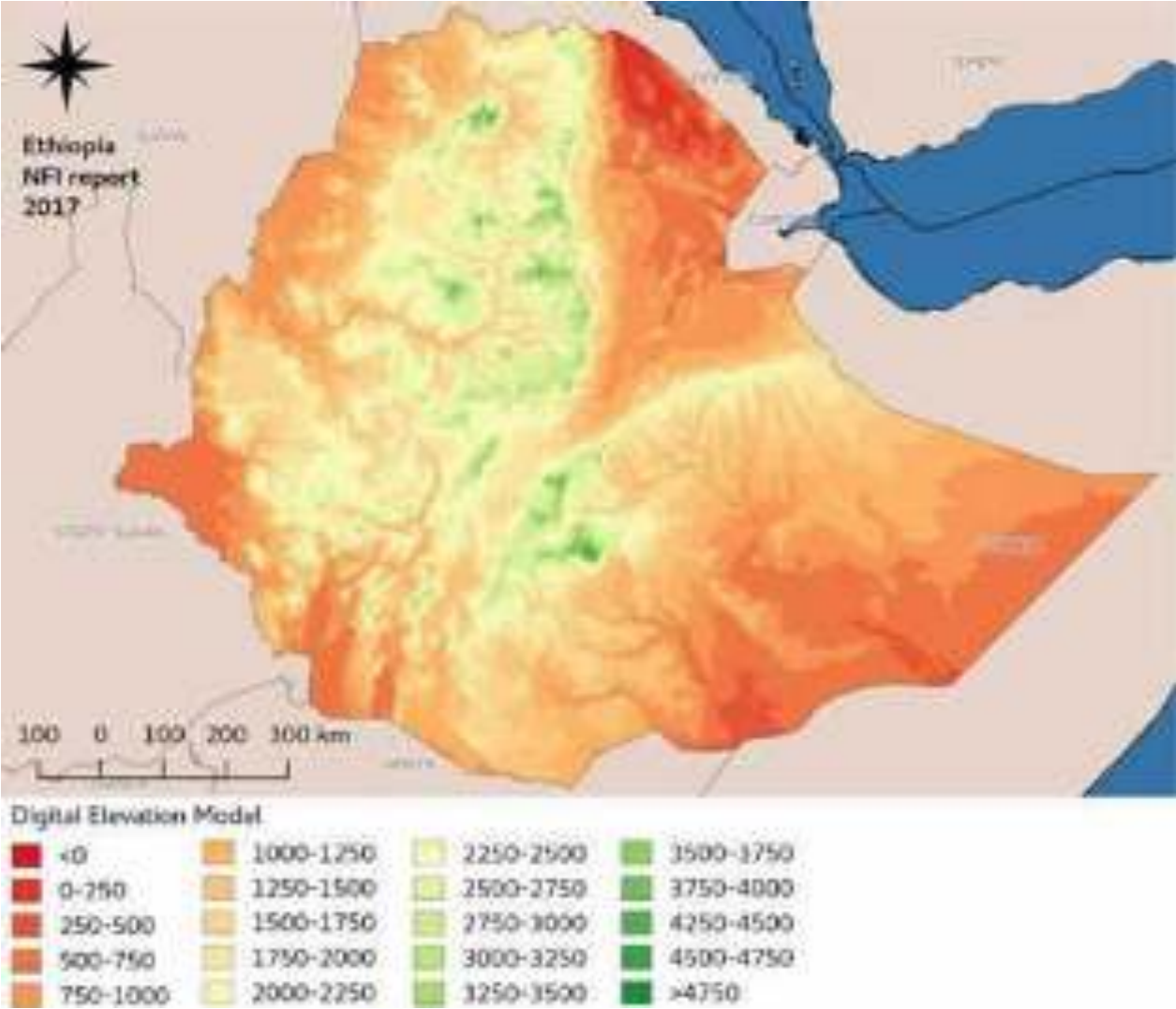
1.1 Biophysical description of Ethiopia

Ethiopia is positioned in the horn of Africa and lies within the tropics between 3°24' and 14°53' latitude; and 32°42' and 48°12' longitude³. With an approximate extent of 1,132,000 km², Ethiopia's topography extremely varies with a range of elevation between -148 m below the sea (Dallol Depression) and 4,620 meters above sea level (Ras Dashenpeak).

Approximately 65% (741,000 km²) of the country is considered as lowlands (i.e., land less than 1,500 m a.s.l.). The high plains or plateaus are located between the Upper Nile valley and the border with Sudan and South Sudan within an altitude of 1,300 m and 3,000 m a.s.l. The northwestern and southeastern highlands are separated by the Main Ethiopian Rift Valley, part of the Great Ethiopian Rift Valley that runs from Syria to Mozambique.

The Rift Valley is characterized by a chain of lakes throughout the greater part of the Valley. In the western part, the plateau extends from Eritrea in the north to Kenya in the south.

Figure 1.1 Digital Elevation Model of Ethiopia



3 Henricksen et al., 1984.

The larger Ethiopian Plateau is also known as the Northwestern Highlands. In the Southeast there is a smaller plateau designated as the Southeastern Highlands (Somalian Plateau). Apart from the major river valleys, the whole part of the Somalian Plateau lies above 1,000 m and about half of the plateau lies above 2,000 m a.s.l.

The Southeastern Plateau is located east and southeast of the Rift system.

The Shewa plateau (2,000 m a.s.l.) occupies the central portion of the Ethiopian Plateau. It extends west to east with a basalt surface.

The extremely varied topography influences the climate in Ethiopia and its climatic belts can be defined as follows⁴:

- Wurch: Frequent frost, usually no rainfed crops would be expected to grow. The altitude is above 3,700 m. a.s.l. and the annual rainfall is normally > 900 mm
- High Dega: Average altitude lies between 3,200 and 3,700 m a.s.l., the temperature varies between 8 °C and 15 °C and the annual rainfall between 900 and 1,100 mm.
- Dega (cool to cold temperature): Typical of the cool highlands with average temperatures between 10 °C and 16 °C. The altitude is above 2,000 meters above the sea level.
- Weina Dega (warm to cool climate): Average annual temperature between 16 °C and 20 °C. This category contains a big part of the highlands.
- Kolla (warm to hot climate): The climate of the hot lowlands has a temperature range in between 20° C to 30° C and altitude ranges from 500 to 1,500 meters above the sea level.
- Bereha (Hot and arid climate): Bereha includes the desert lowlands below 500 meters a.s.l. and the average annual temperature is over 30° C.

Two distinct seasons characterize the Ethiopia's climate, the dry season (from October to May) and rainy season (from June to September) with some relevant variations (3-4 weeks) from east to west.

An upward trend in temperatures (+0.3° C year⁻¹) and a downward trend in rainfall (0.4 mm month⁻¹year⁻¹) has been observed since 1948 and is expected to continue up to 2050⁵.

The rainfall pattern is mainly bimodal in the north part of Ethiopia, with a short season around March or April and a second rainy season that begins in June or July.

The two seasons converge into a unimodal pattern in some areas as in the western and some northern parts of the country where rainfall is generally higher. In the larger part of the central highlands, there is a tendency for the two separate seasons⁶.

Ethiopia has an extremely heterogeneous ecology attributed to the variability in its topography, meteorology and landscapes. These differences have allowed the country to be inhabited by an extremely high diversity of flora and fauna. Ethiopia is one of the top 25 biodiversity rich countries in the world, and it hosts two of the world's 34

4 Hurni, 1998.

5 Jury and Funk, 2013.

6 Ministry of Agriculture, 2000.

biodiversity hotspots⁷: the Eastern Afromontane and the Horn of Africa hotspots⁸. In the Horn of Africa, Ethiopia is recognized as the center of diversity and endemism for several plant species (6,000 species of higher plants of which 10% are endemic⁹). Woody plant species constitute about 1,000 species in high natural forests, woodlands, plantation forests, and bamboo forests.

According to the Vegetation Map of Africa¹⁰ Ethiopia is located in three floristic regions, the Afromontane Region (in the Ethiopia Highlands), the Sudanian Region, and the Somalia- Masai Region (southern Ethiopia excluding the mountains). The Afromontane Region occurring mainly in the Ethiopian Highlands contains undifferentiated woodland and country specific peculiarities like the transition from undifferentiated Ethiopian woodland to Afromontane (i.e. single-dominant *Juniperus procera* areas). These differences are important in order to accept the key indicator characterization assigned by the co- prominence presence of *Podocarpus falcatus*. This kind of single species prominence is also described in other locations as in Afromontane single-dominant *Hagenia abyssinica* forest and Afromontane moist transitional forests.

The Sudanian Region is a natural continuity of the Afromontane Region (north-west), and it is located in the lowland normally below the 750 m a.s.l. It mainly includes areas that are called Ethiopian undifferentiated woodland. The dominating tree genera are *Anogeissus*, *Balanites*, *Combretum*, *Commiphora* and *Terminalia*.

From the steppe slopes of the Afromontane region (south-east) starts the Somalia-Masai region. It is covered with bushes and thickets, and above them deciduous species which grades into a semi-evergreen and evergreen vegetation of the same type in the lower slopes of the mountains. This region belongs mainly to the *Acacia-Commiphora* type and it includes different vegetation types such as scrub forest, riparian forest, secondary grassland and wooded grassland, seasonally waterlogged grassland, semi-desert grassland and shrubland, and desert. A more precise characterization of the forest vegetation of Ethiopia is presented by Friis *et al.* (2010), where the forest types are identified by taking into account the bioclimatic regions, as follows:

1. Upland dry evergreen forest of Eritrea and on the eastern escarpment (including *A. abyssinica* in the south part and *Barbeya oleoides* and *Tarchonantus camphoratus* in the northern part);
2. Upland dry evergreen forest of the plateau (mainly *Juniperus* sp., *Podocarpus* sp. and *Olea* sp.);
3. Upland dry evergreen forest of Sudamo, Bale and Harerge (*J. procera* and *B. oleoides*); Upland Rain Forests;
4. Humid mixed forest around the southern part of the Rift Valley and in Harerge (broadleaves and evergreen);
5. Humid broadleaves forest of the southwestern part of the plateau;

7 WCMC, 1994.

8 Negash and Kanninen, 2013.

9 Million and Leykun, 2001.

10 Hopkins and White, 1987.

6. Humid transitional forest of the southwestern escarpment; Lowland (semi-) evergreen forest of Ilubator;
7. Riverine and riparian forest.

Figure 1.2 Potential Natural Vegetation of Ethiopia (Friis and Sebessebe, 2009)

1.2 Socio-economic description

With an annual population growth of more than 2.5%, Ethiopia will have more than 120 million people by 2030¹². Ethiopia aims to achieve middle-income status by 2025 while developing a green economy. Several annual indicators (total GDP +7.90%, life expectancy of 65 years, urban population growth +4.8%, CO2 emissions +0.12 m) show a

rapid growth of the country ranked +8.5% for the per capita GDP.

Ethiopia hosts a population of more than 629,000 refugees, making it one of the largest hosts of refugees in Africa¹³. More than 57% of the refugees in Ethiopia are migrants from Somalia. Twenty five percent of refugees in Ethiopia are coming from South Sudan and 17% are from Eritrea.

1.3 The Woody Biomass Inventory and Strategic Planning Project

The Woody Biomass Inventory and Strategic Planning Project (WBSIPP) was the first forest and forest biomass inventory in Ethiopia. Since its completion, the WBISPP has been used as a reference for country estimates of the Ethiopian forests and its results are also included in the FAOs Forest Resource Assessment (FRA) for Ethiopia¹⁴. In August 1990, the WBISPP started with the scope to map 350,000 km² (land cover and vegetation maps), and conduct a woody biomass inventory in an area of over 300,000 km², (over a quarter of the country's land area¹⁵).

The main objective of the Project was to strengthen strategic forest resource planning and woody biomass inventory capacity. Other objectives were to identify and manage the country's woody biomass resources and to define a strategy for meeting the future biofuel needs and to decrease the negative impacts of deforestation.

More specifically, the project's tasks included building a woody biomass management information system, as an integral part of forestry and land use strategic planning throughout an inventory and assessment of above-ground woody biomass stock resources, including natural and man-made forests. Included in the task was a survey for the estimation of the consumption and use of woody biomass and other biofuel substances in order to identify the problems and issues with respect to the sustainable development and consumption of woody biomass resources.

WBISPP¹⁶ estimated the forest coverage of Ethiopia to be 59.7 million hectares of which 6.8% is high forest, 49% woodland, 0.2% plantation forest, and 44% shrub land. The WBSIPP's results and the WBSIPP Land Use Land Cover map have been considered in the NFI sampling design phase in order to identify the forested area and to determine the sampling intensity of the sampling points.

Unfortunately, it has not be possible to find a copy of the original database containing the raw data, thus any comparative analysis has not be possible.

13 IOM, 2014.

14 FAO, 2005.

15 Federal Ministry of Agriculture and Rural Development, 2011.

16 Federal Ministry of Agriculture and Rural Development, 2011.

1.4 Forest definitions

In order to correctly frame the objectives of this report it is important to review the definitions of forest adopted in Ethiopia during the period 1994-2015¹⁷.

According to the *Federal negarit gazeta*, "forest" means a community of plants either naturally grown or developed by planting and mainly consisting of trees and other plants having woody character¹⁸. This definition was later changed to better describe the Ethiopian forests for reporting purposes and included in the FRA reporting¹⁹. This definition describes the forest as a land with a tree canopy cover $\geq 10\%$, a 0.5 ha area and a 5m tree height in average.

This definition was adopted to define the forest in the NFI methodology and it is also the definition adopted for this report.

However, Ethiopia adopted a new forest definition²⁰ in February 2015: Land spanning at least 0.5 ha covered by trees (including bamboo) (with a minimum width of 20 m or not more than two-thirds of its length) attaining a height of at least 2 m and a canopy cover of at least 20% or trees with the potential to reach these thresholds in situ in due course.

Currently, Ethiopia is in the process of approving this new forest definition as its national legal definition of forest. The reason for changing the national forest definition is to better capture the natural primary state of forest vegetation. Specifically, the reason to reduce the tree height criteria from 5m to 2m is to capture natural forest vegetation types like the dryland forests which hosts woody species that typically reach a height of around 2-3 m.

The proposed change results in the inclusion areas which previously were classified as Ethiopia's dense woodlands into forest. Because commercial agriculture is expanding mainly on dense woodlands, Ethiopia wishes to enable REDD+ incentives for its conservation of this land type. The reason for increasing the canopy cover threshold from 10% to 20% is to avoid inclusion of highly degraded forest lands into the forest land and in this way to provide incentives for protecting quality forest. This definition of forest is the one adopted for the Forest Reference Emission Level (FREL)²¹.

The new forest definition differs from the one used for reporting GHG emissions and removals from the forestry sector within the framework of the Clean Development Mechanism (CDM)²². This is a minimum of 0.05 ha of land covered by trees attaining a height of more than 2 m and a canopy cover of more than 20%.

The difference between this and the new forest definition is an increase on area threshold. The main reason for increasing the area in FREL is due the limitation of technology for measurement and monitoring to detect changes in small areas of forest.

17 Gyde Lund, 2012.

18 Federal Democratic Republic of Ethiopia, 2007.

19 FAO, 2005.

20 MEFC, 2015.

21 MEFC, 2017.

22 CDM: Designated National Authorities (DNA).

1.5 Objectives and Scopes of the National Forest Inventory

The implementation of NFI started in 2013 as a FAO-TCP project to pilot the NFI and was then scaled up with the aid of FCPF and BioCarbon Fund for broader implementation of REDD+ Readiness at MEFCC. The support included the establishment of a MRV unit and it focused on implementing, in particular, Component 4 of the REDD Readiness Wheel (RRW): Design a Monitoring System for Emissions and Removals and Other Benefits and Impacts by providing the monitoring tools required to measure the impact and revert the effects of this pressure through its main outcome of Ethiopia's MRV capacity to implement REDD+ strengthened. The objectives for a complete assessment of a Monitoring System for Emissions and Removals include the following:

1. The identification of a methodological approach developed by the FAO REDD+ team to support MRV implementation taking into account the national context through a participatory process.
2. The design and set up of a long-term monitoring system of the forests and trees outside of forest based on a network of permanent plots located across the country.
3. The collection of variables that respond to national data needs mainly forest and other land cover area, growing stock, biomass and carbon, state and quality of the forests and trees outside of forests, social economic information about use and management of forest products and services.
4. The establishment of a database to store and manage the field information for analysis and dissemination.

From 2013 to 2016 Ethiopia reached new steps on the REDD+ implementation and the MRV Unit was established at MEFCC. In January 2016, MEFCC with the support of FAO submitted the first FREL report to United Nations Framework Convention on Climate Change (UNFCCC). This document helps the country to respect commitments on the threshold in CO₂ emissions in order to access the result based payments taking into account:

- The reducing of the emissions from deforestation;
- The conservation of forest carbon stocks; and
- The enhancement of forest carbon stocks.

The Emission Factors for the FREL reporting have been calculated using solely the current NFI data.

1.6 NFI in the REDD+ programme

Ethiopia completed Readiness Preparation Proposal (R-PP) in May 2011. In October 2012 after the FCPF approval, Ethiopia received a Readiness preparation grant of USD 3.6

million to prepare the national REDD+ Strategy and to contribute to overall institutional and technical readiness for REDD+.

Ethiopia started the second phase of REDD+ readiness in January 2013, which is focused on implementation. The Governments of Norway and the United Kingdom provided additional funding of USD 10 million for the implementation of the R-PP. In 2013, Ethiopia developed an MRV Roadmap. In addition, FAO and the Ministry of Agriculture (MOA) began work on developing a national forest inventory. After its institutionalization, MEFCC absorbed all MRV system (including NFI).

Ethiopia submitted its first FREL in 2016 on a voluntary basis as for the Decision 12/CP.17, paragraph 13²³. It was the first FREL submission from the Africa region. The FREL submission is in the context of results-based payments for the implementation of reducing emissions from deforestation and forest degradation, sustainable management of forests and enhancement of forest carbon stocks under UNFCCC.

The Emission Factors for this first version were calculated using first phase NFI data collected in the Oromia region since NFI data collection was ongoing in the rest of the country. An adjusted revision calculated using NFI data from the entire country was submitted in January 2017.

1.7 Considerations on the NFI results

The data collected during the NFI were processed to produce results that describe the state of forests and trees outside forests at the country level, and by biome. Because the NFI was initially designed to produce results primarily at the country level (and particularly per NFI strata), all the attempts to downscale the results can reduce the accuracy, as for the case of the results presented per region.

The area estimates presented in this report are based on the analysis of the area proportion within the field samples and therefore cannot substitute the results obtained by using wall-to- wall or extensive remote sensing techniques.

The area estimates provide an overview and the optimal way to use them is to present them per hectare so as to obtain proportions between the land use/ land cover categories and FRA classes.

The results presented in this report are consistent with the Forest Reference Level document submitted to UNFCCC in January 2016 as the emission factor calculation followed the same procedure. The difference is that in this analysis, all SUs have been computed and not just those in forestlands.

23 UNFCCC, 2011.

2 Methodology

2.1 Sampling design

The NFI sampling design was based on a) available map layers which were then used for pre-stratification, b) the available budget and on c) human resources available for the assessment.

The sampling design was constructed to enable periodic monitoring, ensuring statistical reliability of the collected data in the future. Using available geospatial layers of Ethiopia (including Agro-Ecological Zones (AEZ), Potential Vegetation Map (PVM), and Digital Elevation Model (DEM)), the whole country was first classified into five strata. In order to finalize the strata definitions, the forest inventory team conducted a series of consultative meetings with forestry experts, plant ecologists, statisticians and the National MRV Task Force where past experiences on forest inventories was reviewed and discussed. The implementation of the stratification for the systematic sampling design started by overlapping the layer of the AEZ of Ethiopia with the data produced from the WBISPP.

In this first stage, four strata were obtained. A supplementary stratum was included in order to identify the differences between the woodland ecology that encompasses the widest part of North and South western and eastern parts of the country. The woodland stratum was therefore further split into two in order to obtain comparable sampling intensities. The final stratification is therefore composed of five strata that give a macro- ecological characterization of the vegetation of Ethiopia (Table 2.1).

Table 2.1 Strata used classes used for the NFI sampling

Strata	Description
Stratum I	Natural forest and plantation and Bamboo
Stratum II	North and South Eastern part of the woodland mainly <i>Acacia-Commiphora</i> woodland of Somali and Afar region.
Stratum III	North and South Western woodland part of <i>Terminalia-Combretum</i>
Stratum IV	Other land
Stratum V	Desert (Berihā)

The sampling design chosen is stratified systematic cluster²⁴ sampling with different sampling intensities by strata: stratum I has a grid of 8 minutes in latitude and 8 minutes in longitude, stratum II has sampling units every 30 minutes in latitude and 30 minutes in longitude and strata III and IV have sampling units every 25 minutes in latitude and 25 minutes in longitude due to their seamless nature in terms of landscape mosaic. Stratum V includes sampling units every 60 minutes in latitude and 25 minutes in longitude. A final number of 631 sampling units were identified. All the SUs falling in

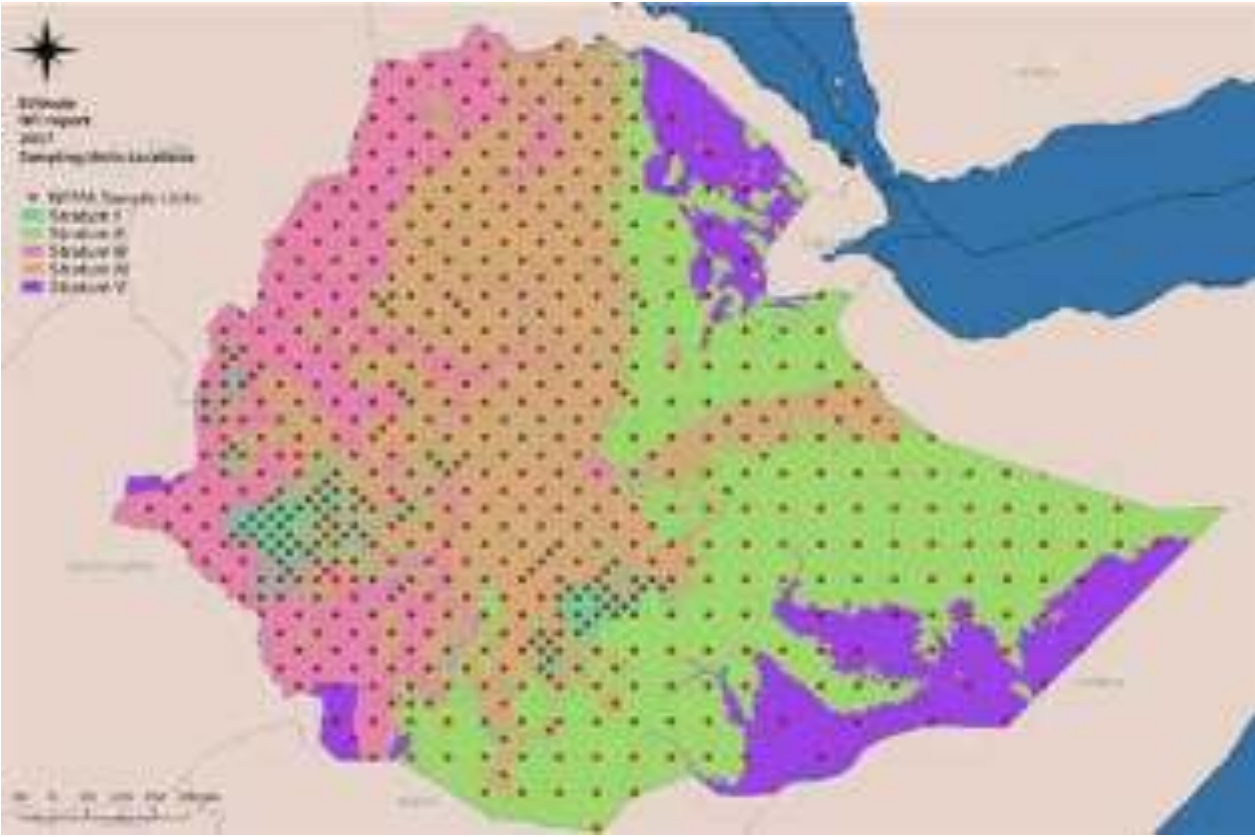
24 Clusters are called as Sampling Unit (SU) in the Ethiopian NFI.

water bodies or falling too close to the country boundaries (4 in total) were removed from the sampling design.

Table 2.2 Sampling units’ distribution within the NFI strata

Stratum	Afar	Amhara	Ben.-Gumuz	Gambela	Oromia	SNNPR	Somali	Tigray	Total
I	0	7	7	12	64	16	0	0	106
II	20	1	0	0	42	1	71	0	135
III	1	32	27	15	16	32	0	13	136
IV	0	71	3	0	99	32	7	18	230
V	6	0	0	0	0	1	13	0	20
Total	27	111	37	27	221	82	91	31	627

Figure 2.1 Sampling units’ distribution within the NFI strata



A sampling unit is a square surface area of 1 km² (1000 m x 1000 m). The coordinates of the south-west corner of the SUs correspond to those of the points selected in the systematic sampling frame. Each SU is organized in plots, subplots, demarcated within the SUs and then further divided into Land Use Cover Section (LUCS) and Land Use Cover Classes (LUCC) recorded in the field forms.

The plots shape is rectangular (of 20 m wide and 250 m long). The plots start at each corner of an inner 500 m² (same center as SUs), and numbered clockwise from 1 to 4.

Three sets of subplots are delimited within each plot corresponding to different data collection levels:

- 3 Rectangular Sub Plot (RSP) 10 m x 20 m (200 m²), corresponding to level 1;
- 3 Circular Sub Plot (CSP), with a radius of 3.99 m (50 m²), corresponding to level 2, located in the left-hand half of the rectangular subplots;
- 3 Litter Sub Plot (LSP), also circular but smaller with a radius of 18 cm (about 0.1 m²), corresponding to level 3, located in the centre of the CSPs.

All these subplot categories are numbered from 1 to 3, from the starting point of the plot to the end of the plot. An edaphic (soil) and topographic Measurement point (MP) is established at the centre of each rectangular subplot. A Fallen Deadwood Transect (FDT) is located at the end of each rectangular subplot. Each plot is divided into LUCS representing homogeneous land use / vegetation cover portions of land (e.g. forest, crops, grassland), with variable section size and shape identified in the field. The classification system adopted to identify the different land use/cover classes is described in appendix (A1.1). Relevant data related to grazing, cropping and forest characteristics, management and resources use and users are collected within the LUCS. Figure 2.1 shows the distribution of SUs in the field and Figure 2.2 shows the structure of the SU, plots and sub plots.

Table 2.3 SU, plots and subplots sizes			
Unit	Shape	Size	Number
Sampling Unit (SU)	Square	1000 m x 1000 m	1
Plot	Rectangle	250 m x 20 m	4/SU
Rectangular Subplot (RSP)	Rectangle	10 m x 20 m	3/plot
Circular Subplot (CSP)	Circular	Radius r = 3,99 m	3/plot
Litter Subplot (LSP)	Circular	Radius r = 18,0 cm	3/plot
Fallen Dead Wood Transect (FDT)	Line	20 m	3/plot
Land Use/Cover Sections (LUCS)	Varies	Varies	Varies
Land Use/Cover Class (LUCC)	Varies	Varies	Varies

The land use/cover classes (LUCC) classification includes different levels as follows:

1. The first level is based on the classification system developed by the FRA Programme of FAO to ensure harmonization between countries for regional or global assessments. These global FRA classes consist of Forests, Other Wooded Land, Other Land and Inland Water.
2. The second, third and fourth levels are country specific, and include additional classes that meet specific federal and regional information needs.
3. The second level applies to all classes and differentiates between land use/land cover.
4. The third level refers to evergreen, semi deciduous and deciduous forests and describes its naturalness.
5. The fourth level mainly applies to vegetation canopy cover and is applicable only to evergreen forest, shrubs and natural grassland types.

A full description of the LUCCs is presented in Appendix A1.

Figure 2.2 Sampling units, plots and subplots structures and positions

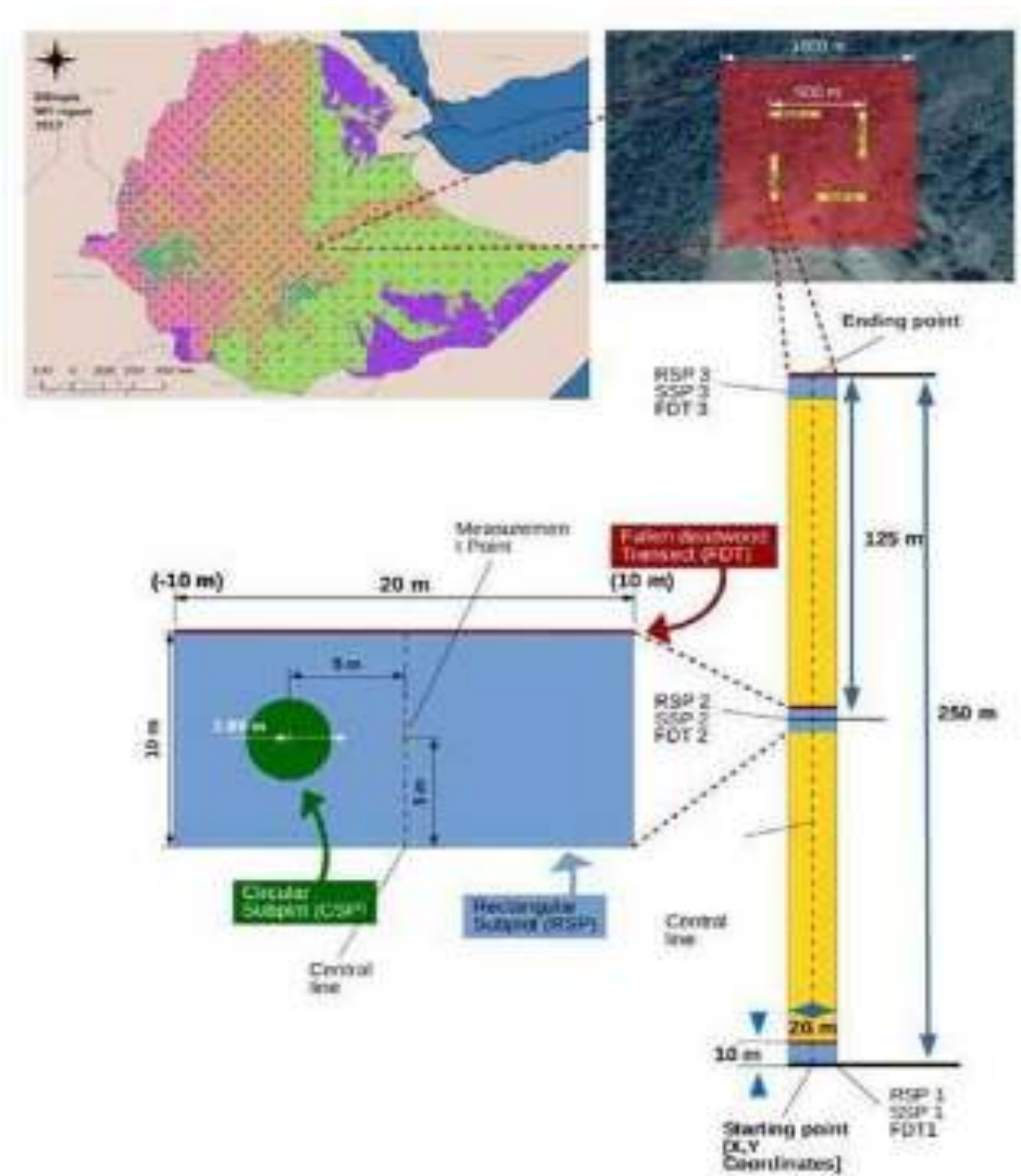
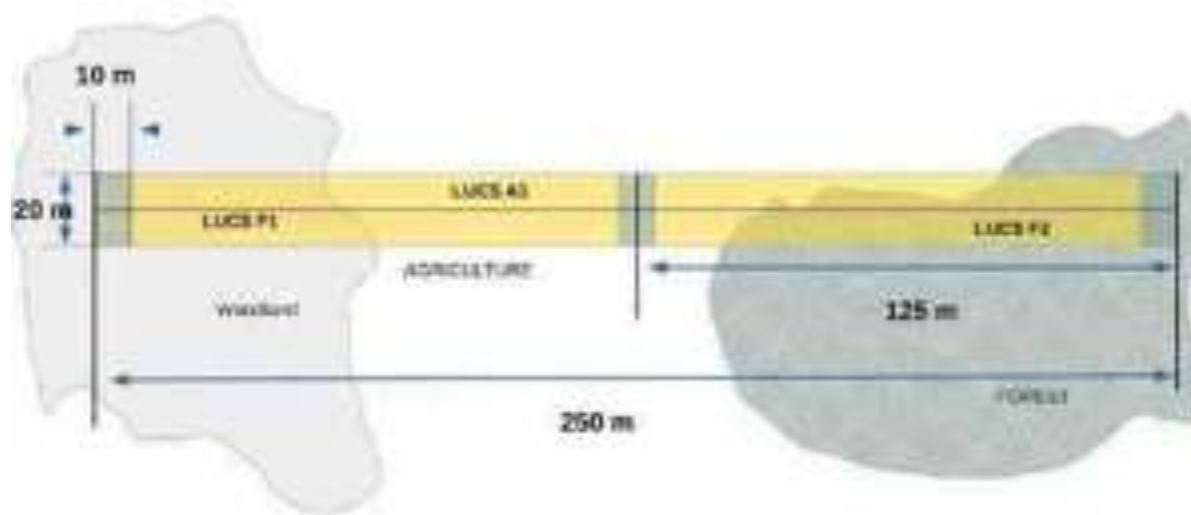


Figure 2.3 Example of land use/cover section identification within the plot



2.2 Data collection

Data collection was organized in the field at different levels and includes data from observations, measurements and interviews. The variables collected through interviews with the local key informants are considered as qualitative data and mainly include general descriptions and general socio-economic data. The variables collected through field observations include LULC description, environmental problems, forest management etc. Measured variables are the ones recorded with the support of specific tools and comprise linear distances and orientation and classic dendrometric data including diameter at the breast height (DBH), tree height, slope, etc. The first section of the field forms is dedicated to the general description of the plot, location, access path and basic socioeconomic description of the nearest village/town. The second section is specific for each plot, and it includes data about locations, photo orientation and measurements. For all the trees and stumps measured, the following variables are collected:

- Position in the plot;
- Tree/stump;
- Species name (scientific names and vernacular names);
- Diameter at 0.3 m level;
- DBH and top height (for trees and stumps greater or equal DBH 10 cm in outside forest and greater or equal to DBH 20 cm in forest) ;
- Bole height;
- Stem quality;
- Tree Health;
- Causative agents;
- Decomposition status.







At every Measurement Point (in the center of each Rectangular Sub Plot) the following variables are recorded:

- Land Use/Cover Section (LUCS);
- Slope/orientation;
- Organic layer thickness;
- Soil drainage.

For the small trees (DBH<10cm) in the circular subplot, species name are collected and the trees counted. Data on shrubs and bushes is collected in the RSP, including the following variables:

- Species;
- Number of stems;
- Average stem diameter;
- Average height.

There are three transects of 10 meters long in each plot (i.e., 12 transects per SU) for the collection of the Deadwood including quality and decomposition status. Characteristic of each LUCS is observed and described in a specific section including:

- Land Use/Land Cover (LULC) type;
- Soil erosion;
- Length of the section (inside the plot);
- Vegetation cover;
- Environmental problems (including disturbances);
- Management status.

Extensive training events were organized in order to secure that the field crews correctly collect the field data. During the first phase of the field campaign from March to September 2014, four teams were trained and engaged. Two additional field teams were added in September 2014. From February until August 2016, two more teams were trained and added. At the end of the NFI field campaign, total of eight teams were operational in the field. Each team was equipped with the following tools to correctly record the data: GPS receiver, compass, camera, caliper, meter tapes, laser range finder, and clinometer.

2.3 Quality Assessment / Quality Control procedures

A set of Quality Assessment/Quality Control (QA/QC) procedures were proposed during a specific training and then implemented in order to ensure an adequate standard in the data collection and data entry procedures. The first step included a

complete

analysis of the procedures considered as critical. The second step covered the identification of two kinds of possible improvements: best practices and supplementary actions.



Procedures to improve precision in the measurements

On the basis of the partial data set collected, a possible cause of uncertainty was identified in the measurement procedures. A series of best practices on the importance of data collection (including double measurement) were compiled and explained to the experts in the field in order to increase the accuracy of the measurements. Three critical tree attributes subject to errors were identified: DBH, height and scientific names. In the dendrometric measurements, height is the variable subject to most common errors. Using the clinometer as a height measurement tool, tree height is measured using the distance between the observer and the tree bole and the angle obtained between the ground slope, the observer eyes, and the top of the tree. The height has a considerable influence on the final Above Ground Biomass estimates²⁵.



Improving the precision on land use identification

Specific training material was prepared in order to improve the correct identification of the forest land use/cover type, by following the definition of forest and by taking into account canopy cover estimates, number of trees per hectare, and other relevant site type indicators.

Re-measurements

Based on a random sub-sampling, 10% of the SUs were re-measured by a semi-independent team (composed of MEFCC experts not involved in the field campaign and specifically trained for QA/QC). At least one randomly selected plot per SU was re-measured entirely and the results were compared with the original values. The QA/QC team used the original data forms to check any irregularities in the records. An error tolerance (10% difference in results between the measured and re-measured sampling units) was introduced and applied in order to reject or accept the collected data. The inventory teams were not aware of which SUs were re-measured. This procedure allowed the QA/QC team to identify the field teams with insufficient or non-standard performances and contact them in order to improve their measurements precision in the data collection.

Correction of tree species names

Occasionally the teams were only able to assign vernacular names to the recorded trees. Therefore, an independent botanist was assigned to evaluate, correct and improve upon the tree names assigned by the teams during the field data collection. The consultant's work was particularly important in translating vernacular names into scientific names. There are two main

reasons to improve upon the precision of this variable: one is to improve the overall accuracy of NFI data, the second is related to the carbon estimates. In the latter case, in the models currently applied in Ethiopia²⁶ the species-specific where dry Wood Density (WD) factor is one variable of applied AGB model.

The botanist's procedure followed these steps:

- Check the specimens collected and dried by the field teams and QA/QC teams during the field procedures;
- Check the consistency of the scientific names according to the current literature; and
- Check and improve matches between vernacular names and scientific names.

2.4 Data entry

Before beginning with data entry phase, all the field forms were scanned for security reasons and the copies were stored in MEFCC hard drives. All collected data was then inputted into a database using Open Foris (OF) Collect software²⁷. OF Collect is a user friendly, open-source and versatile software specifically designed for the data entry and cleansing, and its data can be exported into data analysis and result reporting. Several errors can be avoided by using this type of software, because the database contains a large selection of lookup tables such as a predefined list of tree species names, toponyms, names of team members and validation rules for recorded attributes.

Based on the designed survey the personnel responsible for the data entry can enter the data in an interface organized as the field form and directly populate a database. Because the software can be installed in a server, several experts can enter, modify and manage the field form data at the same time. By applying personal usernames and passwords, the system administrator can follow the progress of data entry. The backup system is automatic and all databases can be restored if necessary.

25 Feldpausch et al., 2012.

26 Chave et al., 2014.

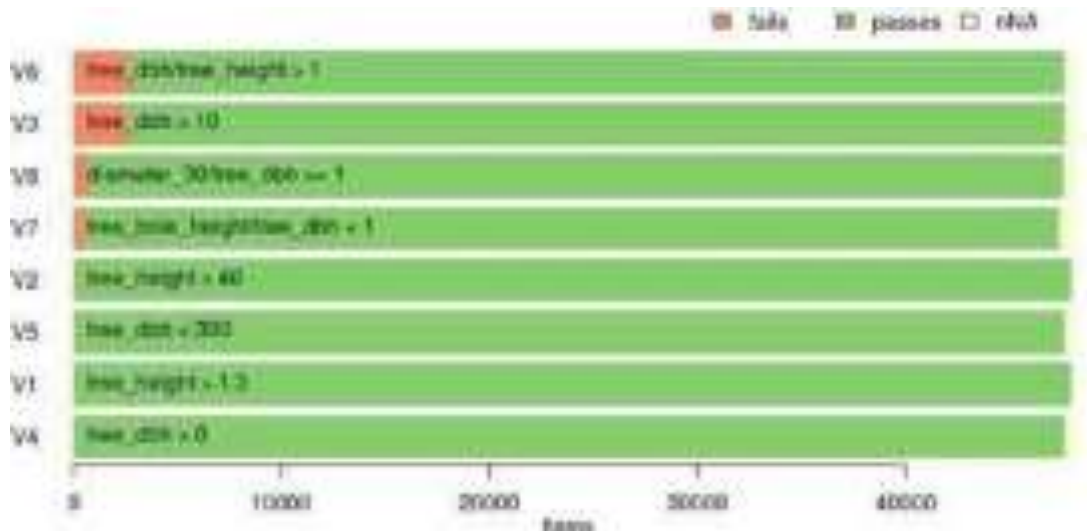


Once the data is entered and validated, then a specific data cleansing reports help the staff to find errors and to fix them. The data entry was started by two experts in MEFCC before the end of the field campaign, allowing the teams to anticipate the data cleaning phase.

2.5 Data cleansing

The data entered into the database was submitted for cleansing procedures in order to filter all the records considered potentially erroneous. Several indicators have been used to identify possibly erroneous values which fall out of the expected range of results (as ratios between DBH - Height, Diameter at 30cm - DBH, DBH - Branchesdiameter).

Figure 2.4 Example of validation graphic used to identify potential error cases



The data cleansing operation was supported by using the `validate`²⁸ package of R²⁹ in RStudio³⁰ that supports this function by introducing rules based on the dataset.

All of the suspect values were first checked in the field forms and then verified and corrected if necessary. In order to avoid decisions that can introduce bias³¹, the outliers were not purged or removed from the final analysis. A similar procedure was adopted for the sample plots' GPS coordinates: all the coordinates have been plotted using GIS software, but in this case some data was edited as not compatible with the plot extents and coordinates.

Regarding the verification and validation of tree species names, two different approaches have been adopted: ex-ante and ex-post methods. In a first phase, all the scientific names were checked using iPlant collaborative portal³² in order to check obsolete names and typos. In the second phase, the BIOMASS³³ package in R was used to introduce the corresponding scientific name, family, genus and dry wood density.

2.6 Data analysis

The data analysis of the field data results has been done using R language scripts and R scripts in OF Calc³⁴. OF Calc is a robust, modular browser-based tool for results calculation that allows expert users to write custom R modules to personalize data analysis.

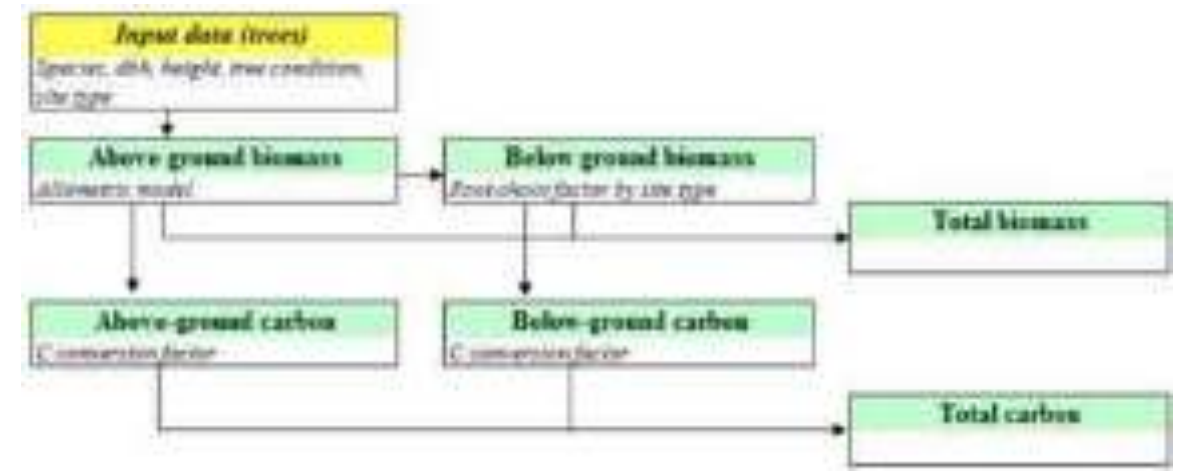
Along with Calc, Saiku³⁵ software was used as a reporting tool to provide a flexible way to produce aggregated results. These systems are perfectly compatible with OF Collect data,

28 Jonge, 2017.
29 R Core Team, 2017.
30 RStudio Team, 2015.
31 Pinkard and Beadle, 2000.
32 Go, Vaughn, and McKay, 2011.
33 Tanguy et al., 2017.
34 FAO, 2018.
35 <http://www.openforis.org/tools/collect-earth/tutorials/saiku.html>

since OF Calc can interact directly with the Collect database and then retrieve the database structure. The whole data processing chain and R scripts are documented in a separate technical project's report (Vesa *et al.* 2017).

Saiku allows an easy way to report the results across the local area network (LAN), offering greatly improved performance as the results are retrieved via LAN instead of reading them from a local hard drive. The Saiku result tables can be exported to MS Excel or R for further analysis, further aggregation or analysis and visualization.

Figure 2.5 Example of analysis chain for the biomass and carbon reporting



2.7 From Strata to Biome estimates

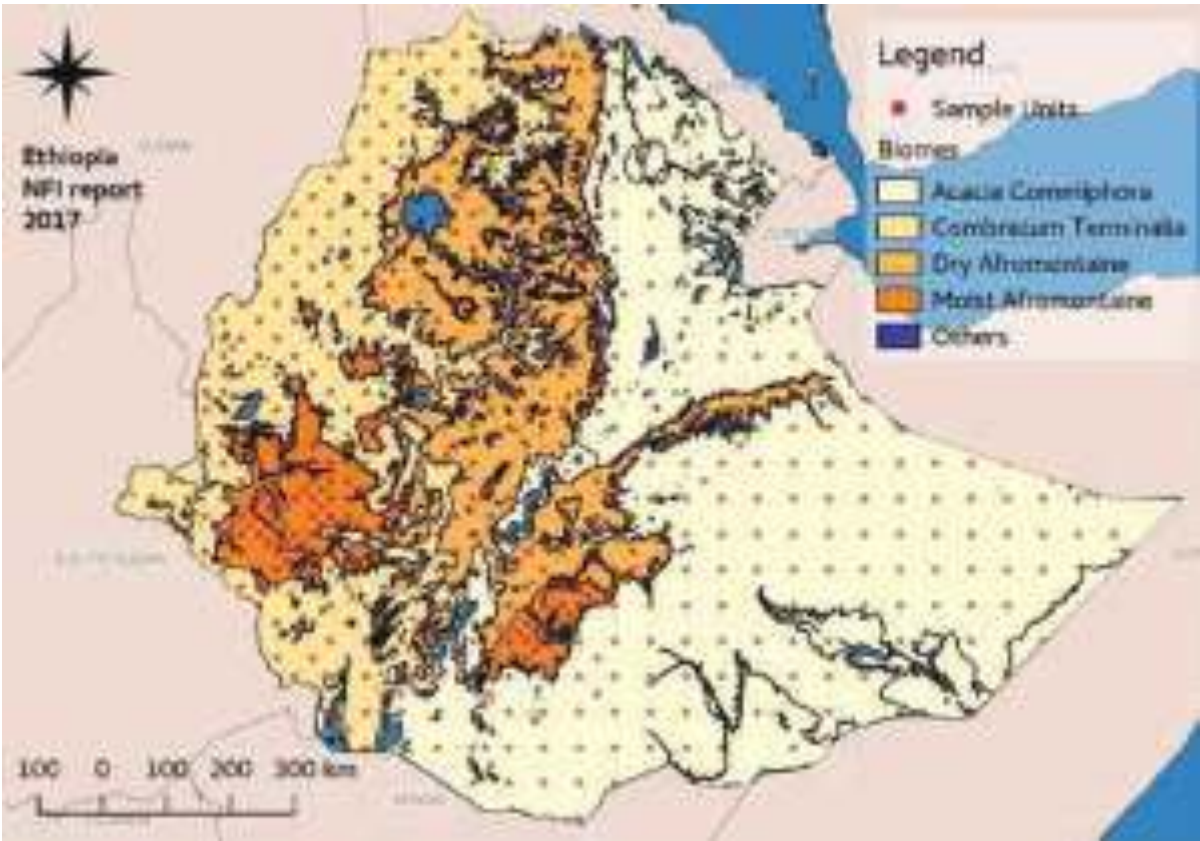
The NFI sampling design has been already described as a stratified systematic sampling, where the whole country is divided into five strata. In order to estimate emission factors for the construction of Ethiopia's national FREL, a new stratification of five biomes was obtained by aggregating the potential vegetation types of Ethiopia (Table 2.4).

Table 2.4 Biomes correspondence with the potential vegetation map of Ethiopia

Biome	Vegetation types (Friis and Sebsebe 2009)
<i>Acacia-Commiphora</i>	<i>Acacia-Commiphora</i> woodland and bushland (ACB); Acacia wooded grassland (ACB/RV); Desert and semi-desert scrubland (DSS)
<i>Combretum-Terminalia</i>	<i>Combretum-Terminalia</i> woodland and wooded grassland (CTW); Wooded grassland of the Western Gambela region (WGG)
Dry Afromontane	Dry evergreen Afro-Montane Forest and Grassland complex (DAF); Afro-Alpine vegetation (AA); Ericaceous Belt (EB);
Moist Afromontane	Moist Evergreen Afro-Montane Forest (MAF); Transitional Rain Forest (TRF);

The biome map (Figure 2.6) is used for post-stratification of the NFI data. The flow of data analysis and results was adapted to this new need, and is described in detail in the FREL report³⁶.

Figure 2.6 Distribution of the SUs within the biomes



36 MEFCC, 2017.

The biome stratification was introduced for the FREL submission when NFI was already in progress. Therefore, a post-stratification methodology was applied in order to correctly estimate the results by the biomes. The post-stratification was needed because the NFI design was based on five strata that did not perfectly overlap with the national biome map.

The number of SUs by biomes and strata is presented in Table 2.5 below.

Table 2.5 Distribution of the sampling units per biome and strata

	<i>Acacia-Commiphora</i>	<i>Combretum-Terminalia</i>	Dry Afromontane	Moist Afromontane	Others	Total
Stratum I	5	13	18	59	-	95
Stratum II	107	-	-	-	-	107
Stratum III	1	93		6	1	101
Stratum IV	38	38	114	29	1	218
Stratum V	15	2	-	-	1	18
Total	164	146	132	94	3	539

In order to ensure consistency of the estimates, a robust statistical procedure was applied based on the method described by Sarndal *et al.* (1992).

The first step implies to use of a ratio estimator, defined by the ratio between the variable *y* (tree variable) and the variable *x* (hectares of forest). Both *y* and *x* have been measured for each plot. The index *d*, which in statistical literature is referred to as a "domain", refers to the biomes.The NFI design is a stratified sampling therefore each stratum has a different sampling intensity defined by the inclusion probability π_k (of each plot). The π_k has been computed by dividing the number of hectares sampled in each stratum by the total area of the strata (when the sampling intensity is higher, inclusion probability is higher).

The domain ratio estimator has been calculated using the following equation:

$$\hat{t}_{dra} = \sum U_d x_k \frac{\sum s_d \hat{y}_k}{\sum s_d x_k} = \left(\sum U_d x_k \right) \hat{B}_d$$

- t_{dra} = ratio estimator of the total (e.g. total biomass)

U_d = domain d ($d=1, ..., D$) (e.g. specific biome)
- x_k = value of variable x for element k (e.g. hectares of forest)

y_k = variable of interest (e.g. biomass)
- s_d = samples that fall into domain d (e.g. plots in a given biome)

B_d = ratio (Total biomass/total hectares)
- π_k = inclusion probability of each plot

Where $\check{y}_k$ is the π_k expanded value of the variable of each plot divided by the plot's inclusion probability:

$$\check{y}_k = \frac{y_k}{\pi_k}$$

$\check{x}_k$ is the π expanded value of hectares of forest of each plot divided by the plots inclusion probability:

$$\check{x}_k = \frac{x_k}{\pi_k}$$

The domain ratio estimator provides the total domain.

The average has been calculated dividing by the size of the domain as follows:

$$\frac{\sum_{k \in s_d} \hat{y}_k}{\sum_{k \in s_d} \hat{x}_k}$$

This calculation has been used to compute the estimates per hectare for each biome. The variance is computed by the Sarndal methodology is:

$$\hat{V}(\hat{f}_{den}) = \left(\frac{\sum_{k \in s_d} U_d \hat{x}_k}{\sum_{k \in s_d} \hat{x}_k} \right)^2 \sum \sum s_d \Delta_{kl} \check{e}_{ks} \check{e}_{ls}$$

where $\check{e}_{ks}$ is the π_k expanded value of the residuals e_{ks} divided by the inclusion probability:

$$\check{e}_{ks} = e_{ks} / \pi_k = (y_k - \hat{y}_k) / \pi_k$$

and:

$$e_{ks} = y_k - \hat{B}_{ls} x_k$$

$\check{e}_{ks}$ and the π_k -expanded co-variance of sampling indicators and variance indicated by the Bernoulli pairwise plot selection probabilities where each pair of plots are here indexed using k and l .

When the plots k and l come from different Strata, their selection has been independent: $\pi_{kl} = \pi_k \pi_l$ and Δ_{kl} is therefore 0. When the plots k and l come from the same Strata, then:

$$\pi_{kl} = \sum_{s \ni k, l} p(s) = \binom{N-2}{n-2} / \binom{N}{n} = \frac{n(n-1)}{N(N-1)}; k \neq l = 1, \dots, N$$

$$\pi_k = \sum_{s \ni k} p(s) = \binom{N-1}{n-1} / \binom{N}{n} = \frac{n}{N}; k = 1, \dots, N$$

As k and l are from the same plot and from the same Stratum $\pi_k = \pi_l$, Δ_{kl} is computed as:

$$\Delta_{kl} = 1 - \frac{n(N-1)}{N(n-1)}$$

For N very big (as it is in this case) Δ_{kl} approximate to 0. For $k = l$, Δ_{kl} approximates to 1. To summarize, the $\Delta_{kl} = 1 - \pi_k$ equals 0 whenever $k \neq l$.

The variance formula can be then approximated to:

$$variance = \frac{\sum_{k \in s_d} (\hat{y}_k - \hat{B}_{ls} \hat{x}_k)^2}{\sum_{k \in s_d} \hat{x}_k}$$

Where B is a biome-specific factor akin to a slope in a regression between y and x . The comparative advantage of this approach is that the result is not approximated or inaccurate conversed because the biomes and NFI strata do not perfectly spatially intersect. The variance values differs in the new estimates because they combine the variance from the NFI strata and the biomes.

2.8 DBH-Height models

Field conditions do not always allow field crews to successfully use the clinometer for tree height measurement because the visibility of the top of tree that can sometimes be partially or totally prevented by branches or other trees. For this reason, not all the tree heights are measured in the field. In these cases tree height models that can be applied to estimate the missing tree heights. Given that height is a fundamental factor in estimating tree volume and biomass in the tropics³⁷, it is important to ensure a proper and consistent DBH/Height model to avoid losses on the results, mainly in the REDD+ countries where the stock sequestration performances are a part of the national economic strategy. Three different models, i.e. Naslund (1937), Schumacher (1939) and Curtis (1967), were tested for the Ethiopia NFI dataset using the `lmfor`³⁸ package in R, and by using only healthy living trees as input data. Curtis' model (1967) was ultimately selected since its better fit with the big trees in Forest Land:

$$h = 1.3 + a * \left(\frac{dbh}{1 + dbh} \right)^b$$

h = estimated top height [m];
 dbh = diameter at the breast height (DBH)[cm];
 a, b = parameters.

The parameters were estimated using nonlinear estimation techniques, and height model parameters were computed by FRA classes by calibrating the model by sampling unit.

37 Chave, 2005.

38 Mehtatalo, 2017.

2.9 Biomass models

Several Biomass models³⁹ have been developed for Ethiopia (at least 64⁴⁰). However, all of the models identified are species-specific (for *Dichrostachys cinerea*, *Euclea shimperi*, *Eucalyptus camadulensis*, *Eucalyptus globulus*, *Grewia bicolor* and *Otostegia integrifolia*) and site-specific (National Regional State of Tigray, Ker Jarso, Jufi, Ethiopian Heritage Forestry Park, Weldeab Ager, Addis Abeba, Munessa Shashamane).

In the absence of applicable biomass models for every Ethiopian ecosystem/biome consistent with international requirements, the choice between the models has been restricted to the pan-tropical model of Chave *et al.* (2014).

$$AGB = 0.673 (WD \cdot dbh^2 \cdot h)^{0.976}$$

AGB = Above ground biomass [kg];

WD = Dry wood density [t m⁻³];

The default value⁴¹ for the *WD* is 0.615 t · m⁻³.

In order to correctly compute the below-ground biomass (BGB) estimates, root-shoot ratios from the Intergovernmental Panel on Climate Change (IPCC)(2006) by the ecological zones have been adopted. Table 2.6 shows the distribution of SU by biomes and Table 2.7 shows the applied conversion factors correspondent to each ecological zone.

Table 2.6 Distribution of the sampling units by biomes and ecological zones

Biome	Tropical desert	Tropical dry forest	Tropical moist deciduous forest	Tropical mountain system	Tropical shrubland	Water	Total Result
<i>Acacia-Commiphora</i>	18	0	0	54	123	0	195
<i>Combretum-Terminalia</i>	0	43	13	92	31	0	179
Dry Afromontaine	0	1		148	2	0	151
Moist Afromontaine	0	1	7	85	0	0	93
Others	1	1	0	2	4	1	9
Total	19	46	20	381	160	1	627

39 Henry et al., 2013.

40 <http://www.globalloometree.org/>

41 Secondary Data to support Forest Reference Emission Level/Forest Reference Level for REDD+ in Ethiopia.

Table 2.7 Conversion factors applied to calculate the BGB

Ecological zone	Root-shoot ratio
Tropical shrubland	0.4
Tropical desert	0.5
Tropical mountain system	0.27
Tropical dry forest	0.56
Tropical moist deciduous forest	0.2

2.10 Wood densities

Wood density data of over 400 tree species found in Ethiopia has been analyzed. The wood density data was compiled in order to support precise wood biomass and carbon stock assessment; hence to assist the derivation of accurate emission factors for the FREL reporting. Wood density is the expression of the ratio of solid material to air/water space in the wood. It becomes an important parameter for improving model-based estimation of biomass and carbon content of woody plants. In forest inventories, wood density is usually expressed as $\text{t}\cdot\text{m}^{-3}$.

Most wood density reports in the scientific literature are reporting basic density. However, a large number of scientific papers also report air dry density, and options exist to convert between air dry density and basic density. One such option is multiplying the basic density by a factor of 1.22 to obtain air-dry density or multiplying air-dry density by a factor of 0.82 to obtain basic density. Wood density is affected by both tree and environmental factors such as age, sampling position (radially and longitudinally within the tree), and edaphic and climatic factors that affect growth rate. Therefore, results from individual samples, depending on site of sampling, position of sample from the stem and age of the sampled tree may yield different results for a single tree. This is the reason why different values are often observed for the same species in different documents. For the NFI analysis, the ones with the highest quality have been selected and applied. In order to assign a reliable value for each of the tree species inventoried in Ethiopia and missing in the country databases, the Global Wood Density Database (GWDDDB)⁴² has been adopted by using an R script through the RStudio environment and the BIOMASS⁴³ package. After the operation from the National Data Base, only the values identified as Low Quality Data were excluded and replaced by the GWDDDB data. As the result is out of 360 species identified during the NFI cycle, wood densities of 341 species have been selected using a validated value.

⁴² Zanne et al., 2009.

⁴³ Tanguy et al., 2017.

2.11 Deadwood

For the fallen deadwood volume, De Vries⁴⁴ formula given below was used. This formula requires the length of transect and the log diameter at the point of intersection, as follows:

$$V = \frac{\pi^2 \sum d^2}{8L}$$

V = volume per hectare of deadwood;
 d = log diameter at the point of intersection of the transect perpendicular to the axis of the log;
 L = length of transect.

Two decomposition classes are recorded for deadwood particles: sound and rotten. Be- cause a rotten wood is lighter than a sound wood, the dry wood density of deadwood is scaled down using lower wood densities than for live trees, as follows:

- ζ Sound deadwood WD: 90% * Default WD;
- ζ Rotten deadwood WD: 50% * Default WD.

44 De Vries, 1986.

3 Results

The NFI design was based on a stratification scheme described in the methodology section (Chapter 2) and the data collection followed that approach. These strata have then been replaced in Ethiopia with biomes for reporting purposes. Therefore, in this document the results have been reported at the country level and per biomes.

The reporting levels regarding the vegetation/land use categories follow the methodology described in the NFI Field Manual, i.e. containing FRA categories (Level1), Major LUCC types (Level2) and LUCC (Level3). In order to meet the national information needs of regional states, the data has been also analyzed and reported also byregions (inAppendices).

NFI consisted of 539 accessible SUs (containing a total of 2,077 plots, and 2,853 plot cover sections). Remaining 88 SUs have been identified as inaccessible due to different factors including excessive remoteness, topography and temporary security problems.

A supplementary crosscheck of all the inaccessible SUs was conducted using Google Collect Earth survey data. After this analysis, for the SUs classified as possibly accessible, a new attempt to reach the sites was accomplished.

Finally, 49,829 trees and 2029 stumps were recorded and analyzed in NFI and a cumulative plot area of 1,029 ha was surveyed by the field crews during approximately 27 months (Table 3.1).

Table 3.1 Timeline of the NFI’s sampling units completed per month

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
2014			2	5	30	24	28	24		23	28	27	191
2015	15	30	22	3	50	54	43	29	4	12	18	30	310
2016		9	5	2	4	12	6						38
Total	15	39	29	10	84	90	77	53	4	35	46	57	539

3.1 Area results

Area results reported in this section have been calculated by analyzing LULC data observed and recorded in the field sample plots. Thus, the resulting figures are different from the maps and estimates derived from the remote sensing (RS) material. The area data derived from the field observations were used to obtain the per hectare statistics which can be multiplied by the official area estimates to obtain total estimates.

Therefore, the results shared in this report are presented as per hectare statistics and proportions between the different LULC types. The results can be used to calculate proportions between the LULC, but cannot substitute the country level area estimates obtained with RS analysis. The official area estimates of RS analysis are not reported here.

3.1.1 Forest and non-forest areas

The area estimates derived from the NFI and used to calculate the per hectare statistics, show a forest area of 6,520,900 ha, equal to 5.7% of the country’s area, and consisting of natural forests (6,147,500 ha) and plantations (373,400 ha). The woodland area is estimated to cover a total of 57,458,700 ha, corresponding to 49.9% of the total land area.

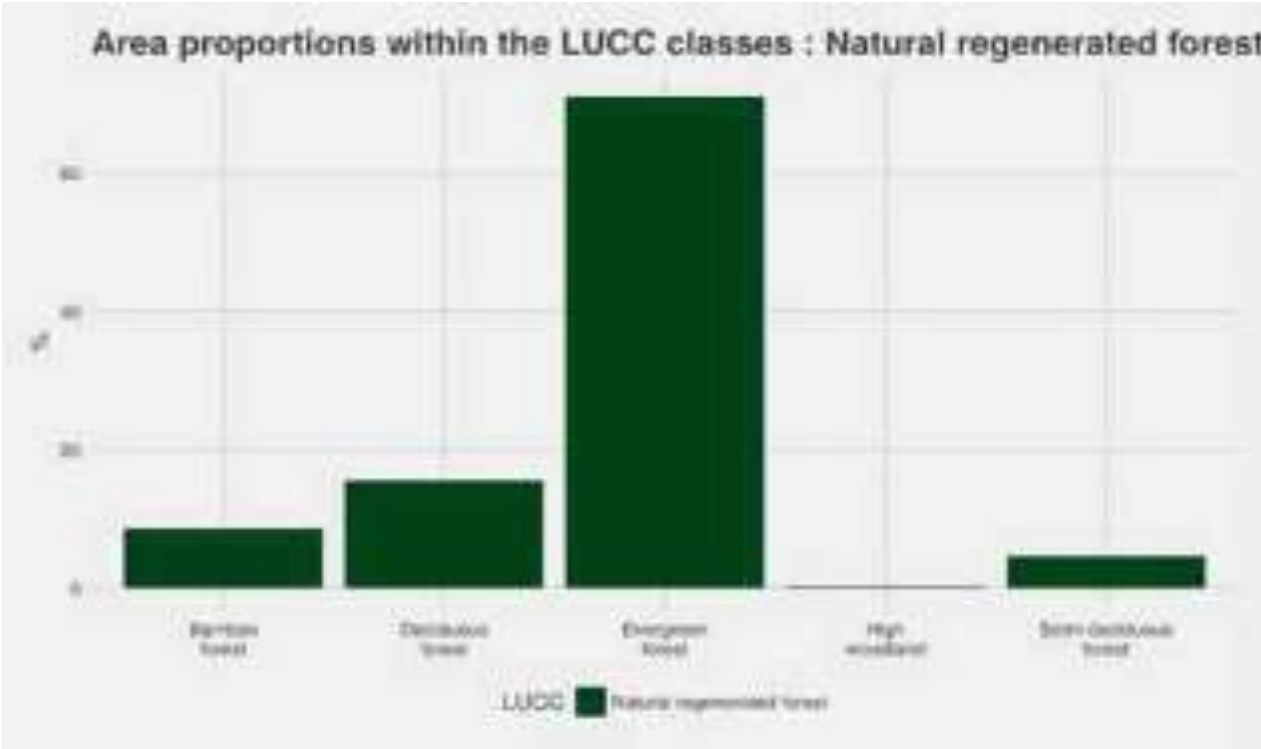
The other two FRA classes, Other Land and Inland Water, occupy 50,703,900 ha (44.1% of the total country area) and 360,400 ha (0.3%), respectively. The complete area results are presented in Appendix A2.

Figure 3.1 Area distribution by FRA classes



In the natural regenerated forests (Figure 3.2), the evergreen forest represents 71.0%, followed by deciduous forest (15.6%) and bamboo forest (i.e. LUCS composed of trees and bamboo, 8.6%).

Figure 3.2 Forest type areas in the natural regenerated forest



In the forest plantations, broadleaved planted forests occupy 76.1%, followed by coniferous forest plantations (19.5%) and forest plantation mixed with bamboo (4.4%).

Figure 3.3 Forest types areas proportion in plantations

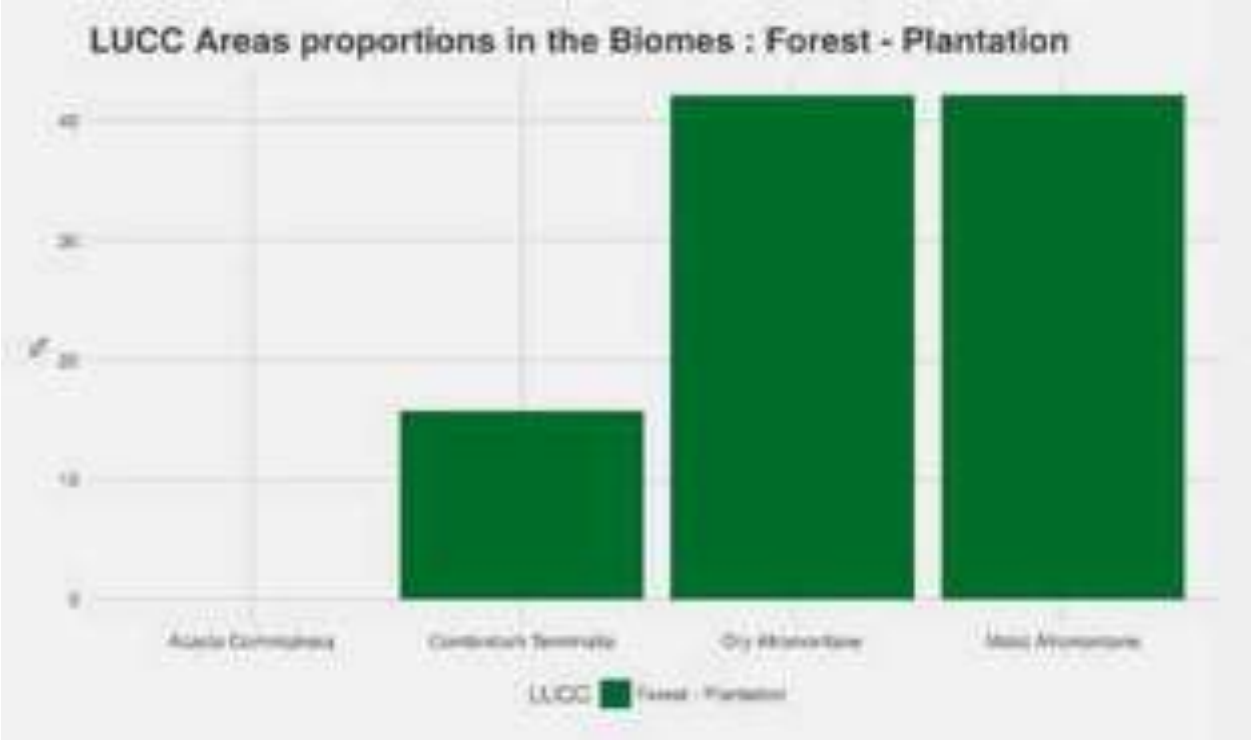
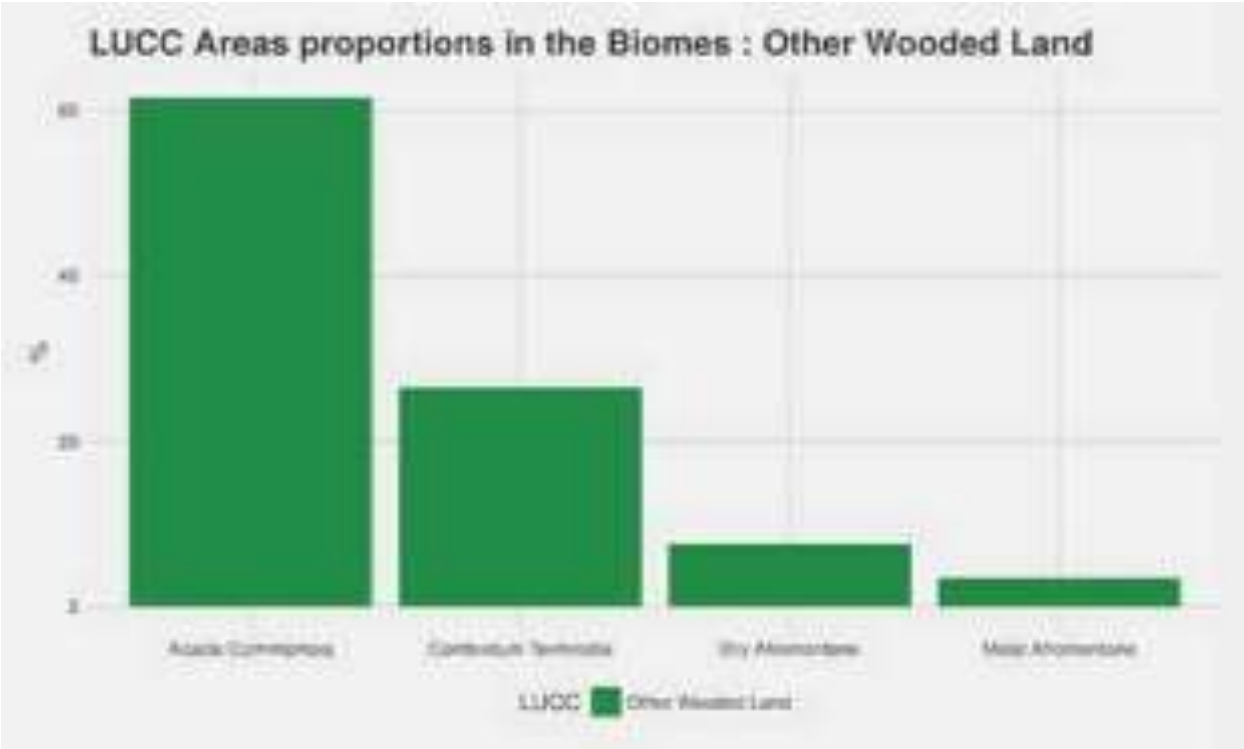


Figure 3.4 Forest types areas proportion in Other Wooded Land



In the Woodland (i.e. non-forest strata according to the forest definition), the pure woodland occupies more than 90.7% of all Woodlands, followed by the wooded grassland (9.0%). The wooded wetlands represent less than 0.3% of the total woodland area.

3.1.2 Land cover area by biomes

In the following figures (Fig. 3.5–3.8), area proportions are presented by the Major LUCC types in each biome⁴⁵. There are total of four biomes in Ethiopia (cf. Chapter 2). The *Acacia- Commiphora* biome area is dominantly covered by Other Wooded Land, 73.1% of the total area in the biome, followed by Natural Other Land (15.7%). In *Combretum-Terminalia*, the dominating Major LUCC class is Other Land - Cultivated (42.5%), followed by Other Land – Natural (8.9%) and Natural regenerated forest (8.3%). The area proportions observed in the Dry Afromontane biome shows a Cultivated Other Land occupies 61.6% of the area, followed by Other Land – Natural (18.7%) and Other Wooded Land (11.2%). The Moist Afromontane is the biome that is covered by Other land - Cultivated (48.2%), followed by the Natural regenerated forest (26.3%).

45 LUCC areas 3% are not displayed

Table 3.2 Area proportions by biomes and Major LUCC types

	<i>Acacia-Commiphora</i>	<i>Combretum-Terminalia</i>	Dry Afromontane	Moist Afromontane	Other
Natural regenerated forest	1.4%	8.3%	4.4%	26.3%	
Forest plantation		0.3%	0.9%	1.3%	
Other wooded land	73.1%	38.7%	11.2%	11.8%	80.0%
Other land – Natural regenerated forest	15.7%	8.9%	18.7%	10.6%	14.5%
Other land - Cultivated	9.1%	42.5%	61.6%	48.2%	5.5%
Other land – Built-up	0.3%	1.0%	3.0%	1.8%	
Other land - Quarry/Mining		0.0%			
Inland water	0.5%	0.2%	0.2%	0.1%	0.0%
Total	100.0%	100.0%	100.0%	100.0%	100.0%

Figure 3.5 Major LUCC areas in the *Acacia-Commiphora* biome

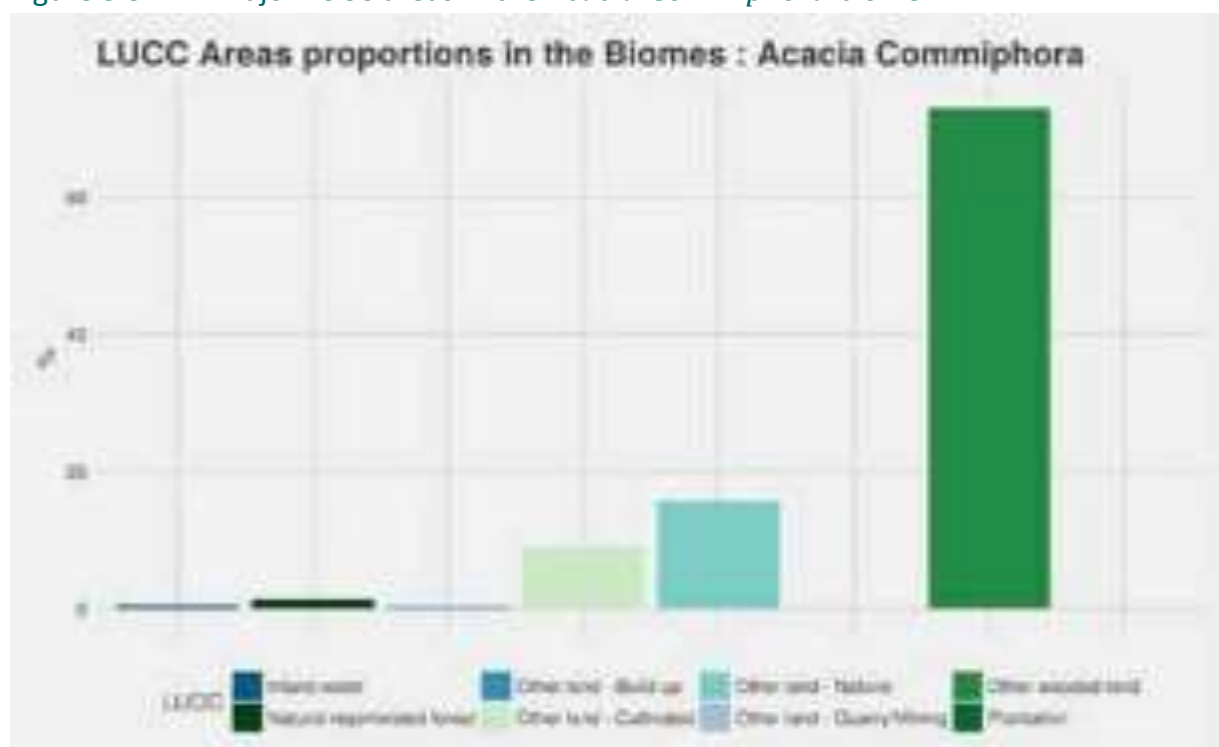


Figure 3.6 Major LUCC areas in the *Combretum-Terminalia* biome

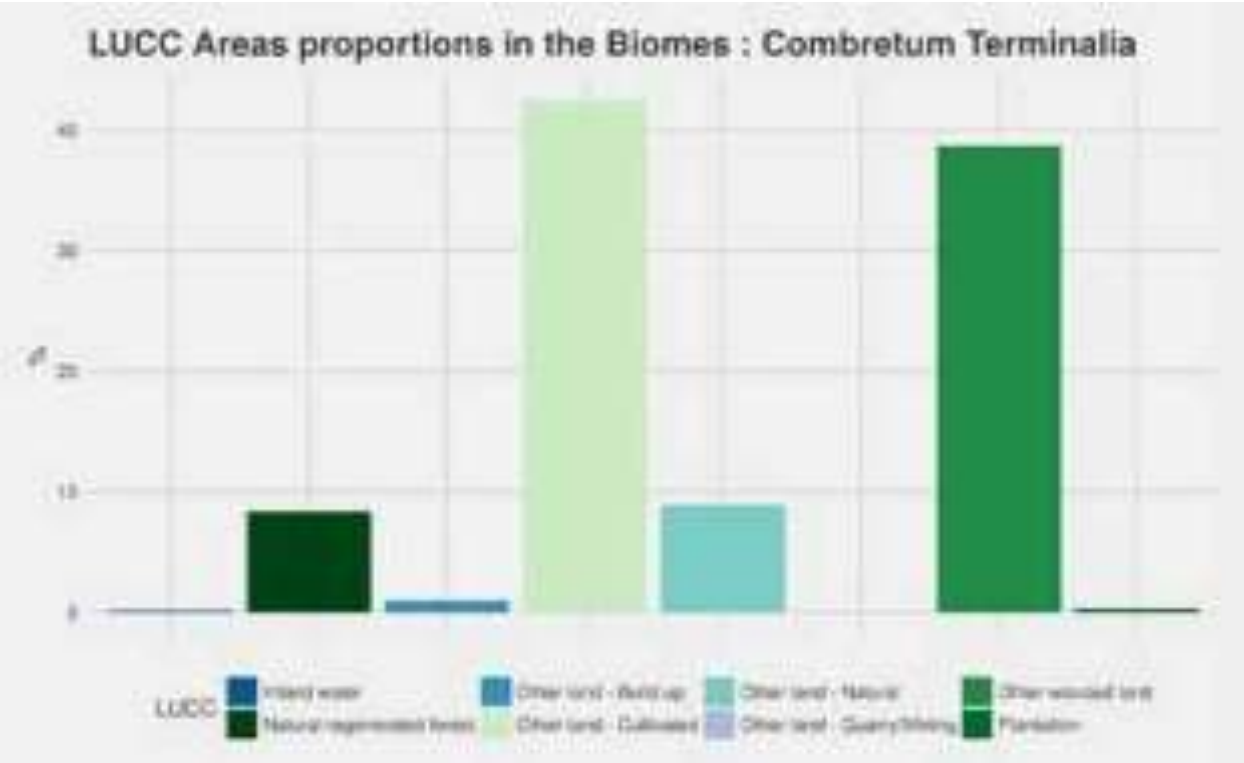


Figure 3.7 Major LUCC areas in the Dry Afromontane biome

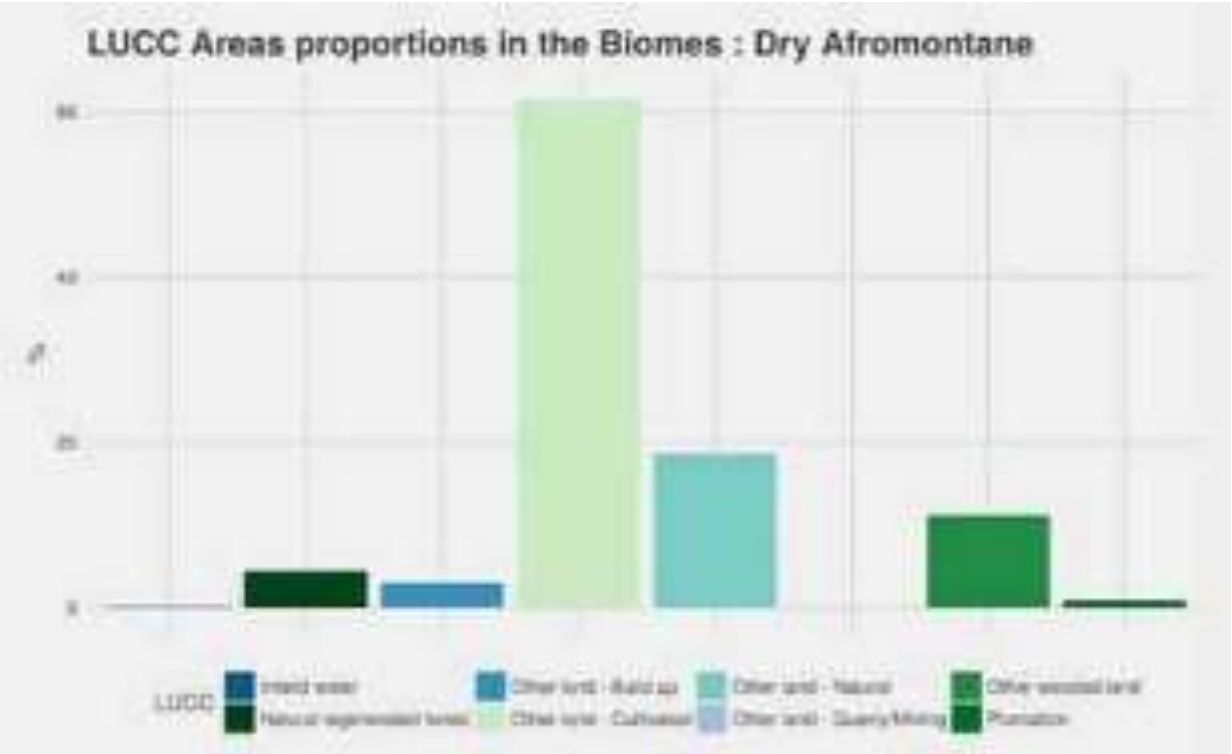


Figure 3.8 Major LUCC areas in the Moist Afromontane biome

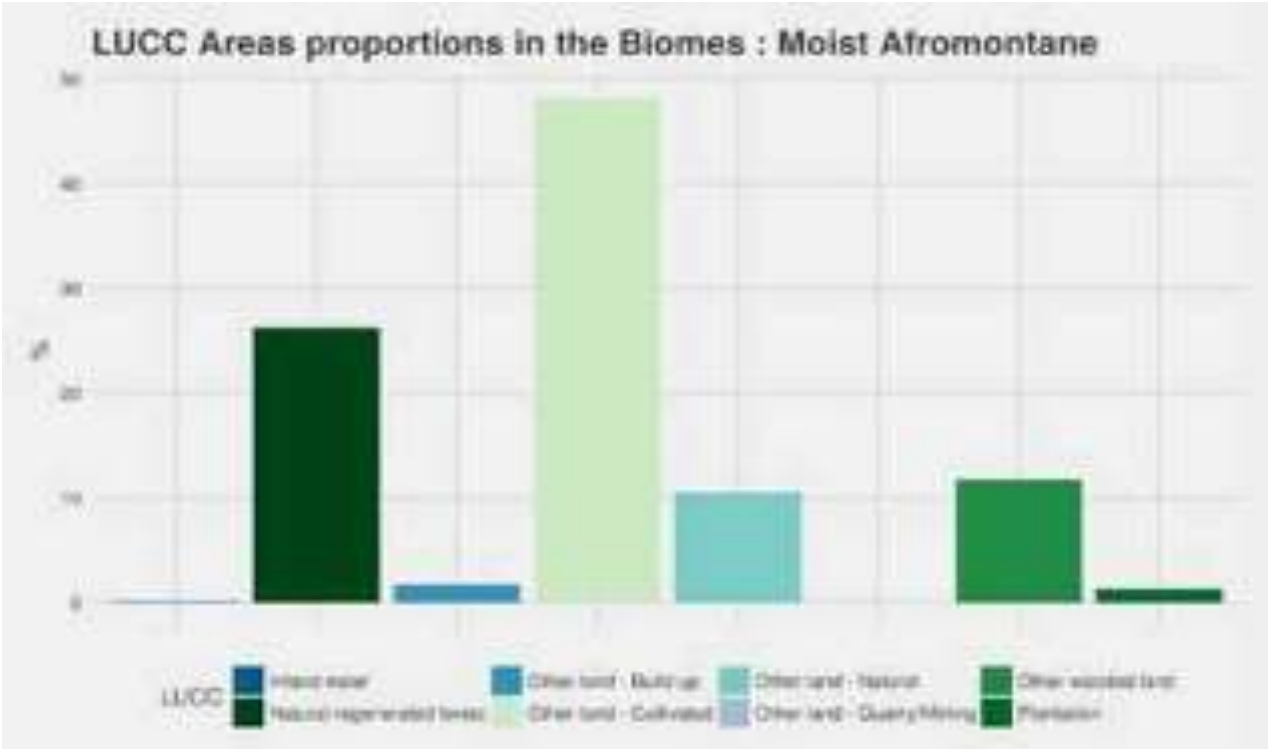
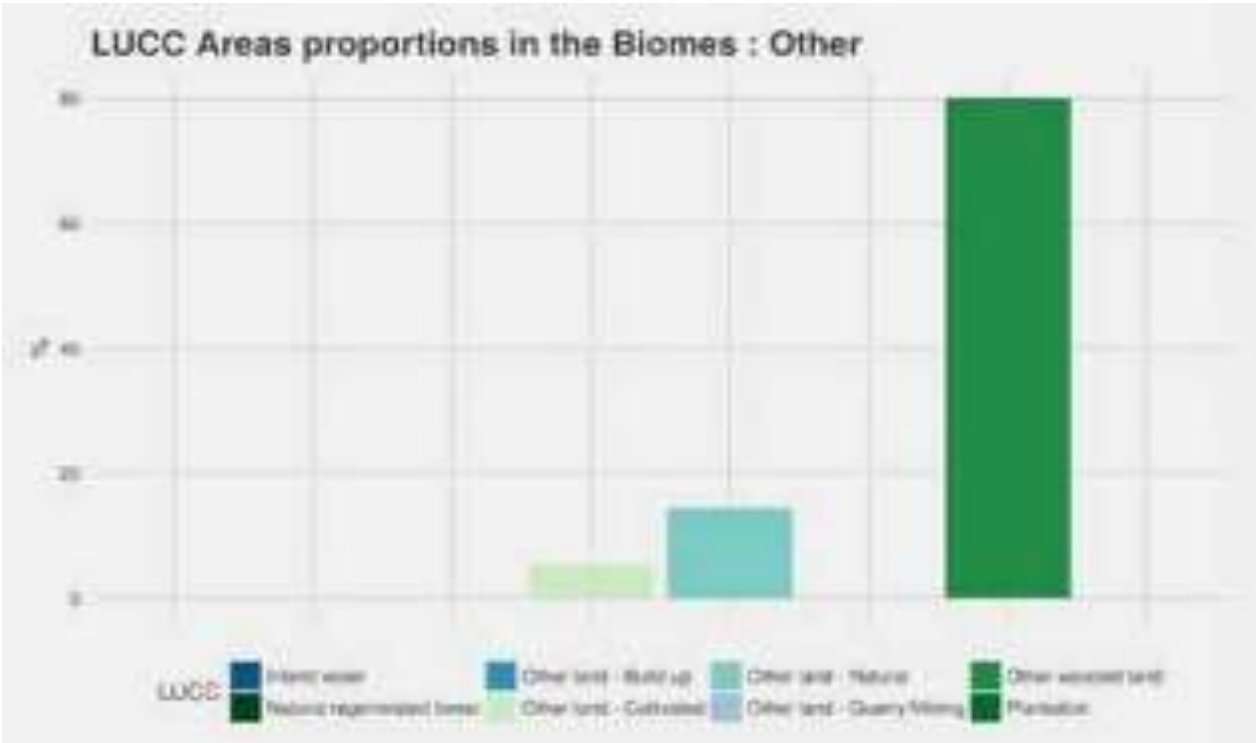


Figure 3.9 Major LUCC areas in “Other” biome class



3.2 Tree density

This section includes the results for the number of trees per hectare in the FRA classes, LUCCs and biomes. The mean numbers of trees per ha in forests and outside of forest is 124.7 and 21.1 respectively.

3.2.1 Tree density by forest and non-forest

On average, forests have a tree density of 124.7 trees ha⁻¹, while non-forests have a tree density of 21.1 trees ha⁻¹. When tree densities are examined by LUC types (Table A3.1), the highest densities are located in semi-deciduous (147.9 trees ha⁻¹) and evergreen forests (124.9 trees ha⁻¹). An ongoing inventory in planted forests will provide more reliable data on tree densities in planted forests LUC.

Table 3.3 Number of trees per hectare in Forest and Non-forest lands

FRA class	N of trees (N ha ⁻¹)
Forest	124.7
Non-forest	21.1

3.2.2 Tree density by biomes

The highest mean density by biomes (containing all land types) is in Moist Afromontane biome (76.4 trees ha⁻¹), followed by *Combretum-Terminalia* (55.9) and *Acacia- Commiphora* (46.0).

Table 3.4 Number of trees by biomes

Biome	N. of trees (N ha ⁻¹)	Relative CI (%)
<i>Acacia-Commiphora</i>	46.0	17%
<i>Combretum-Terminalia</i>	55.9	16%
Dry Afromontane	27.2	23%
Moist Afromontane	76.4	50%
Other	16.2	254%

3.2.3 Tree density by diameter classes

Tree densities by DBH classes typically follow a reversed J-shape curve in natural forests⁴⁶. Therefore the highest number of trees is found in the smaller DBH classes. Big trees (DBH ≥ 40cm) constitute 16% of all trees in the forest land.

46 Leak, 2002.

Table 3.5 Number of trees per hectare by FRA and diameter classes

FRA class	10-14.9 cm	15-19.9 cm	20-24.9 cm	25-29.9 cm	30-34.9 cm	35-39.9 cm	40+ cm
Forest	26.4	22.2	23.8	15.8	9.7	6.5	20.2
Other Wooded Land	29.6	15.9	8.2	3.5	1.9	1.0	1.3
Other Land	5.5	4.3	3.6	2.3	1.7	1.1	2.6
Water	3.4	8.2	5.5	3.3	2.1	0.5	2.3

Table 3.6 Number of trees per hectare by biomes and diameter classes

Biome	10-14.9 cm	15-19.9 cm	20-24.9 cm	25-29.9 cm	30-34.9 cm	35-39.9 cm	40+ cm
<i>Acacia-Commiphora</i>	23.9	11.7	5.8	2.1	1.1	0.5	0.8
<i>Combretum-Terminalia</i>	15.2	12.4	10.2	6.7	4.3	2.7	4.5
<i>Dry Afromontane</i>	8.8	5.9	4.4	2.6	1.7	1.0	2.6
<i>Moist Afromontane</i>	17.5	16.7	12.7	7.2	5.0	3.5	13.7
Other	14.5	1.4	0.3				

3.2.4 Tree density by height classes

Mean tree density by 5cm range tree height classes show that the highest density of trees (DBH $\geq$ 10cm) is in the height class 10-14.9m in Forest and OWL, and in the class 5-9.9m in Other Land.

The results by biomes show that the highest proportion of trees taller than 20m in Forest are found in Moist Afromontane biome (29% of trees), whereas Other Wooded Land contains relatively few trees taller than 20m.

Table 3.7 Number of trees per hectare by FRA and height classes

FRA Class	<5m	5-9.9m	10-14.9m	15-19.9m	20-24.9m	25-29.9m	30-34.9m	35-39.9m	40+ m
Forest	0.3	28.3	40.6	24.8	12.7	11.5	4.8	1.3	0.4
Other Wooded Land	3.0	44.6	10.1	2.6	0.6	0.2	0.1		
Other Land	0.4	9.5	6.4	3.2	0.9	0.4	0.1	0.1	
Water	0.2	15.7	5.3	2.5		1.5			

3.3 Regeneration count

3.3.1 Regeneration count by FRA classes

Small tree and tree regeneration data contains observations where tree height is more or equal to 1.3 m and DBH less than 10 cm. In natural regenerated forest the highest mean density of saplings is located in Bamboo forest (4,759 trees ha⁻¹), followed by Evergreen and Deciduous forest types, where the mean densities are 3,553 and 2,886, respectively.

Table 3.8 Regeneration count and AGB by LUCC types

FRA class	Major LUCC	LUCC	Regeneration count (N ha ⁻¹)	AGB (t ha ⁻¹)
Forest	Natural regenerated forest	Evergreen forest	3552.9	5.7
Forest	Natural regenerated forest	Deciduous forest	2885.8	4.6
Forest	Natural regenerated forest	Semi-deciduous forest	2137.3	3.4
Forest	Natural regenerated forest	Bamboo forest	4759.2	7.6
Forest	Natural regenerated forest	High woodland	800.0	1.3
Forest	Plantation	Broadleaved planted forest	4445.8*	7.1*
Forest	Plantation	Coniferous planted forest	11778.7*	18.9*
Other Wooded Land	Other wooded land	Woodland	2067.5	3.3
Other Wooded Land	Other wooded land	Wooded grassland	2479.5	4.0
Other Wooded Land	Other wooded land	Wooded wetland	2297.0	3.7

*Results not sufficiently representative. Separate inventory undertaken for plantations

3.3.2 Regeneration count by biomes

The number of trees per hectare by biomes in regeneration data shows the highest stocking in the Dry Afromontaine biome (6,423), followed by the biome identified as 'Other' (5,382, this biome is a very small by area and it falls dominantly into non-permanent rivers). The third highest density is found in the *Acacia Commiphora* biome (4,481).

Table 3.9 **Regeneration count and AGB by biomes**

Biome	FRA class	Count (N ha ⁻¹)	AGB (t ha ⁻¹)	BGB (t ha ⁻¹)
<i>Acacia-Commiphora</i>	Forest	4,481.1	5.7	2.2
<i>Acacia-Commiphora</i>	Other Wooded Land	2,071.2	2.6	1.0
<i>Combretum-Terminalia</i>	Forest	2,667.5	3.4	1.2
<i>Combretum-Terminalia</i>	Other Wooded Land	2,000.8	2.5	0.9
Dry Afromontane	Forest	6,422.7	8.1	2.2
Dry Afromontane	Other Wooded Land	2,032.4	2.6	0.7
Moist Afromontane	Forest	3,076.3	3.9	1.0
Moist Afromontane	Other Wooded Land	1,568.5	2.0	0.5
Other	Other Wooded Land	5,382.4	6.8	2.7

3.4 Basal area

3.4.1 Basal area by FRA classes and forest type

The basal area by FRA classes shows a similar pattern to tree density and volume. The mean basal area in Ethiopian forest is 13.3 m² ha⁻¹, and 1.7 m² ha⁻¹ in Other Wooded Land. Table 3.11 shows that evergreen forest has the highest basal area (15.6 m² ha⁻¹), followed closely by coniferous planted forests (14.8 m² ha⁻¹).

Table 3.10 **Tree basal area by FRA classes**

FRA Class	Basal area (m ² ha ⁻¹)
Forest	13.3
Other Wooded Land	1.7
Other Land	1.9
Water	1.3

Table 3.11 Tree basal area by LUCC types

FRA class	Major LUCC	LUCC	Basal area (m ² ha ⁻¹)
Forest	Natural regenerated forest	Evergreen forest	15.6
Forest	Natural regenerated forest	Deciduous forest	10.6
Forest	Natural regenerated forest	Semi-deciduous forest	7.1
Forest	Natural regenerated forest	Bamboo forest	5.8
Forest	Natural regenerated forest	High woodland	3.9
Forest	Plantation	Broadleaved planted forest	7.5
Forest	Plantation	Coniferous planted forest	14.8
Forest	Plantation	Forest plantation of bamboo	3.8
Other Wooded Land		Woodland	1.7
Other Wooded Land		Wooded grassland	1.9
Other Wooded Land		Wooded wetland	1.5
Other Land	Other land - Natural	Natural forest coffee	46.7*
Other Land	Other land - Natural	Barren land	0.1
Other Land	Other land - Natural	Natural grassland	1.0
Other Land	Other land - Natural	Marsh	0.5
Other Land	Other land - Cultivated	Coffee plantation	20.5*
Other Land	Other land - Cultivated	Annual crop	1.5
Other Land	Other land - Cultivated	Perennial crop	8.8
Other Land	Other land - Cultivated	Mixed annual and perennial crop	3.3
Other Land	Other land - Cultivated	Fallow	1.1
Other Land	Other land - Cultivated	Wood lot of bamboo	1.0
Other Land	Other land - Cultivated	Wood lot	5.0
Other Land	Other land – Built-up	Built-up area	2.6
Other Land	Other land - Quarry/ Mining	Quarry/Mining site	1.7
Water	Inland water	Perennial river	2.2
Water	Inland water	Intermittent river (seasonal)	1.0

*Results not sufficiently representative.

3.4.2 Basal area by biomes

The basal area results (in all LUCC) by biomes shows that the highest basal area is found in the Moist Afromontane biome (10.3 m² ha⁻¹). All other four types have basal area less than 4 m² ha⁻¹.

Table 3.12 Tree basal area by biomes

Biome	Basal area (m ² ha ⁻¹)
<i>Acacia-Commiphora</i>	1.2
<i>Combretum-Terminalia</i>	3.1
Dry Afromontane	1.9
Moist Afromontane	10.3
Other	0.2

3.4.3 Basal area by DBH and FRA classes

Table 3.13 Tree basal area (m² ha⁻¹) by DBH and FRA classes

FRA class	10-14.9 cm	15-19.9 cm	20-24.9 cm	25-29.9 cm	30-34.9 cm	35-39.9 cm	40+ cm
Forest	0.3	0.5	0.9	0.9	0.8	0.7	9.2
Other Wooded Land	0.3	0.4	0.3	0.2	0.1	0.1	0.3
Other Land	0.1	0.1	0.1	0.1	0.1	0.1	1.2
Water	0	0.2	0.2	0.2	0.2	0.1	0.5

3.5 Volume

3.5.1 Volume by Forest and Non-forest areas

The mean stem volume in Forest in Ethiopia is 128.3 m³ ha⁻¹. Although Other Wooded Land dominates in terms land area, Forest land still contains more timber than OWL. Other Land contains a high volume of timber (707 million m³), especially when compared to forests (836 million m³).

Table 3.14 Volume by FRA classes

FRA class	Volume (m ³ ha ⁻¹)	Relative CI (%)
Forest	128.3	22%
Other Wooded Land	7.4	19%
Other Land	13.9	45%
Water	7.2	133%

The highest total volume in forest found in Evergreen forest type, with an average volume per hectare of 158.1 m³ ha⁻¹. All other forest types constitute just 17% of the total volume in Forest. Woodlands and Annual crop lands also contain remarkable growing stocks, 378 million m³ and 248 million m³, respectively.

Table 3.15 Tree Volume per FRA classes and LUCC

FRA class	Major LUCC	LUCC	Volume (m ³ ha ⁻¹)	Relative CI (%)
Forest	Natural regenerated forest	Evergreen forest	158.1	27%
Forest	Natural regenerated forest	Deciduous forest	87.0	40%
Forest	Natural regenerated forest	Semi-deciduous forest	27.7	36%
Forest	Natural regenerated forest	Bamboo forest	52.3	82%
Forest	Natural regenerated forest	High woodland	34.3	-
Forest	Plantation	Broadleaved planted forest	69.0	58%
Forest	Plantation	Coniferous planted forest	95.3	84%
Forest	Plantation	Forest plantation of bamboo	28.7	-
Other Wooded Land		Woodland	7.3	21%
Other Wooded Land		Wooded grassland	9.0	33%
Other Wooded Land		Wooded wetland	7.4	130%
Other Land	Other land - Natural	Natural forest coffee	601.2*	87%
Other Land	Other land - Natural	Barren land	0.2	88%
Other Land	Other land - Natural	Natural grassland	5.8	35%
Other Land	Other land - Natural	Marsh	3.6	229%
Other Land	Other land - Cultivated	Coffee plantation	181.4*	95%
Other Land	Other land - Cultivated	Annual crop	8.8	20%
Other Land	Other land - Cultivated	Perennial crop	84.6	81%
Other Land	Other land - Cultivated	Mixed annual and perennial crop	31.5	74%

Table 3.15 Tree Volume per FRA classes and LUCC

FRA class	Major LUCC	LUCC	Volume (m ³ ha ⁻¹)	Relative CI (%)
Other Land	Other land - Cultivated	Fallow	6.5	78%
Other Land	Other land - Cultivated	Wood lot of bamboo	9.6	-
Other Land	Other land - Cultivated	Wood lot	32.3	147%
Other Land	Other land - Build up	Built-up area	17.5	68%
Other Land	Other land - Quarry/Mining	Quarry/Mining site	7.2	-
Water	Inland water	Perennial river	17.0	199%
Water	Inland water	Intermittent river (seasonal)	3.5	55%

*Results not sufficiently representative

3.5.2 Tree volume by biomes

The highest mean volume in biomes is found in Moist Afromontane type, 100 m³ ha⁻¹ and 854millionm³, respectively. Acacia Commiphora biome contains only 5.5m³ha⁻¹.

Table 3.16 Tree volume by biomes

Biome	Volume (m ³ ha ⁻¹)	Relative CI (%)
<i>Acacia-Commiphora</i>	5.5	35%
<i>Combretum-Terminalia</i>	19.5	22%
Dry Afromontane	13.8	40%
Moist Afromontane	100.0	39%
Other	0.4	115%

3.5.3 Volume by DBH classes

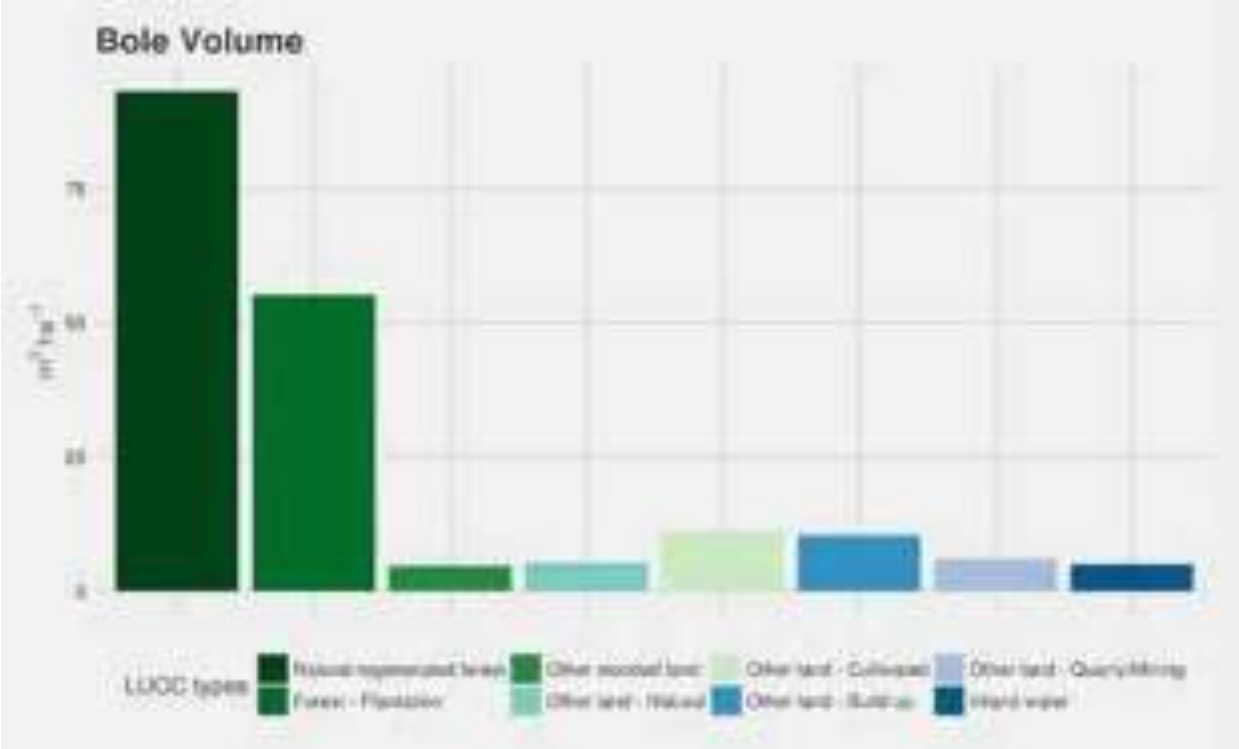
Table 3.17 Volume (m³ ha⁻¹) by height and FRA classes

FRA Class	<5m	5-9.9m	10-14.9m	15-19.9m	20-24.9m	25-29.9m	30-34.9m	35-39.9m	40+ m
Forest	0	3	10.3	16.1	18.5	30.4	23.8	14.7	11.5
Other Wooded Land	0.1	2.8	2.1	1.3	0.6	0.4	0.3	0	
Other Land	0	1.1	2.1	3.3	1.7	2.3	1.4	0.9	0.9
Water	0	1.6	1.4	1.2		3			

3.6 Bole volume

Bole volume refers to the merchantable part of stem, up to trees first big branch. Because bole vole rather a technical measure of the stem, it cannot give exact information on the commercial value of the forest. However bole volume can serve as an indicator for timber harvesting potential.

Figure 3.10 Tree bole volume by Major LUCC types



The highest mean bole volume can be found in the Natural regenerated forest (93.1 m3 ha⁻¹), following the forest plantation (55.1 m3 ha⁻¹). The forest plantations in Ethiopia contain approximately 12.7% of the total bole volume stock, mainly located in the coniferous planted forests (6.8%), followed by the broadleaved planted forests (4.1%) and planted forests with bamboo (1.7%). The mean bole volumes are rather low in other land categories varying from 4.7 to 11.3 m3 ha⁻¹. The 79% of the bole volume in the natural forest strata is stored in the trees with DBH greater than 40 cm.

Table 3.18 Bole volume (m³ ha⁻¹) by DBH and FRA classes

FRA Class	10-14.9 cm	15-19.9 cm	20-24.9 cm	25-29.9 cm	30-34.9 cm	35-39.9 cm	40+ cm
Forest	1.3	2.9	4.6	5.5	4.9	4.8	87.7
Other Wooded Land	0.7	0.9	0.9	0.6	0.5	0.4	1.4
Other Land	0.2	0.4	0.6	0.6	0.7	0.5	8.3
Water	0.1	0.7	0.8	1.0	1.6	0.5	2.5

Within the forest plantation type, bole volume seems to be more evenly distributed along the DBH classes than in the other land types, and only the 37% of it is stored in the bigger diameter classes (i.e. trees DBH ≥ 35 cm). The general overview suggests a possibility to increase more rational management planning in the forest plantations, and at the same time there are possibilities to support silvicultural practices in the Natural regenerated forests, at least areas with low stockings.

Table 3.19 Tree bole volume (m³ ha⁻¹) by DBH classes in biomes and FRA classes

FRA class	Major LUCC	10- 14.9 cm	15- 19.9 cm	20- 24.9 cm	25- 29.9 cm	30- 34.9 cm	35- 39.9 cm	40+ cm
Forest	Natural regenerate d forest	1.0	2.6	4.1	5.0	4.7	4.7	91.1
Forest	Plantation	6.1	7.1	12.7	14	6.9	6.7	31.5
Other Wooded Land		0.7	0.9	0.9	0.6	0.5	0.4	1.4
Other Land	Natural	0.2	0.2	0.2	0.2	0.2	0.2	5.8
Other Land	Cultivated	0.3	0.4	0.7	0.7	0.8	0.7	9.6
Other Land	Built-up	0.7	1.6	2.5	2.4	2.2	1.2	2.9
Other Land	Quarry/ Mining		0.6					
Water	Inland water	0.1	0.7	0.8	11.6	0.5	2.5	

3.7 Biomass and carbon stocks

The AGB estimates have been calculated by using allometric equations, the BGB as fraction of the ABG. The carbon fraction of the biomass is obtained by using default values. The deadwood fraction is calculated separately. The carbon in litter using the measurement collected in the field is described in a separate report together with the soil organic carbon (SOC) estimates.

Forest land types contain the highest mean above-ground biomass in Ethiopia: Natural regenerated forest has 125 t ha⁻¹ and Forest plantations 71.3 t ha⁻¹. All other major LUCC types contain less than 18 t ha⁻¹.

Figure 3.11 AGB by Major LUCC types

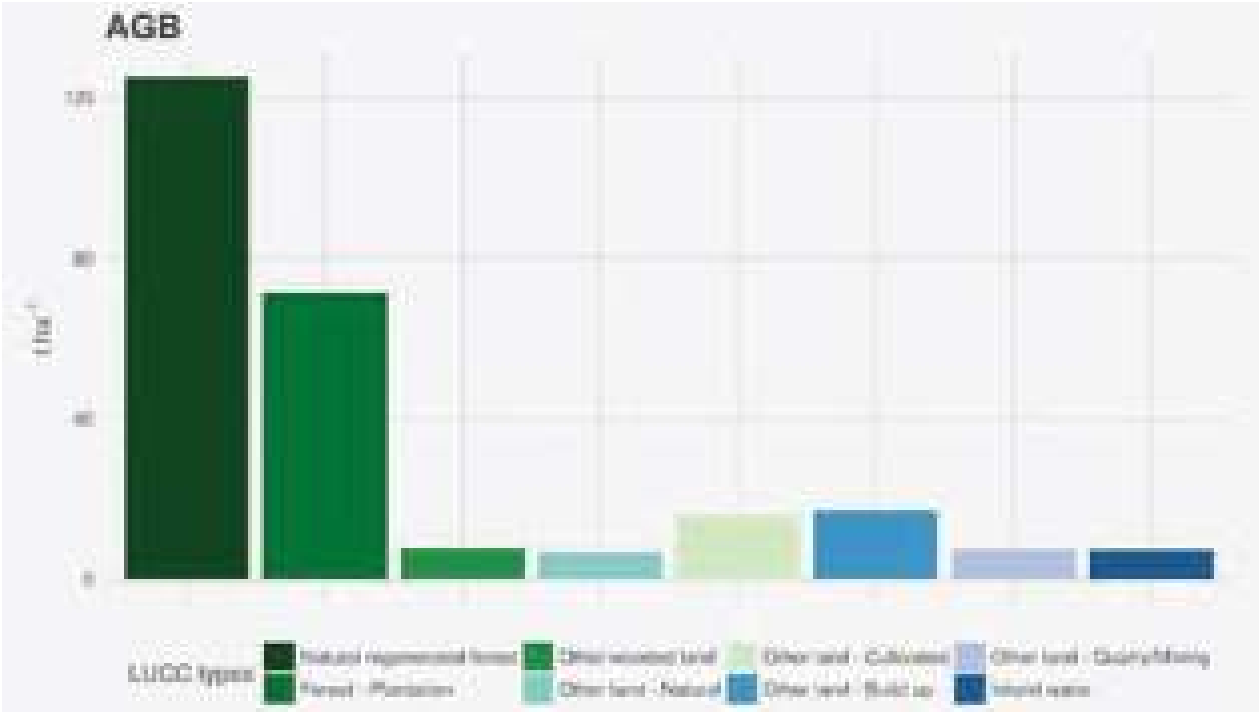
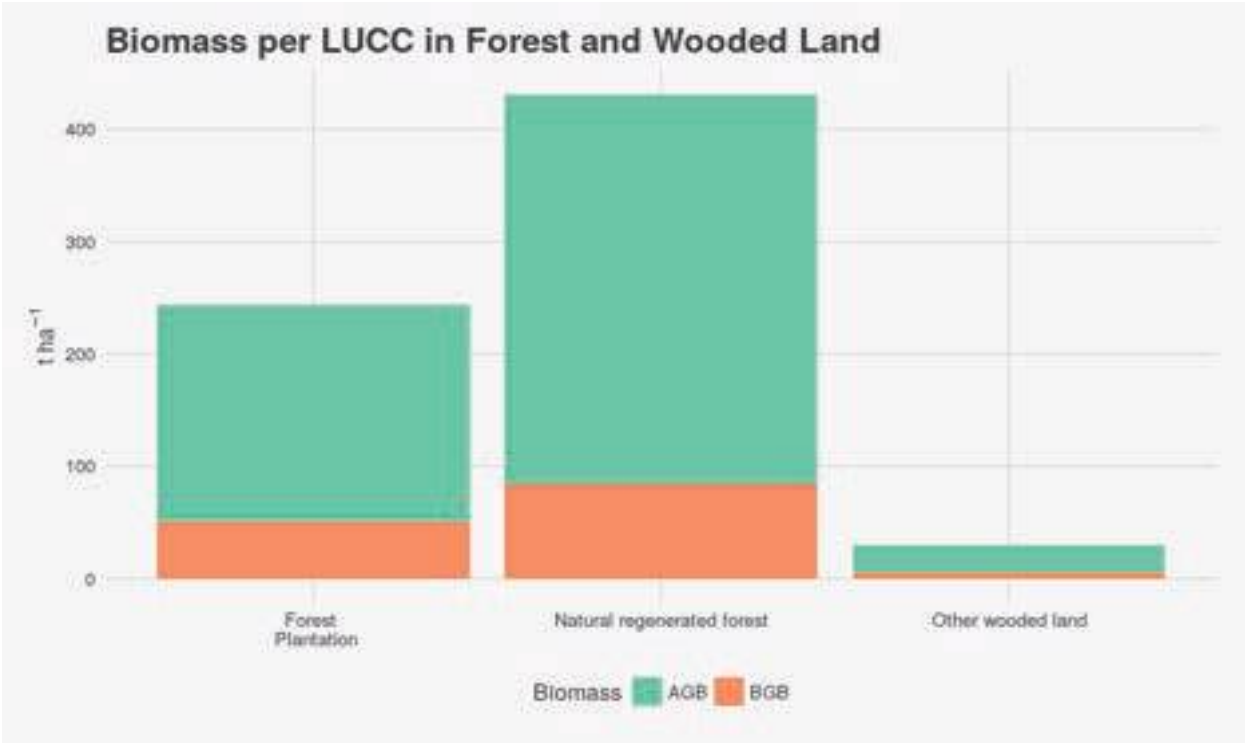


Figure 3.12 AGB and BGB in Forest and Other Wooded Land



The combined mean biomass (AGB+BGB) by Major LUCC types is presented in the chart 3.11. The biomass in natural regenerated forest is 155.4 t ha⁻¹, in Forest plantations 90.5 t ha⁻¹ and in OWL 9.5 t ha⁻¹. The below ground-biomass is approximately 20% of the total biomass.

Table 3.20 AGB by Major LUCC types

FRA class	Major LUCC	AGB (t ha ⁻¹)	Relative CI (%)
Forest	Natural regenerated forest	125.1	25%
Forest	Plantation	71.3	44%
Other Wooded Land		7.6	19%
Other Land	Other land - Natural	6.7	77%
Other Land	Other land - Cultivated	16.4	56%
Other Land	Other land – Built-up	17.1	70%
Other Land	Other land - Quarry/Mining	7.3	-
Water	Inland water	7.2	133%

The stump contribution to the AGB, BGB and carbon is presented in Table 3.21. Stumps with diameter of more or equal to 10 cm were recorded in Forest Land. In general, stumps represent a minor part of forest biomass and carbon in Ethiopian forests and lands outside of forests. The estimated biomass of stumps is approximately 1.1 t ha⁻¹ in both Major LUCC types in the forest land.

Table 3.21 Stump biomass and carbon by Major LUCC types

FRA class	Major LUCC	AGB (t ha ⁻¹)	BGB (t ha ⁻¹)	Biomass (t ha ⁻¹)	AGC (t ha ⁻¹)	BGC (t ha ⁻¹)	Carbon (t ha ⁻¹)
Forest	Natural regenerated forest	0.2	0.9	1.1	0.1	0.5	0.6
Forest	Plantation	0.2	0.9	1.1	0.1	0.5	0.6
Other Wooded Land		0	0.1	0.1	0	0	0
Other Land	Natural	0	0.1	0.1	0	0	0.1
Other Land	Cultivated	0.1	0.1	0.2	0	0.1	0.1
Other Land	Built-up	0.2	0.4	0.6	0.1	0.2	0.3
Water	Inland water	0	0	0	0	0	0

Bamboo was a rather rare plant within the NFI sampling plots. It only occurred in two major LUCC types: Natural regenerated forest and OWL. In Natural regenerated forest the mean bamboo AGB is 6.6 t ha⁻¹ which is about 4% of the AGB of trees in this forest type.

Table 3.22 Bamboo AGB, biomass and carbon by Major LUCC types

Major LUCC	AGB (t ha ⁻¹)	Biomass (t ha ⁻¹)	Carbon (t ha ⁻¹)
Natural regenerated forest	6.6	8.3	4.2
Other wooded land	0.7	0.9	0.4

Table 3.23 Carbon in deadwood by Major LUCC types

FRA class	Major LUCC	Carbon (t ha ⁻¹)
Forest	Natural regenerated forest	15.8
Forest	Plantation	0.5
Other Wooded Land	Other wooded land	1.9
Other Land	Cultivated	2.6
Other Land	Natural	0.9

3.8 Results by biomes – Forest only

The combined main result variables by biomes were reported earlier across all land types. This Section presents forest-related results by biomes based solely on data collected in forest LUCSs.

Table 3.24 Main result variables by biomes, forest only data

Biome	Count (N ha ⁻¹)	Basal area (m ² ha ⁻¹)	Volume (m ³ ha ⁻¹)	Bole volume (m ³ ha ⁻¹)	AG biomass (t ha ⁻¹)	BG biomass (t ha ⁻¹)	Biomass (t ha ⁻¹)	AG carbon (t ha ⁻¹)	BG carbon (t ha ⁻¹)	Carbon (t ha ⁻¹)
<i>Acacia-Commiphora</i>	97.3	6.5	58.2	36.3	56.8	20.0	76.8	28.4	10.0	38.4
<i>Combretum-Terminalia</i>	117.4	8.4	66.8	44.2	65.5	26.9	92.4	32.7	13.4	46.2
Dry Afromontane	116.2	12.4	120.0	87.7	114.4	30.9	145.3	57.2	15.5	72.7
Moist Afromontane	145.1	20.7	216.1	156.8	202.9	53.9	256.9	101.5	27.0	128.4

The results for mean density (forest only) and volume by diameter class and biome are presented in Tables 3.25 and 3.26. The Moist Afromontane biome contains a high timber stock (189 m³ ha⁻¹) among trees with DBH greater than 40 cm at DBH, which is a double when compared to the second category (Dry Afromontane, 96.3 m³ ha⁻¹).

Table 3.25 Mean tree density by diameter class and biome (in forests)

Biome	Count (N ha ⁻¹)						
	10-14.9 cm	15-19.9 cm	20-24.9 cm	25-29.9 cm	30-34.9 cm	35-39.9 cm	40+ cm
<i>Acacia-Commiphora</i>	23.1	12.9	25.1	12.5	9.2	4.8	9.6
<i>Combretum-Terminalia</i>	20.8	16.7	26.6	20.4	12.0	7.0	13.9
Dry Afromontane	26.5	20.8	21.8	14.0	8.0	5.4	19.6
Moist Afromontane	33.1	31.5	21.5	13.3	8.5	7.0	30.3

Table 3.26 Mean volume by diameter class and biome (in forests)

Biome	Volume (m ³ ha ⁻¹)						
	10-14.9 cm	15-19.9 cm	20-24.9 cm	25-29.9 cm	30-34.9 cm	35-39.9	40+ cm
<i>Acacia-Commiphora</i>	1.1	1.6	5.5	4.2	5.1	3.6	37.1
<i>Combretum-Terminalia</i>	1.0	1.8	5.6	7.8	7.4	6.3	36.9
Dry Afromontane	1.7	2.2	5.0	5.6	4.6	4.5	96.3
Moist Afromontane	1.7	4.7	4.9	5.2	4.8	5.8	189.0

3.9 Tree species and biodiversity

A total of 360 tree species has been identified during the data collection and the quality control operations. The tree species have been then ranked following two different approaches: species names are sorted in descending order of total frequency and by total stem volume. The first criterion requires counting the total number of stems by species in the category of interest, and the second one is calculation of the total tree volumes accordingly. Besides the ranking, the Shannon index estimates are computed.

The unknown species have not been included in the analysis. However, unknown species represent approximately 20% of the total number of trees observed in the NFI.

Figures 3.13 and 3.14 show the 30 most abundant tree species and the 30 tree species with highest volume stock in Ethiopia, respectively.

Figure 3.13 30 most abundant tree species by frequency in the Ethiopia

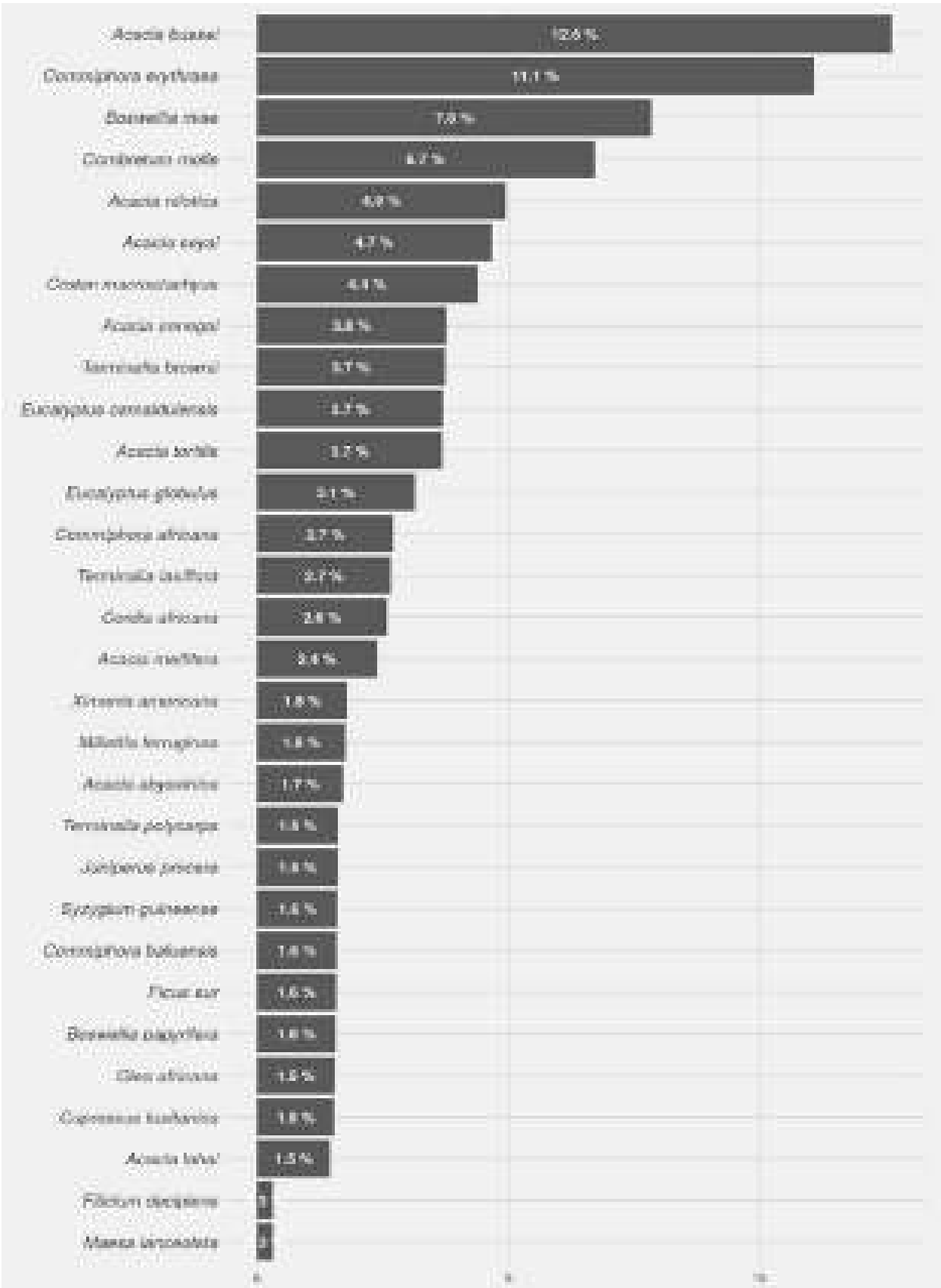
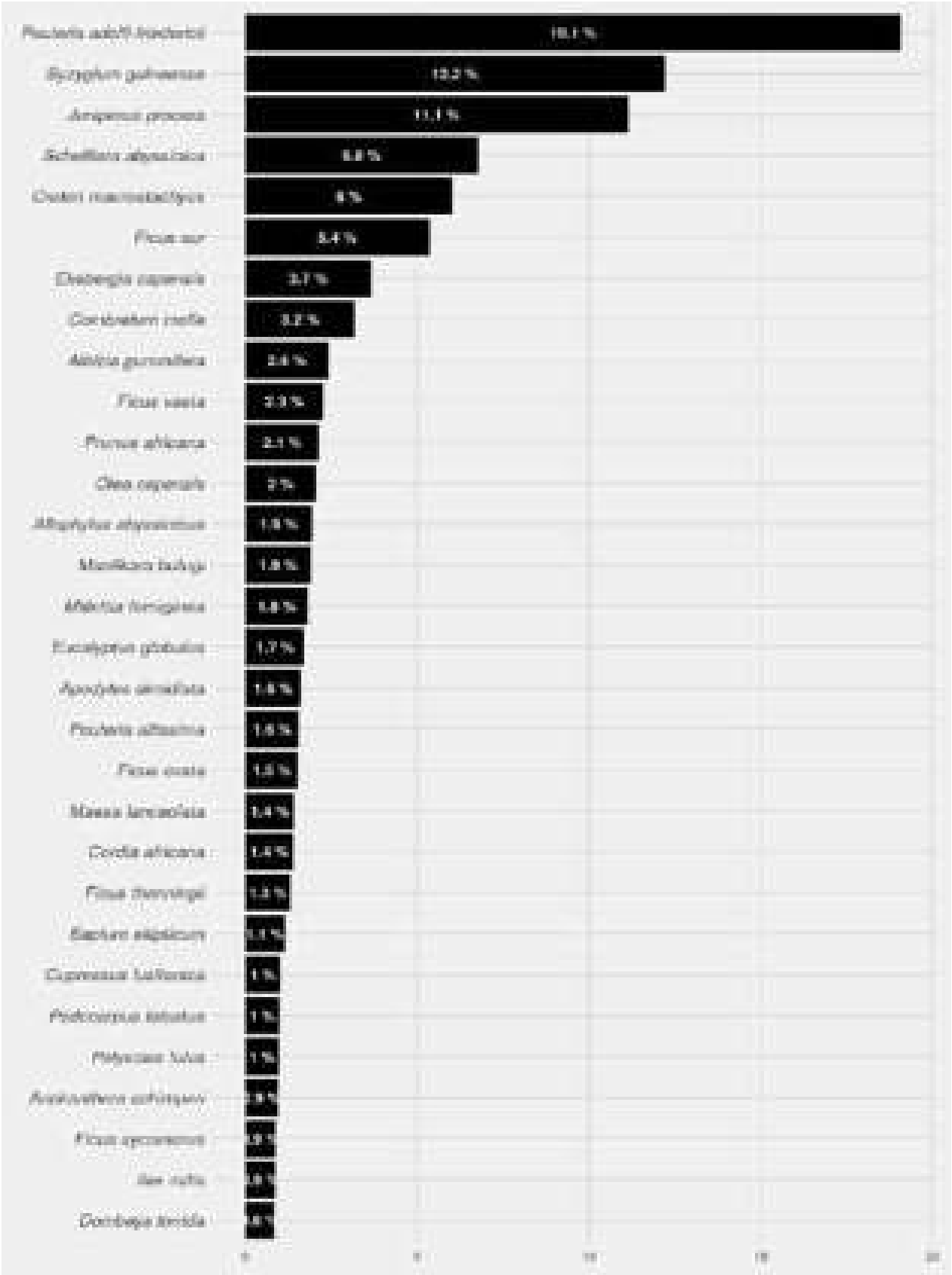


Figure 3.14 30 most abundant tree species by stem volume in Ethiopia



Considering the occurrence of the species in the different FRA classes, the most frequent species in Forest are *Croton macrostachyus* (4.9 trees ha⁻¹), *Syzygium guineense* (4.6 trees ha⁻¹) and *Juniperus procera* (3.0 trees ha⁻¹). In OWL, *Acacia bussei* (5.7 trees ha⁻¹) is the most frequent, followed by *Commiphora erythraea* (4.8 trees ha⁻¹) and *Boswellia rivae* (3.7 trees ha⁻¹).

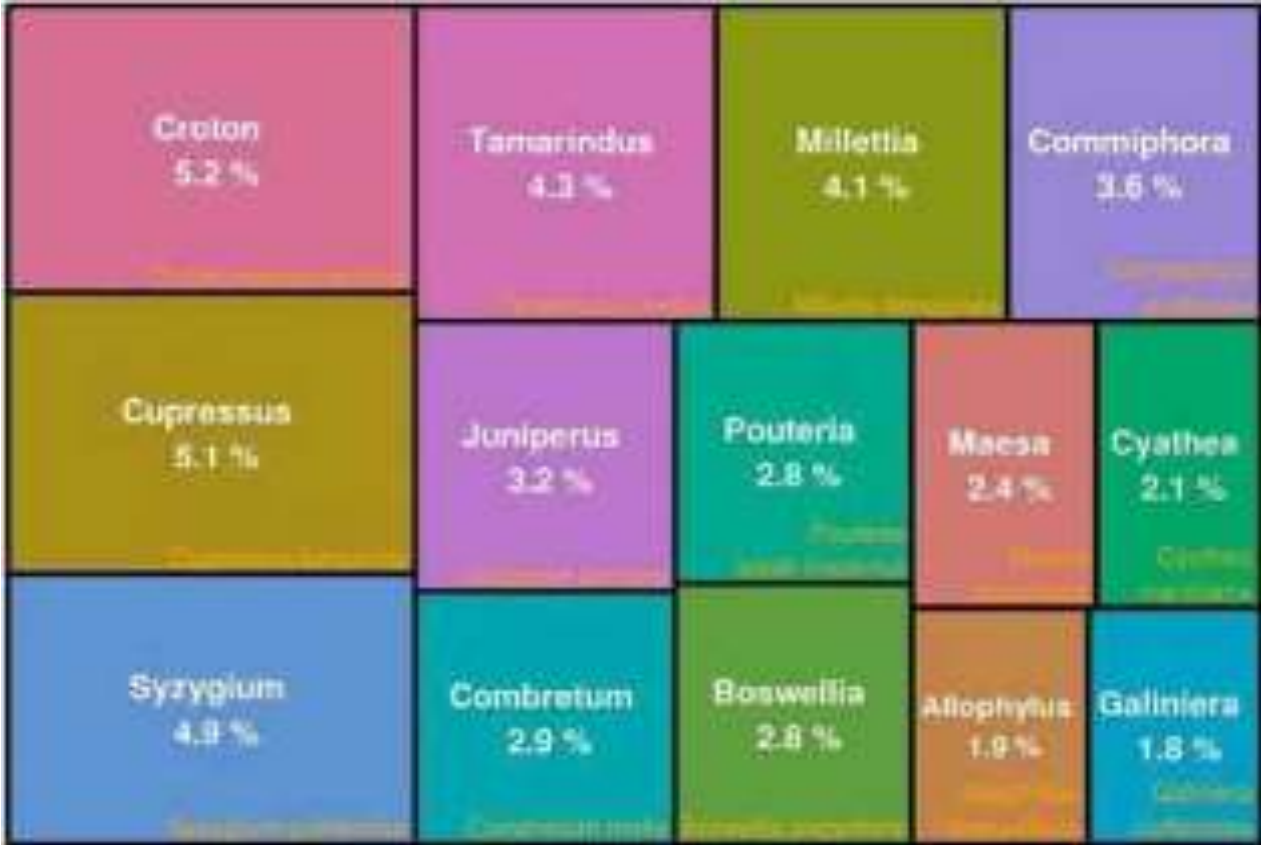
Table 3.27 Three most abundant species in Forest and Other Wooded Land categories

FRA class	Species	Count (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
Forest	<i>Croton macrostachyus</i>	4.9	5.0
Forest	<i>Cupressus lusitanica</i>	4.8	0.8
Forest	<i>Syzygium guineense</i>	4.6	10.1
Other Wooded Land	<i>Acacia bussei</i>	5.7	0.5
Other Wooded Land	<i>Commiphora erythraea</i>	4.8	0.4
Other Wooded Land	<i>Boswellia rivae</i>	3.7	0.2

Table 3.28 Three species with the highest mean stem volume by biomes

Biome	Species	Count (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
<i>Acacia-Commiphora</i>	<i>Commiphora erythraea</i>	5.1	0.51
<i>Acacia-Commiphora</i>	<i>Acacia bussei</i>	5.6	0.47
<i>Acacia-Commiphora</i>	<i>Cordia africana</i>	0.1	0.02
<i>Combretum-Terminalia</i>	<i>Ficus vasta</i>	0.2	1.35
<i>Combretum-Terminalia</i>	<i>Combretum molle</i>	3.1	1.01
<i>Combretum-Terminalia</i>	<i>Cordia africana</i>	0.8	0.31
Dry Afromontane	<i>Juniperus procera</i>	1.7	2.91
Dry Afromontane	<i>Eucalyptus globulus</i>	3.7	1.29
Dry Afromontane	<i>Eucalyptus camaldulensis</i>	3.1	0.65
Moist Afromontane	<i>Pouteria adolfi-friedericii</i>	1.9	13.53
Moist Afromontane	<i>Syzygium guineense</i>	2.6	5.71
Moist Afromontane	<i>Ekebergia capensis</i>	0.4	4.49

Figure 3.15 The 14 most abundant tree species in Forest in Ethiopia



The Shannon⁴⁷ biodiversity index is an index that is commonly used to characterize species diversity in a community. It accounts for both abundance and evenness of the species present. The Shannon index was computed per plot and then averaged by Major LUCC type. Results shows a generally higher tree species diversity, as expected, in the natural regenerated forest and lower in the forest plantations. Other land and Other woodland have a high level of tree species biodiversity as well as, even in built-up areas.

Table 3.29 Shannon index by Major LUCC types

Major LUCC	Shannon index (Average per plot)
Natural regenerated forest	1.4
Forest plantation	0.8
Other wooded land	1.0
Other land - Natural	0.8
Other land - Cultivated	0.9
Other land – Built-up	0.8
Inland water	1.2

NOTE: the inland water value has been derived by the data collected in only a few SUs in non-permanent rivers. It is reported here for consistency with the methodology.

47 Ortiz-Pulido and Pavon, 2009.

Table 3.30 Average number of tree species by Major LUCC types

Major LUCC type	Number of species (per plot)
Natural regenerated forest	7.6
Forest plantation	4.6
Other wooded land	4.5
Other land - Natural	2.1
Other land - Cultivated	3.2
Other land - Built-up	3.3
Inland water	2.0

3.10 Comparisons on the results

The average AGB of trees in Ethiopian natural forest land is estimated at 125 tons ha⁻¹ and in OWL 7.6 tons ha⁻¹. Approximately 40.3% of the total growing stock (across all land types) in terms of AGB is located in the natural regenerated forests, followed by the Other land – Cultivated (28.5%). Growing stock is high in cultivated lands due to the fact that a considerable number of TOF are an important part of the landscape in this land category.

The mean figures (per hectare) show the highest stocking in natural regenerated forest land. The built-up land type contains a considerable number of trees per hectare when compared to the other land classes. The trees assigned to inland water are the ones collected in non-perennial rivers or riverine vegetation sites. These sampling plots contained also a remarkable number of trees in the data.

Table 3.31 Main combined results by Major LUCC types

FRA – Major LUCC	N. of trees (N ha ⁻¹)	Bole volume (m ³ ha ⁻¹)	Volume (m ³ ha ⁻¹)	AGB (t ha ⁻¹)
Forest - Natural regenerated forest	118	113.2	131.6	125
Forest - Plantation	178	84.9	72.4	71.3
Other Wooded Land	59	5.3	7.4	7.6
Other land - Natural	10	7.1	7.1	6.7
Other land - Cultivated	25	13.2	17.2	16.4
Other land - Built-up	49	13.4	17.5	17.1
Other land - Quarry/Mining	35	8.0	7.2	7.3
Water - Inland water	24	7.2	7.2	7.2

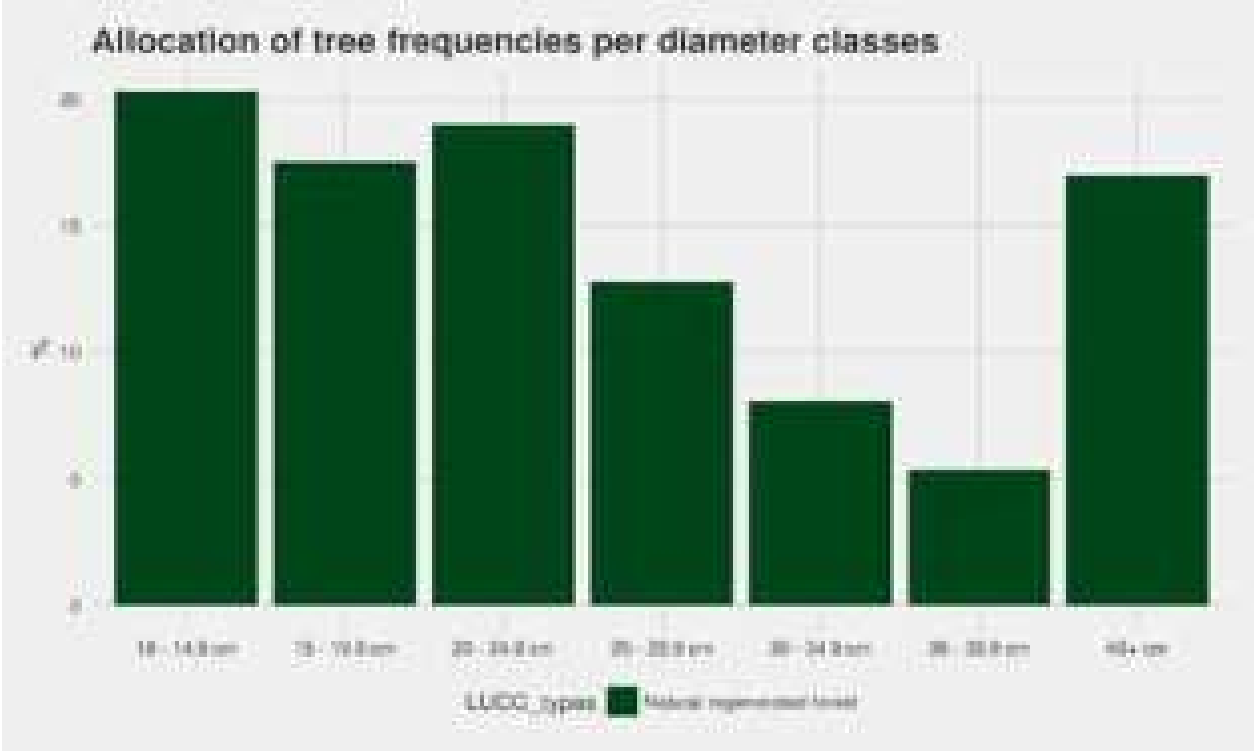
The results by biomes (in both forest and non-forest land) show that the trees in the Moist Afromontane biome contain 43.3% of the total volume. The results per hectare show that the highest stocking in terms of biomass is in the Moist Afromontane biome (93.7 t ha⁻¹), followed by *Combretum-Terminalia* (19.3 t ha⁻¹).

Table 3.32 Main combined results by biomes

Biome	N. of trees (N ha ⁻¹)	Bole volume (m ³ ha ⁻¹)	Volume (m ³ ha ⁻¹)	AGB (t ha ⁻¹)
<i>Acacia-Commiphora</i>	46.0	3.6	5.5	5.5
<i>Combretum- Terminalia</i>	55.9	12.4	19.5	19.3
Dry Afromontane	27.2	9.7	13.8	13.3
Moist Afromontane	76.4	70.9	100.0	93.7
Other	16.2	0.3	0.4	0.5

The distribution of the trees by diameter classes is fundamental in order to understand the structure of the forests. In the natural regenerated forest the DBH distribution shows that the trees with DBH less than 30 cm represent almost 70% of the total number trees. Trees with DBH less than 15 cm (supposedly young trees) represent approximately 20% of the density. In the forest plantations, the trees with DBH less than 30 cm represent more than 84% of all standing trees, and those with DBH less than 15 cm are 30% of the total growing stock. Probably due to the timber management operations, in the forest plantation the trees with DBH greater than 40 cm are only 7.3% of the total number of trees.

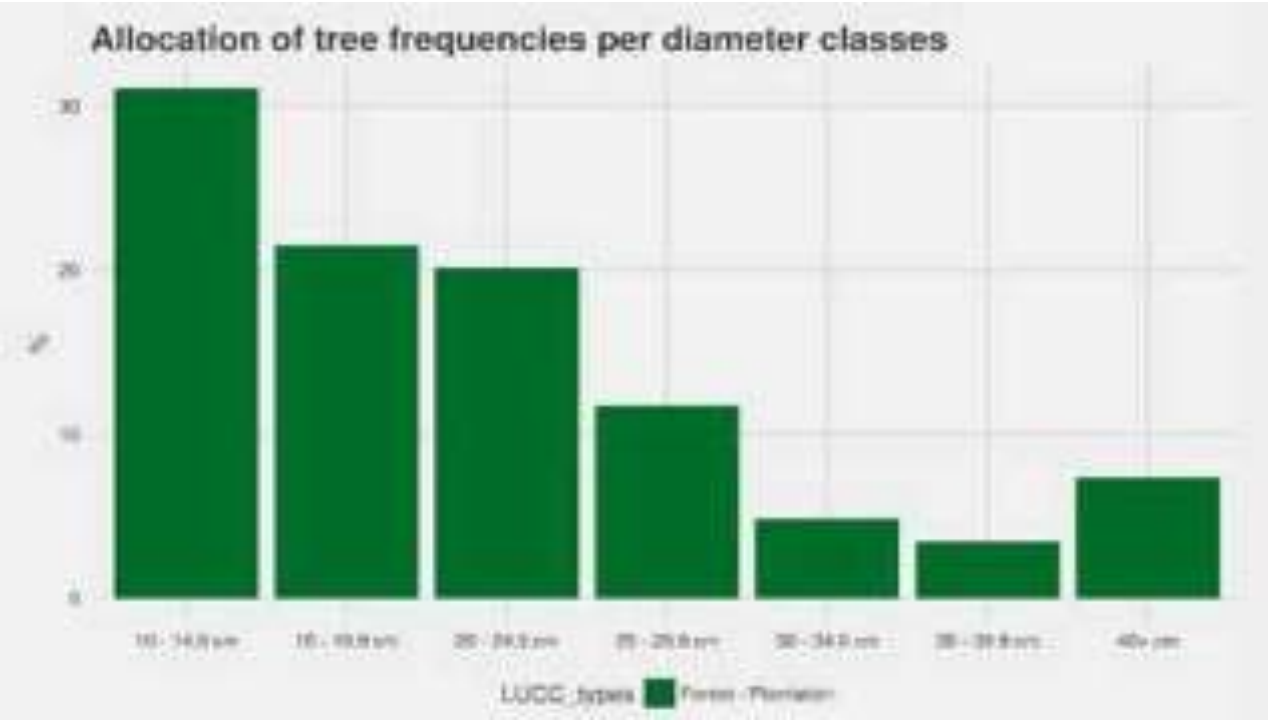
Figure 3.16 Tree frequencies by diameter classes in the Natural regenerated forest



A slightly different trend is observed in the Woodland where the trees with DBH less than 30 cm constitute 93% of the total number of trees, and the big trees with DBH greater than 40 cm are only 2%.

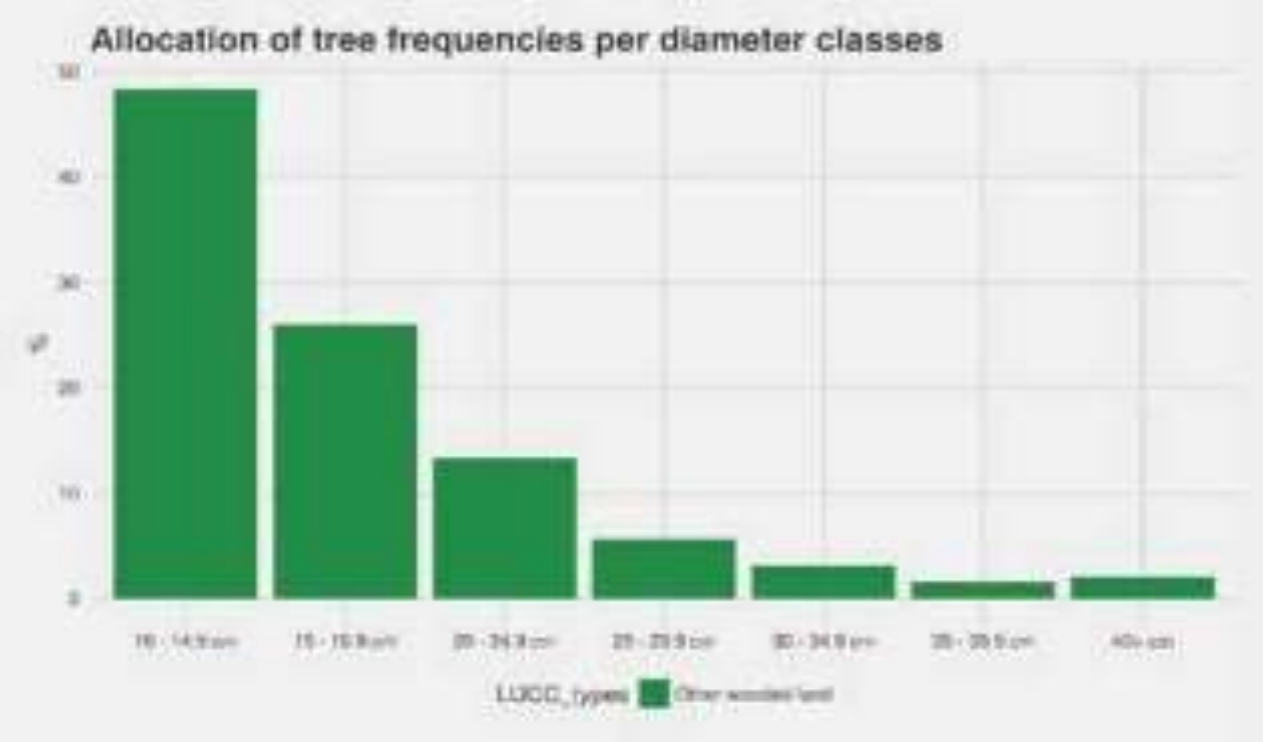
The results for stem volume (i.e. percentages of the stem volumes by DBH classes) show that the big trees with DBH greater than 40 cm constitute 80% of the total standing stock.

Figure 3.17 Tree frequencies by diameter classes in Forest plantations



This shows a typical dominance of big trees, but also that these trees are rather rare too when compared to the corresponding frequency results.

Figure 3.18 Tree frequencies by diameter classes in Other Wooded Land



A similar behavior is observed in the forest plantations where the big trees constitute almost 50% of the total volume, but the trees with DBH less than 30 cm are 37%.

Figure 3.19 Tree volume by diameter classes in Other Wooded Land

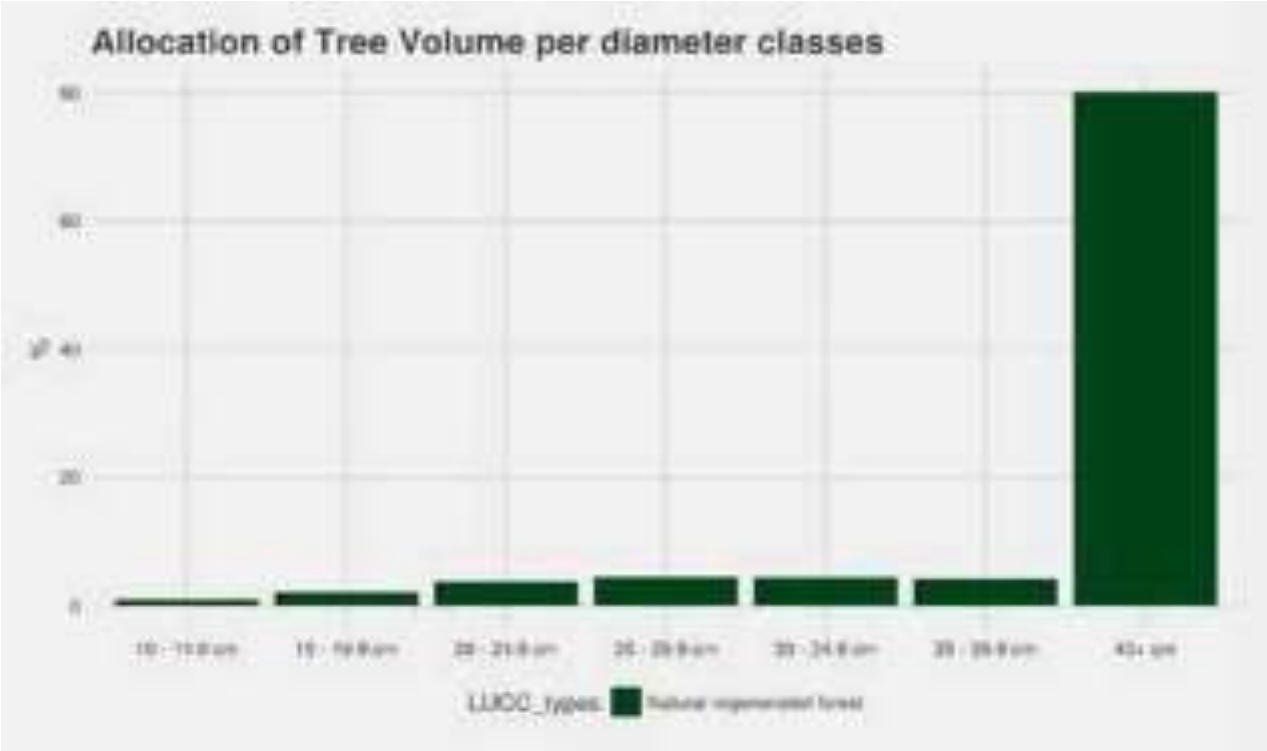


Figure 3.20 Tree volume by diameter classes in Forest plantation

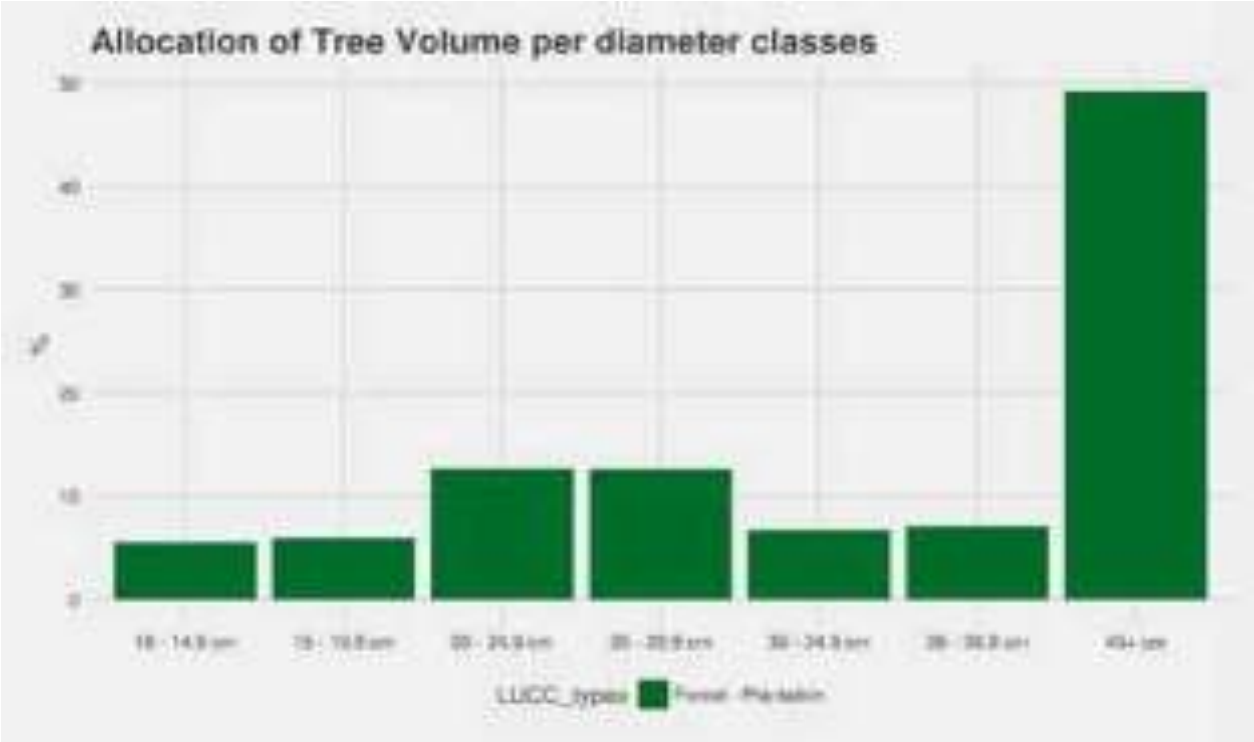


Figure 3.21 Tree volume by diameter classes in Other Wooded Land

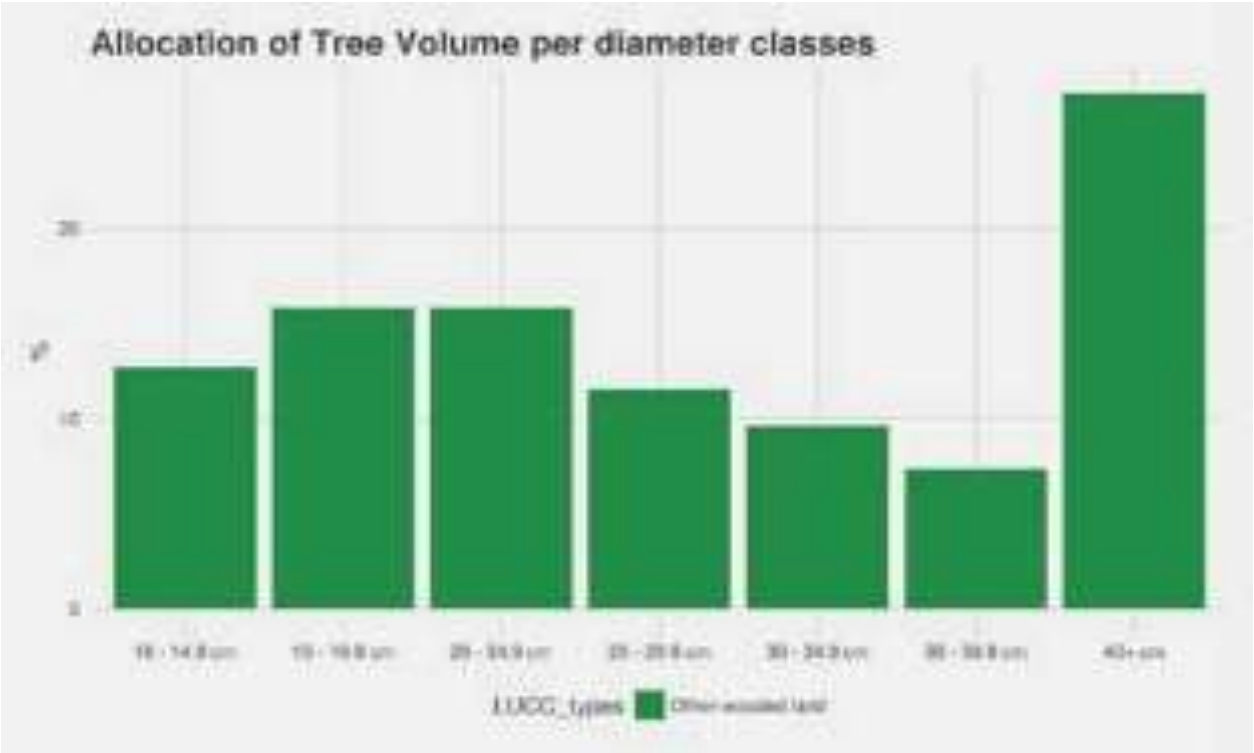
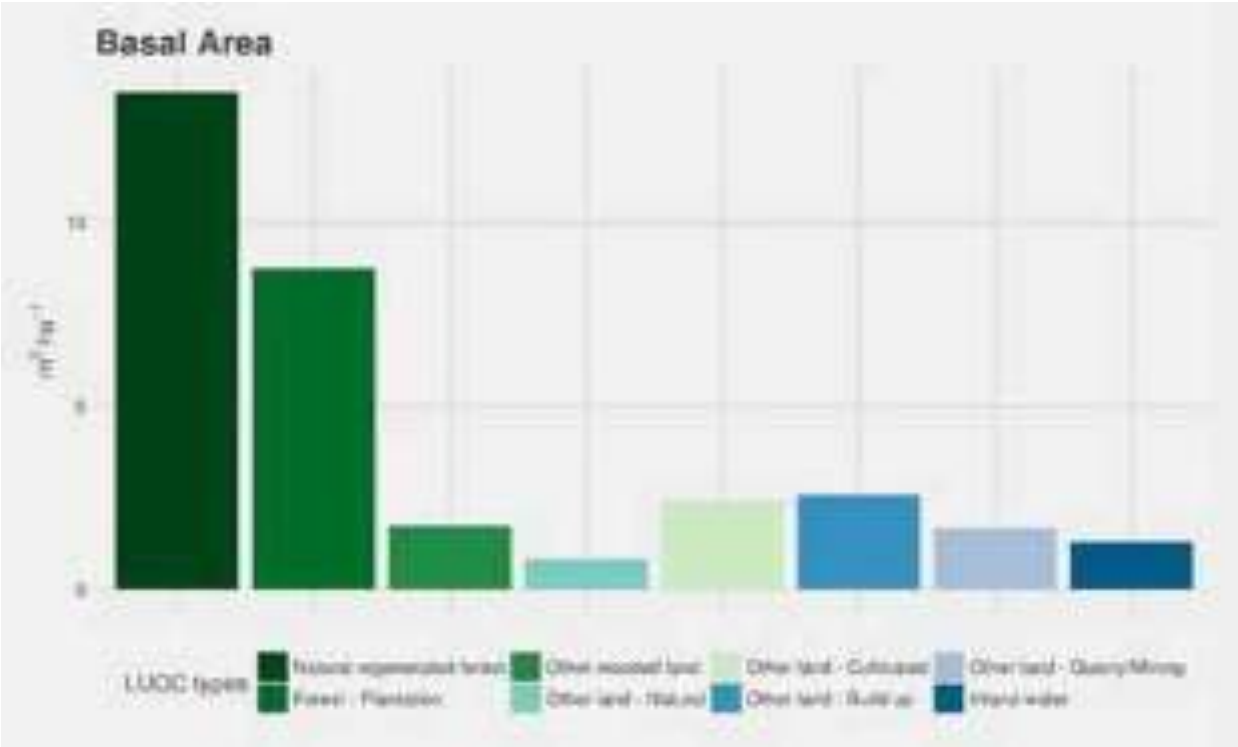


Figure 3.22 Tree basal area by Major LUCC types



The results for Other Wooded Land show a different pattern, with a majority of the volume (56%) stored in trees with less than 30 cm of DBH, whereas big trees with DBH > 40 cm contain 27% of the growing stock.

The mean DBHs and heights have been computed with two methods: as arithmetic averages and as basal area weighted averages.

Table 3.33 Mean DBH and height by Major LUCC types

Major LUCC	Average DBH (cm)	Weighted Average DBH (cm)	Height (m)	Weighted Height (m)
Natural regenerated forest	26.1	45.7	11.8	13.8
Plantation	19.1	24.6	11.8	12.7
Other wooded land	18.0	25.1	8.5	9.6
Other land - Natural	21.7	35.0	12.6	14.2
Other land - Cultivated	27.1	49.9	11.4	13.5
Other land - Built-up	24.3	35.3	11.5	12.8
Other land - Quarry/Mining	23.5	27.8	9.0	10.2
Inland water	23.5	30.5	10.8	12.1

Some more detailed results are presented for volume in different LUCC types in the Forest and Other Wooded Land (FRA) classes. In the natural regenerated forest, the highest mean volume per hectare is also in Evergreen forest and Deciduous forest types.

Figure 3.23 Tree volume by Major LUCC types

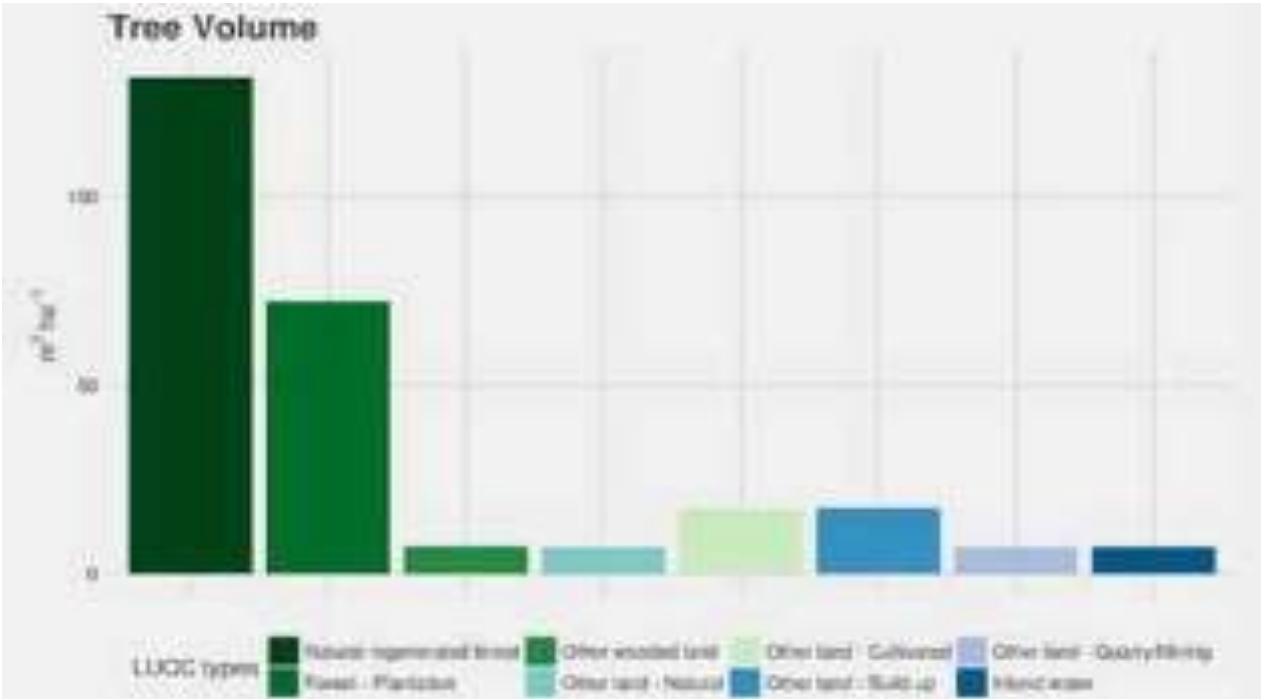


Figure 3.24 Tree volume in Natural Regenerated Forest

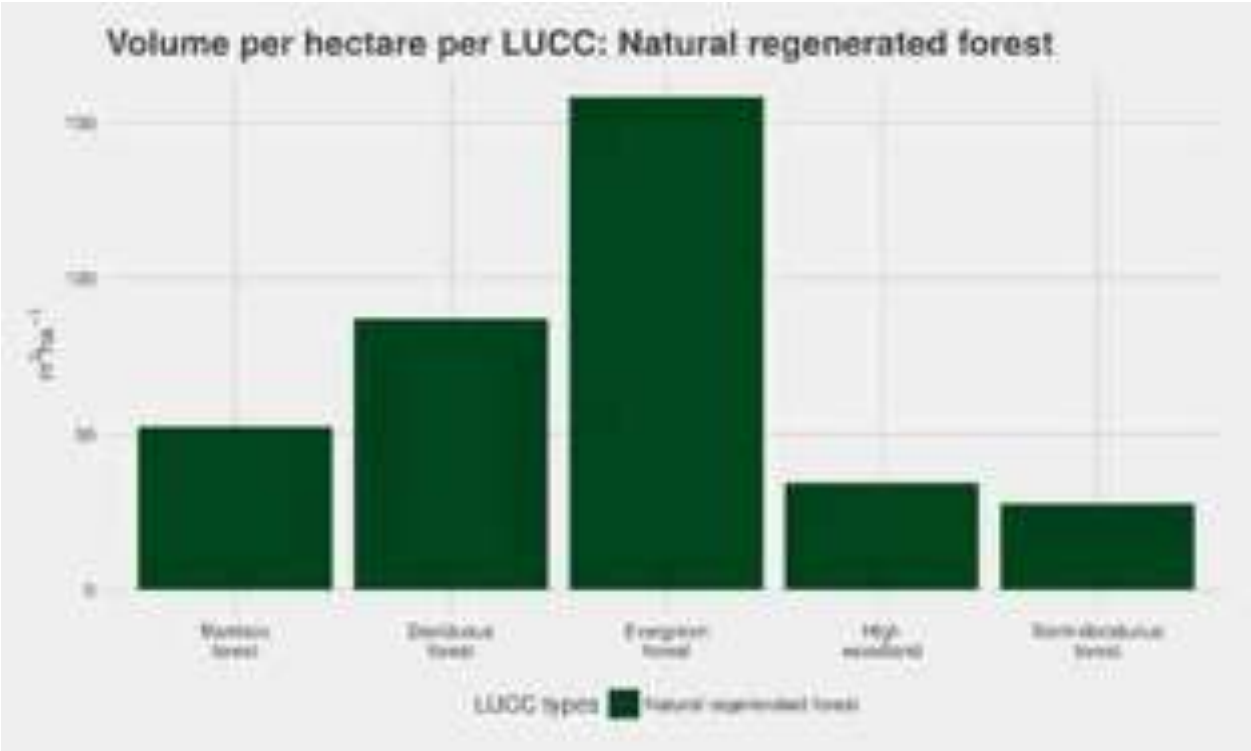
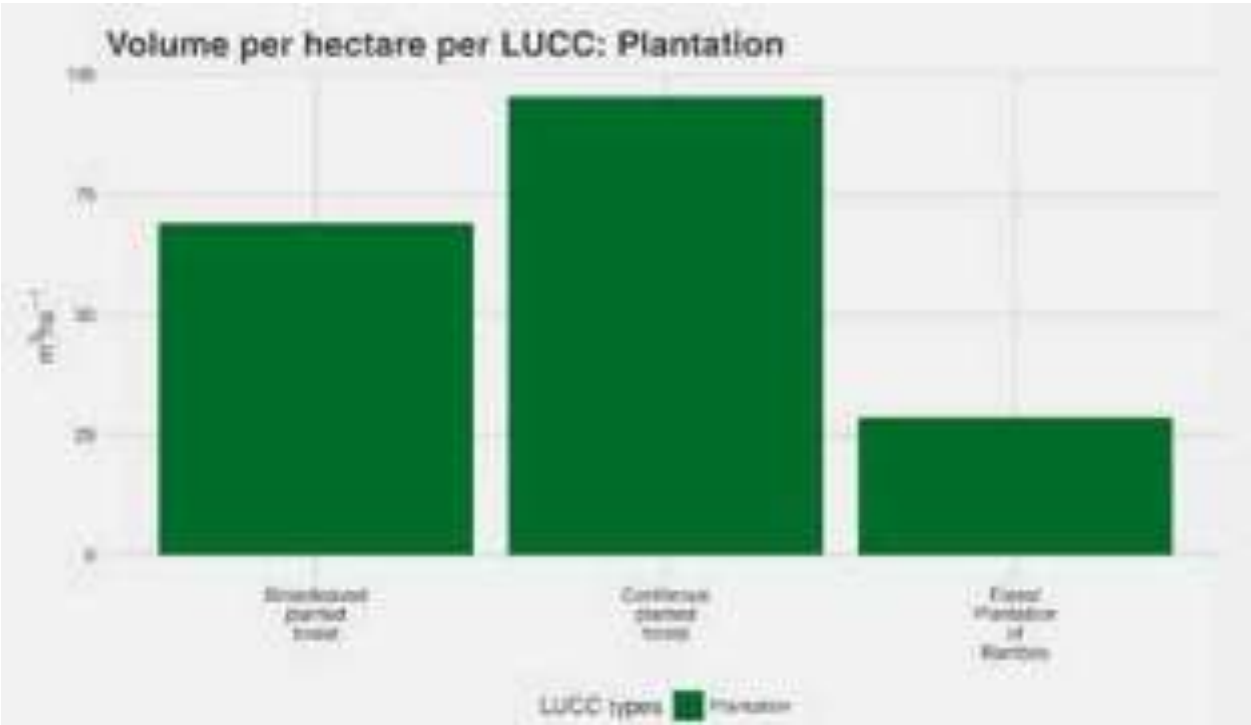
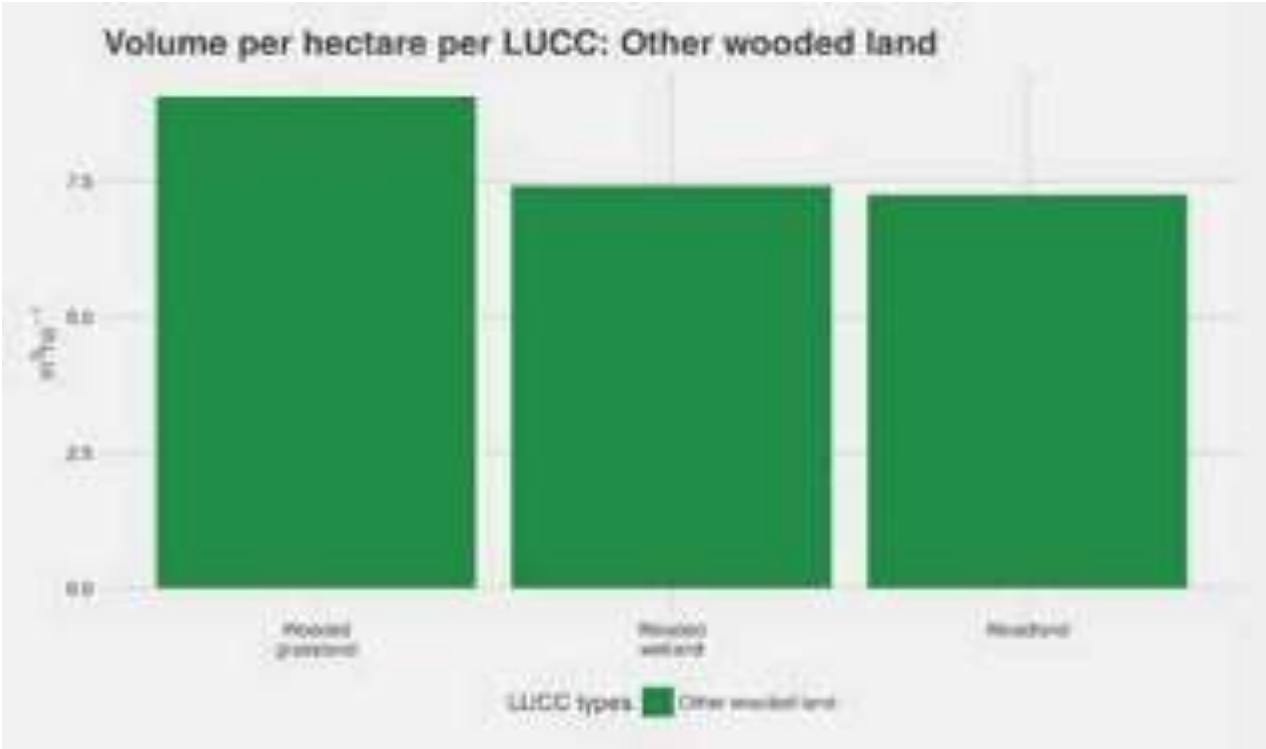


Figure 3.25 Tree volume in Forest plantation



In forest plantations, the highest mean volume is found in the coniferous planted forests (95 m³ ha⁻¹), followed by the broadleaves planted trees (69 m³ ha⁻¹) and plantation with bamboo forest (29 m³ ha⁻¹).

Figure 3.26 Tree volume in Other Wooded Land



OWL shows quite equal mean volumes between LUC types (Woodland 7.2 m³ ha⁻¹, Wooded grassland 9 m³ ha⁻¹ and Wooded wetland 7.4 m³ ha⁻¹). The highest mean volume is found in the Wooded grassland it is probably related to abundance of the TOF species and to the trees in agroforestry lands.

3.11 Forest disturbance and health status

In NFI data about the types of human disturbances affecting the forest or other wooded land were collected. Based on the NFI data, 65.8% of forest lands are somehow disturbed, and 58.8% of OWLs are disturbed by human activity. These are rather high figures but typical to countries with high population pressure on forest resources. In forest land the main disturbance is attributed to the encroachment by agriculture.

Table 3.34 Human disturbances in Forest and Other Wooded Land

	Undisturbed (%)	Encroachment by agriculture (%)	Over exploitation (%)	Settlements (%)	Quarry and mineral exploration (%)	Other (%)
Forest	34.2	28.8	10.6	6.0	0.5	19.8
Other Wooded Land	41.2	14.7	25.8	3.3		15.0

In addition, a relatively high frequency of *Croton macrostachyus* (usually present in the degraded forests) may suggest a loss of biomass in the Ethiopian forests if used as an ecological indicator. Similar conclusion based on a natural regeneration pattern in areas recolonized by trees, can be derived by a relatively high frequency of *Acacia bussei* in OWL areas. This requires further analysis and more precise investigation to confirm the observation.

Wildlife disturbance was estimated to assess the impact level of wildlife in the forest resources. The results show that in forest lands less than 20% of the areas are somehow disturbed by wildlife.

Table 3.35 Wildlife disturbance by Major LUCC types						
FRA class	Major LUCC	Not disturbed (%)	Slightly disturbed (%)	Moderately disturbed (%)	Heavily disturbed (%)	Total (%)
Forest	Natural regenerated forest	80.2	12.5	3.6	3.7	100
Forest	Plantation	95.6	0	4.4	0	100
Other Wooded Land	Other Wooded Land	90.1	4.5	2.9	2.5	100
Other Land	Natural	90.6	6.4	2.6	0.4	100
Other Land	Cultivated	71.8	17.7	4.4	6.0	100
Other Land	Build up	82.2	13.9	0	4.0	100

The fire is another important agent of damage in the forest lands, usually human caused.

Table 3.36 Evidence of fire by Major LUCC types				
FRA class	Major LUCC	No evidence of fire (%)	Recent fire (%)	Old fire (%)
Forest	Natural regenerated forest	73.3	23.5	3.3
Other	Wooded Land	92.7	6.7	0.6
Other Land	Natural	99.7	0.3	0
Other Land	Cultivated	93.3	6	0.7

In NFI tree health data was collected to assess health conditions of trees and their crowns. The general tree health status (computed based of stem density) shows a generally good tree health level: more than 94% of the trees are healthy, with no remarkable differences between the site types.

Figure 3.27 Tree health status by Major LUCC types

FRA class	Major LUCC	Healthy (%)	Slightly affected (%)	Severely affected (%)	Dead or Dying standing tree (%)	Dead or Dying fallen tree (%)	Total (%)
Forest	Natural regenerated forest	94.4	2.2	0.4	1.4	1.7	100
Forest	Plantation	96.8	2.8	0	0.2	0.2	100
Other Wooded Land		94.5	1.4	0.5	2.3	1.3	100
Other Land	Other land - Natural	96.7	1.6	1.1	0.3	0.4	100
Other Land	Other land - Cultivated	92.3	3.3	1.7	1.6	1.1	100
Other Land	Other land - Build up	99.1	0.6	0	0	0.3	100
Other Land	Other land - Quarry/ Mining	100.0	-	-	-	-	100
Water	Inland water	91.7	1.9	2.1	2.1	2.1	100

Figure 3.28 Canopy health status by Major LUCC types

FRA Class	Major LUCC	Healthy (%)	Declining health (%)	Unhealthy (%)	Dying (%)	Dead (%)	Total (%)
Forest	Natural regenerated forest	94.8	1.8	0.4	1.0	2.0	100
Forest	Plantation	97.2	2.1	0.3	0.2	0.1	100
Other Wooded Land		94.2	2.0	0.6	1.0	2.2	100
Other Land	Natural	97.1	1.6	1.1	0	0.2	100
Other Land	Cultivated	92.2	3.5	1.7	0.8	1.8	100
Other Land	Build up	99.3	0.4	-	-	0.3	100
Other Land	Quarry/ Mining	100.0	-	-	-	-	100
Water	Inland water	91.7	4.0	0	0	4.2	100

3.12 Governance, management and ownership

Based on the NFI field data, the majority of forest lands belong to the State (i.e. that land is owned by the central government, or by government-owned institutions or

corporations) or to the local government level institutions (as district, municipalities). However, 55% of forest plantations are under private ownership. According to the NFI data, in Other Wooded Land approximately 80% of all lands are under public ownership.

Table 3.37 Land ownership by Major LUCC types

Major LUCC	N/A	Private Individual (%)	Private Industries (%)	Private - Local comm. (%)	Private - Other private (%)	Public State (%)	Public - Local gov. (%)	Indig. or Tribal comm. (%)	Other (%)	Total (%)
Natural regenerated forest	0.5	4.8	0.4	7.5	6.1	9.4	66.1	5.2	-	100
Plantation	5.8	55.2	-	-	1.4	3.2	34.4	-	-	100
Other wooded land	0.4	4.2	0.5	9.7	0.2	4.8	79.9	0.3	0.1	100
Other land - Natural	100.0	-	-	-	-	-	-	-	-	100
Other land - Cultivated	99.8	0.1	-	-	-	-	0.1	-	-	100
Other land - Build up	100.0	-	-	-	-	-	-	-	-	100
Other land - Quarry/ Mining	100.0	-	-	-	-	-	-	-	-	100
Inland water	100.0	-	-	-	-	-	-	-	-	100

Land management refers here to the management agreement and arrangement between the land owner and other groups. Majority of Forest and Other Wooded Lands belong to *‘Joint management - with communities’* category. In most cases in the Other Land categories the owner retains management rights and responsibilities within the limits specified by the legislation and the owner is the exclusive manager.

Table 3.38 Land management by Major LUCC

Major LUCC	Owner only	J. management -	J. management -	Dev. of communities (%)	Dev. of manage- (%)	Not known or	Other (%)	Total (%)
Natural regenerated forest	7.5	62.7	1.7	4.7	0.2	22.9	0.3	100
Plantation	52.0	30.5	0	5.8	0	11.7	0	100
Other wooded land	6.4	67.0	0	4.4	0.1	21.2	1.1	100
Other land - Natural	30.7	30.6	0.3	3.1	0	34.8	0.5	100
Other land - Cultivated	86.8	7.8	1.1	0	1.1	3.0	0.2	100
Other land - Build up	74.1	12.0	0	0	1.8	14.2	0	100
Other land - Quarry/Mining	100.0	-	-	-	-	-	-	100
Inland water	2.5	79.4	0	0	0	18.1	0	100

4 Conclusions and way forward

The Ethiopian NFI was designed by following the FAO-NFMA methodology with technical support provided by FAO. The objective of the NFI is to obtain countrywide baseline information for the forest sector. The NFI was further enriched when the country decided to participate in the REDD+ mechanism. NFI is an essential part of the national forest monitoring and MRV systems for REDD+. Conducting the NFI helped Ethiopia to increase its capacity and gain valuable experiences in implementing MRV of forest resources through extensive capacity building and awareness raising of topics such as reference/baseline scenario construction, NFI design and implementation, monitoring of forest area change, statistics, data processing and reporting. More than 150 Ethiopian technicians have received training in data collection, analysis and reporting methods at the federal and regional levels during the project.

The NFI and MRV systems will allow the Government to assess its own progress in reducing emissions from deforestation and degradation and in increasing forest carbon stocks. In addition, a strengthened forest information database and improved capacity will place Ethiopia in a better position for formulation of policy for sustainable forest management, implementing best practices in forest management and establishing a complete national forest assessment monitoring system. Thus, the NFI provides baseline information that can support Ethiopia to improve management of its lands and particularly forest lands, to improve the strategic planning of the fuel wood production and to further improve the national and subnational forest information systems. Since the NFI collected data in all land types, including also areas outside of forests, the NFI results will give a preliminary but exhaustive overview on the status of trees outside of forests at the national and subnational levels too.

The NFI was designed to provide information for REDD+, and therefore new methods, tools and collected data stemming from this initiative have improved the knowledge on current forest carbon stock, emission factor(s) and other relevant forest information on forest use and forest cover. The information derived from the NFI was used to calculate and report the emission factors for the first Ethiopian forest reference level submission in 2016.

In 27 months of field data collection, a total of 49,400 trees were recorded in 1,029 plots in 539 sampling units in all the regions of the country. The results show that the population pressure is high, because 65.8% of forest lands are somehow disturbed and 58.8% of OWLs are disturbed by human activity. In forest land the main cause of forest disturbance is encroachment by agriculture. Results on the basal area in forest lands also indicate that forests appear to be degraded, with BA in forests $13.3 \text{ m}^2\text{ha}^{-1}$ and in High woodland only $3.8 \text{ m}^2\text{ha}^{-1}$. The basal area results by biome (forest only plots), in *Acacia-Commiphora*, *Combretum-Terminalia*, Dry Afromontane and Moist Afromontane estimates are 6.5, 8.4, 12.4 and $20.7 \text{ m}^2\text{ha}^{-1}$, respectively. Only Moist Afromontane forest has a basal area greater than $20 \text{ m}^2\text{ha}^{-1}$.

All the established sampling units can be considered as permanent and can be revisited during the next NFI cycles. Remeasurements of the sampling units will allow for better understanding of the status of forests and especially forest dynamics in the country. A continuous NFI is recommended and means that a certain proportion of land (or the sampling units) is re-measured every year (as for example in Finland and Poland). The continuous NFI method would allow the country to have a non-interrupted flow of up-to-date information on its forests continuously. Similarly, new variables can be added to the inventory, if needed. At the moment, Ethiopia is planning to repeat the NFI in 5-year cycles, but it is recommended that a continuous NFI method be considered as an alternative option.

All collected data are stored in an Open Foris Collect database and managed by the MRV Directorate. This database can be used for further analysis and scientific research purposes. However, it should be noted that the first NFI was designed to give results primarily for the national level, despite results being reported by region. Therefore, an intensification of NFI sampling in the future can expand the use of NFI data to the lower (subnational) levels and improve the accuracy of the estimates.

Furthermore, the NFI methodology can be adapted and improved, by adding new variables or stratification schemes. In 2016, Ethiopia already institutionalized MRV in a decentralized way, thus allowing all the administrative levels to be part of the process. MEFC has several options to improve NFI mechanisms, through formalization of NFI as part of MRV regularly carried out by (decentralized) institutions.

For complete and correct adoption of this strategy it is recommended that Ethiopia improve the following aspects of the NFI:

- Formal approval of the new national forest definition;
- Take into account the change in the national forest definition in order to harmonize the data analysis and reporting consistently;
- Improve the stratification system;
- Improve the forest type classification;
- Decentralize the field operations so that they are coordinated technically at the central level in order to secure faster and more efficient field data collection phase;
- Establish clear protocols for data flow between the NFI levels (i.e. operators);
- Reinforce links between remote sensing analysis, inventory planning and field data collection;
- Use NFI field data as validation of RS results (forest cover/change detection analysis);
- Include forest degradation indicators into the list of attributes to be collected;
- Include additional temporary sample plots in the inventory;
- Further training and supervision to reduce the number of measurement errors and improve the QA/QC procedures;
- Assign permanent teams to the NFI in order to create specialized and professional teams;
- Inclusion of a botanist in each team.

Finally, it is strongly recommended to reinforce links with the academia by allowing the research teams to access the NFI data and to improve the quality of the variables collected. This will yield an increased number of topical publications, thesis, and books that can emphasize the importance, complexity and uniqueness of the Ethiopian forests.

5

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6 Appendices

Appendix A1 Land Use Land Cover

Table A1.1 Land use/land cover description

Level 1	Level 2	Level 3	Brief description	Code
Forest	«Area ≥ 0.5 ha; Tree canopy cover ≥10%; Tree height ≥ 5 m at maturity in situ; Width > 20 m. Excludes land that is predominantly under agricultural or urban land use (orchards, agroforestry systems...).»			F
	Natural regenerated forest	Forest predominantly composed of trees established through natural regeneration.		
		Evergreen forest	Naturally regenerated forest composed of more than 75% of evergreen trees species. Includes : Moist forest Dry forest	FE
		Deciduous forest	«Naturally regenerated forest composed of more than 75% of deciduous trees species. Includes : Moist forest Dry forest Secondary young	FD
		Semi-deciduous forest	«Naturally regenerated forest where trees are at least 25% each of evergreen and deciduous species. Includes : Moist forest Dry forest Secondary young	FSD
		Bamboo forest	Naturally regenerated forest predominantly composed of bamboo vegetation.	FB
		Raffia/Palms	Naturally regenerated forest predominantly composed of palm or raffia vegetation.	FRP
		High woodland		HW
	Plantation	Forest predominantly composed of trees established through planting and/or deliberate seeding. Includes coppice from trees that were originally planted or seeded.		
		Broadleaved planted forest	Planted forest composed of more than 75% of broadleaved species.	FPB
		Coniferous planted forest	Planted forest composed of more than 75% of coniferous species.	FPC
		Mixed planted Forest	Planted forest of at least 25% each of coniferous and broadleaved species.	FPM
		Forest Plantation of Bamboo	Planted forest composed of more than 75% of Bamboo	BFP

Table A1.1 Land use/land cover description

Level 1	Level 2	Level 3	Brief description	Code
Other wooded lands	Area ≥ 0.5 ha, tree crown cover 5- 10% or shrubs/bushes canopy cover ≥10%			W
	Woodland	Others (bushes,..)		W
	Wooded grassland	Land covered by natural growth of graminea and herbaceous vegetation, with some scattered trees (tree canopy cover between 5-10%); Land not covered seasonally or permanently by water.		WG
	Wooded wetland	Land seasonally or permanently covered by water with natural growth of graminea and herbaceous vegetation and some scattered trees (canopy cover between 5-10%).		WW
Other Land	Land not classified as forest or other wooded land, as described above (Includes land with tree canopy cover <5% or with shrubs/bushes <10% or with predominant agricultural/urban land use or with shrubs/ trees<0.5ha).			O
	Natural	Natural Forest Coffee		ONC
		Barren Land	Land where vegetation cover is less than 2%. Includes land covered of sand, soil and rocks.	OX
		Natural Grassland	Land covered with natural growth of graminea and herbaceous vegetation.	OG
		Marsh	Land seasonally or permanently covered by water and dominated by natural growth of graminea, reed and other herbaceous.	OM
	Cultivated	Coffee plantation		OCC
		Improved pastures	Land sown with introduced grass and leguminous for the grazing of livestock.	OP
		Annual Crop	Area covered by crops that are sown and harvested during the same production season/ agricultural year.	OCA
		Perennial crop	Crops that are sown or planted once and need not to be replanted after each annual harvest. Includes trees (e.g. apples or other fruit trees), bushes and shrubs (e.g. berries, coffee...), palms (e.g. dates), vines (e.g., grapes), herbaceous stems (e.g. bananas) and stemless plants (e.g. pineapples).	OCP
		Mixed annual and perennial crop	Association of annual and perennial crops.	OCM
		Fallow	Previously cultivated land kept free from crops or weeds during at least one growing season, where woody vegetation is and will not reach 5m height.	OF
		Wood lot of Bamboo	Bamboo areas spanning between 0.2 and 0.5 ha, with trees >5m at maturity mainly used is for wood stock	OWB
		Wood lot	Other areas spanning between 0.2 and 0.5 ha, with trees >5m at maturity mainly used is for wood stock	OW

Table A1.1 Land use/land cover description

Level 1	Level 2	Level 3	Brief description	Code
Other Land	Built up area	Populated areas with significant constructions. Includes homes scattered in the field. Notes: a road is considered as a distinct Land Use/Cover Section (built-up area) if wider than 15 meters (from bottom of ditch on one side to the bottom of ditch on the other side when ditches exists, otherwise the width of the road bank) and if not a forest road.»		OB
	Quarry/ Mining site	Areas used for extraction of minerals, rocks, sands, clay... Includes: quarry, mining, extraction areas, oil/gas wells.		OQ
Inland water	Area occupied by major rivers (width ≥ 15m), lakes, ponds and reservoirs.			IW
	Perennial River	Rivers (width ≥ 15m) that maintains water in its channel throughout the year.		IRP
	Intermittent River (seasonal)	Rivers (width≥ 15m) that flows only at certain times of the year.		IRS
	Lake	Large body of salt or fresh water surrounded by land.		IL
	Dam	Reservoir created by a barrier constructed to hold back the water and raise its level.		ID
	Pond	Small body of still water formed naturally or by hollowing or embankment.		IP

Appendix A2 Inventory Area results

Table A2.1 Area estimates by biomes, FRA, Major LUCC and LUCC

Biome	FRA Class	Major LUCC	LUCC	Area
<i>Acacia-Commiphora</i>	Forest	Natural regenerated forest	Evergreen forest	793 798
	Other Wooded Land	<i>Other wooded land</i>	Woodland	40 606 507
			Wooded grassland	2 236 138
			Wooded wetland	68 351
	Other Land	Other land - Natural	Natural Forest Coffee	47 242
			Barren Land	7 967 932
			Natural Grassland	1 207 115
			Marsh	0
		Other land - Cultivated	Annual Crop	4 126 458
			Perennial crop	306 044
			Mixed annual and perennial crop	146 921
			Fallow	774 296
		Other land - Build up	Built up area	177 805
	Water	Inland water	Perennial River	47 377
			Intermittent River (seasonal)	217 130
<i>Combretum-Terminalia</i>	Forest	Natural regenerated forest	Evergreen forest	982 275
			Deciduous forest	499 506
			Semi-deciduous forest	249 423
			Bamboo forest	501 964
		Plantation	Broadleaved planted forest	80 087
	Other Wooded Land	Other wooded land	Woodland	8 191 417
			Wooded grassland	2 125 610
			Wooded wetland	43 620
	Other Land	Other land - Natural	Barren Land	376 264
			Natural Grassland	1 968 364
			Marsh	43 620
		Other land - Cultivated	Coffee plantation	312 669
			Annual Crop	9 904 670
			Perennial crop	382 386
			Fallow	752 726
			Wood lot	25 125
		Other land - Build up	Built up area	258 408
		Other land - Quarry/Mining	Quarry/Mining site	11 711
	Water	Inland water	Perennial River	26 504
			Intermittent River (seasonal)	26 298

Table A2.1 Area estimates by biomes, FRA, Major LUCC and LUCC

Biome	FRA Class	Major LUCC	LUCC	Area
Dry Afromontane	Forest	Natural regenerated forest	Evergreen forest	746 629
			Deciduous forest	88 233
			Semi-deciduous forest	26 870
			Bamboo forest	16 251
		Plantation	Broadleaved planted forest	116 204
			Coniferous planted forest	49 498
			Forest Plantation of Bamboo	16 576
	Other Wooded Land	Other wooded land	Woodland	1 857 958
			Wooded grassland	361 376
	Other Land	Other land - Natural	Barren Land	372 342
			Natural Grassland	3 327 490
		Other land - Cultivated	Coffee plantation	57 242
			Annual Crop	11 447 972
			Perennial crop	180 816
			Mixed annual and perennial crop	204 817
			Fallow	290 519
			Wood lot	16 750
		Other land - Build up	Built up area	604 227
	Water	Inland water	Perennial River	16 052
			Intermittent River (seasonal)	17 623
			Dam	0
Moist Afromontane	Forest	Natural regenerated forest	Evergreen forest	1 843 050
			Deciduous forest	368 568
			Semi-deciduous forest	6 330
			Bamboo forest	12 737
			High woodland	11 811
		Plantation	Broadleaved planted forest	87 876
			Coniferous planted forest	23 196
	Other Wooded Land	Other wooded land	Woodland	513 277
			Wooded grassland	444 982
			Wooded wetland	45 820
	Other Land	Other land - Natural	Natural Forest Coffee	70 863
			Natural Grassland	787 292
			Marsh	43 507
		Other land - Cultivated	Coffee plantation	703 124
			Annual Crop	2 804 011
			Perennial crop	280 587
			Mixed annual and perennial crop	42 175
			Fallow	192 444
			Wood lot of Bamboo	43 620
			Wood lot	50 948
		Other land - Build up	Built up area	152 979
	Water	Inland water	Perennial River	7 503
			Intermittent River (seasonal)	1 890

Table A2.1 Area estimates by biomes, FRA, Major LUCC and LUCC

Biome	FRA Class	Major LUCC	LUCC	Area
Other	Other Wooded Land	Other wooded land	Woodland	963 662
	Other Land	Other land - Natural	Natural Grassland	174 481
			Marsh	0
		Other land - Cultivated	Fallow	65 931
	Water	Inland water	Lake	0

Table A2.2 Area estimates by FRA, Major LUCC and LUCC

FRA Class	Major LUCC	LUCC	Area	Proportion	Relative SE
Forest	Natural regenerated forest	Evergreen forest	4 365 753	3.8%	4.3%
		Deciduous forest	956 307	0.8%	14.2%
		Semi-deciduous forest	282 624	0.2%	31.2%
		Bamboo forest	530 952	0.5%	14.3%
		High woodland	11 811	0.0%	39.7%
	Plantation	Broadleaved planted forest	284 167	0.2%	3.9%
		Coniferous planted forest	72 694	0.1%	10.8%
		Forest Plantation of Bamboo	16 576	0.0%	NA
Other Wooded Land	Other wooded land	Woodland	52 132 821	45.3%	2.1%
		Wooded grassland	5 168 105	4.5%	7.8%
		Wooded wetland	157 791	0.1%	19.6%
Other Land	Other land - Natural	Natural Forest Coffee	118 106	0.1%	23.6%
		Barren Land	8 716 538	7.6%	5.9%
		Natural Grassland	7 464 741	6.5%	4.0%
		Marsh	87 127	0.1%	4.1%
Other Land	Other land - Cultivated	Coffee plantation	1 073 035	0.9%	6.8%
		Annual Crop	28 283 110	24.6%	3.1%
		Perennial crop	1 149 832	1.0%	13.1%
		Mixed annual and perennial crop	393 913	0.3%	4.9%
		Fallow	2 075 917	1.8%	4.4%
		Wood lot of Bamboo	43 620	0.0%	4.0%
		Wood lot	92 824	0.1%	4.0%
	Other land - Build up	Built up area	1 193 418	1.0%	5.8%
	Other land - Quarry/Mining	Quarry/Mining site	11 711	0.0%	47.3%
Water	Inland water	Perennial River	97 435	0.1%	16.2%
		Intermittent River (seasonal)	262 940	0.2%	34.3%

Table A2.3 Area estimates by regions, biomes and FRA classes

Region	Biome	FRA Class	Area
Afar	Acacia-Commiphora	Other Wooded Land	3 327 085
		Other Land	5 848 667
		Water	254 588
	Combretum-Terminalia	Other Land	824 138
	Other	Other Wooded Land	205 455
Amhara	Acacia-Commiphora	Other Wooded Land	380 876
		Other Land	183 903
		Water	5 905
	Combretum-Terminalia	Forest	115 671
		Other Wooded Land	2 319 284
		Other Land	3 842 367
		Water	24 038
	Dry Afromontane	Forest	256 204
		Other Wooded Land	1 071 954
		Other Land	6 452 170
		Water	29 662
	Moist Afromontane	Forest	17 797
		Other Wooded Land	6 979
		Other Land	1 022 108
	Other	Water	0
Beneshangul Gumu	Combretum-Terminalia	Forest	1 286 777
		Other Wooded Land	1 801 759
		Other Land	1 338 700
		Water	17 464
Gambela	Combretum-Terminalia	Other Wooded Land	1 027 275
		Other Land	616 365
	Moist Afromontane	Forest	160 765
		Other Land	54 801
Oromia	Acacia-Commiphora	Forest	431 237
		Other Wooded Land	11 149 959
		Other Land	3 728 188
	Combretum-Terminalia	Forest	205 087
		Other Wooded Land	645 693
		Other Land	3 116 631
	Dry Afromontane	Forest	488 946
		Other Wooded Land	694 253
		Other Land	7 029 220
		Water	0
	Moist Afromontane	Forest	1 643 917
		Other Wooded Land	867 005
		Other Land	2 747 305
		Water	6 252
	Other	Other Land	0
		Water	0

Region	Biome	FRA Class	Area
SNNPR	Acacia-Commiphora	Forest	43 620
		Other Wooded Land	139 420
		Other Land	942 107
		Water	4 013
	Combretum-Terminalia	Forest	511 565
		Other Wooded Land	3 473 942
		Other Land	2 052 572
	Dry Afromontane	Forest	224 557
		Other Wooded Land	64 034
		Other Land	1 585 157
	Moist Afromontane	Forest	519 747
		Other Wooded Land	130 094
		Other Land	1 203 215
		Water	3 141
Somali	Acacia-Commiphora	Other Land	174 481
		Water	0
		Forest	318 942
	Other	Other Wooded Land	27 811 236
		Other Land	3 978 887
Tigray	Acacia-Commiphora	Other Wooded Land	758 207
		Other Land	65 931
	Combretum-Terminalia	Other Wooded Land	102 420
		Other Land	72 061
		Forest	194 155
		Other Wooded Land	1 092 694
		Other Land	2 245 171
		Water	11 300
	Dry Afromontane	Forest	90 555
		Other Wooded Land	389 092
		Other Land	1 435 627
		Water	4 013
	Moist Afromontane	Forest	11 341
		Other Land	144 121

Appendix A3 Tree density

Table A3.1 Number of trees per hectare by FRA classes and LUCC

FRA Class	Major LUCC	LUCC	N of Trees (N ha ⁻¹)
Forest	Natural regenerated forest	Evergreen forest	124.9
		Deciduous forest	114.7
		Semi-deciduous forest	147.9
		Bamboo forest	93.3
		High woodland	54.7
	Plantation	Broadleaved planted forest	116.6*
		Coniferous planted forest	453*
		Forest Plantation of Bamboo	31.6
Other Wooded Land	Other wooded land	Woodland	61.7
		Wooded grassland	57.1
		Wooded wetland	53.5
Other Land	Other land - Natural	Natural Forest Coffee	101.6*
		Barren Land	1.8
		Natural Grassland	18.3
		Marsh	41.1
	Other land - Cultivated	Coffee plantation	125.1*
		Annual Crop	20.9
		Perennial crop	45.7
		Mixed annual and perennial crop	38.6
		Fallow	16.5
		Wood lot of Bamboo	6.0
		Wood lot	142.9
	Other land - Build up	Built up area	49.4
	Other land - Quarry/ Mining	Quarry/Mining site	35.1
Water	Inland water	Perennial River	26.9
		Intermittent River (seasonal)	24.6

*Results not sufficiently representative. Separate inventory undertaken for plantations

Table A3.2 Number of trees per hectare by DBH classes, biomes and FRA classes

Biome	FRA Class	N of Trees (N ha ⁻¹)								
		< 5m	5 - 9.9m	10 - 14.9m	15 - 19.9m	20 - 24.9m	25 - 29.9m	30 - 34.9m	35 - 39.9m	40+ m
<i>Acacia-Commiphora</i>	Forest	0.3	19.4	44.2	22.2	3.6	2.5	1.9	2.1	1.1
	Other Wooded Land	3.8	46.7	7.5	0.9	0.0	0.0			
	Other Land	0.3	3.6	1.3	0.5	0.3	0.1	0.0	0.1	0.1
	Water		20.3	0.7						
<i>Combretum-Terminalia</i>	Forest	0.1	38.2	31.8	19.5	9.3	13.5	4.6	0.5	0.0
	Other Wooded Land	0.5	39.4	19.7	9.6	3.1	1.2	0.6	0.1	
	Other Land	0.4	16.0	10.5	4.4	0.8	0.2	0.1	0.0	0.0
	Water		3.9	5.8	3.9		1.9			
Dry Afromontane	Forest	0.1	21.4	40.6	26.4	11.8	9.5	4.6	1.7	0.1
	Other Wooded Land	0.7	25.5	8.5	1.9	0.7	0.2	0.0		
	Other Land	0.4	9.1	5.8	3.2	1.0	0.5	0.1	0.0	0.0
	Water		2.6	7.8	2.6					
Moist Afromontane	Forest	0.5	24.8	48.1	30.3	19.3	13.5	6.1	1.8	0.7
	Other Wooded Land	1.4	74.0	39.3	4.4	1.0	0.2			
	Other Land	0.4	10.1	12.4	8.0	2.8	1.4	0.7	0.4	0.1
	Water	9.3		120.7	65.0		46.4			
Other	Other Wooded Land		18.5							
	Other Land		6.9							

Table A3.3 Number of trees per hectare by regions and FRA classes

Region	FRA Class	N of Trees (N ha ⁻¹)
Afar	Other Wooded Land	13.8
	Other Land	2.2
	Water	21.9
Amhara	Forest	97.8
	Other Wooded Land	51.8
	Other Land	26.8
	Water	14.2
Beneshangul Gumu	Forest	127.9
	Other Wooded Land	125.4
	Other Land	25.0
	Water	23.5
Gambela	Forest	182.4
	Other Wooded Land	29.8
	Other Land	2.9
Oromia	Forest	126.9
	Other Wooded Land	94.5
	Other Land	23.5
	Water	362.8
SNNPR	Forest	121.4
	Other Wooded Land	57.8
	Other Land	34.6
	Water	
Somali	Forest	82.3
	Other Wooded Land	50.0
	Other Land	2.9
Tigray	Forest	155.4
	Other Wooded Land	67.4
	Other Land	26.2
	Water	

Table A3.4 Number of trees per hectare by regions and biomes

Region	Biome	N of Trees (N ha ⁻¹)
Afar	<i>Acacia-Commiphora</i>	6.8
	<i>Combretum-Terminalia</i>	6.0
	Other	0.5
Amhara	<i>Acacia-Commiphora</i>	8.1
	<i>Combretum-Terminalia</i>	54.3
	Dry Afromontane	23.8
	Moist Afromontane	9.5
	Other	
Beneshangul Gumu	<i>Combretum-Terminalia</i>	95.5
Gambela	<i>Combretum-Terminalia</i>	18.6
	Moist Afromontane	144.9
Oromia	<i>Acacia-Commiphora</i>	75.0
	<i>Combretum-Terminalia</i>	53.5
	Dry Afromontane	26.0
	Moist Afromontane	82.2
	Other	
SNNPR	<i>Acacia-Commiphora</i>	23.2
	<i>Combretum-Terminalia</i>	51.1
	Dry Afromontane	38.3
	Moist Afromontane	95.9
	Other	
Somali	<i>Acacia-Commiphora</i>	45.1
	Other	23.5
Tigray	<i>Acacia-Commiphora</i>	83.5
	<i>Combretum-Terminalia</i>	48.6
	Dry Afromontane	35.3
	Moist Afromontane	2.2

Table A3.5 Number of trees per hectare by regions, DBH and FRA classes

Region	FRA Class	N of Trees (N ha ⁻¹)						
		10-14.9 cm	15-19.9 cm	20-24.9 cm	25-29.9 cm	30-34.9 cm	35-39.9 cm	40+ cm
Afar	Other Wooded Land	8.0	3.0	1.6	0.4	0.5	0.1	0.3
	Other Land	0.6	0.4	0.3	0.3	0.2	0.4	
	Water	3.8	8.3	5.3	3.0			1.5
Amhara	Forest	32.5	9.0	14.0	9.8	4.7	3.8	24.0
	Other Wooded Land	15.1	13.8	8.5	5.9	3.0	2.4	3.1
	Other Land	7.1	5.7	4.6	3.3	2.2	1.4	2.4
	Water	2.9	3.4		2.9	3.4	1.5	
Beneshangu l Gumu	Forest	26.1	15.6	26.7	21.6	13.9	9.4	14.6
	Other Wooded Land	36.5	35.7	23.5	12.3	7.6	4.1	5.8
	Other Land	3.4	3.8	4.4	3.5	3.2	2.5	4.2
	Water		5.9	11.8		5.9		
Gambela	Forest	9.5	128.6	15.7	8.8	4.0	3.1	12.8
	Other Wooded Land	7.0	9.4	5.2	4.8	2.0	0.7	0.7
	Other Land	0.0	0.2	0.3	0.2	0.2	0.2	1.6
Oromia	Forest	25.7	19.8	23.7	15.0	9.9	6.6	26.2
	Other Wooded Land	44.1	26.4	13.6	5.0	2.5	1.3	1.7
	Other Land	5.7	4.6	4.0	2.3	2.0	1.1	3.8
	Water	14.0	83.7	69.8	41.9	69.8	14.0	69.8
SNNPR	Forest	28.8	24.2	19.0	12.4	8.3	6.7	22.1
	Other Wooded Land	19.8	13.1	10.6	5.4	3.7	2.1	3.1
	Other Land	8.9	7.3	5.7	3.7	2.6	1.5	4.8
Somali	Forest	32.3	21.5	18.1	6.5	2.6	1.3	
	Other Wooded Land	29.1	12.2	5.2	1.8	1.0	0.4	0.3
	Other Land	1.3	0.5	0.6	0.2		0.1	0.2
Tigray	Forest	19.4	25.7	56.6	35.4	14.3	2.7	1.3
	Other Wooded Land	29.6	16.5	8.8	5.2	2.6	1.9	2.6
	Other Land	9.6	6.0	4.3	2.3	1.8	1.0	1.4

Table A3.6 Number of trees per hectare by regions, height and FRA classes

Region	FRA Class	N of Trees (N ha ⁻¹)								
		< 5m	5 - 9.9m	10 - 14.9m	15 - 19.9m	20 - 24.9m	25 - 29.9m	30 - 34.9m	35 - 39.9m	40+ m
Afar	Other Wooded Land	0.2	11.9	8.4	4.6	0.9	0.4	0.2	0.1	0.0
	Other Land		4.9	7.8	1.5					
	Water	0.0	28.7	35.0	22.3	12.6	21.9	6.8	0.5	
Amhara	Forest	0.5	45.4	33.9	25.1	11.8	5.3	2.8	0.6	
	Other Wooded Land	0.1	15.0	7.7	1.8	0.2	0.1			
	Other Land			5.9	11.8		5.9			
	Water		4.7	10.5	33.8	83.1	29.6	12.1	4.5	4.1
Beneshangu l Gumu	Forest		3.0	10.8	12.1	3.0	0.9			
	Other Wooded Land	0.0	0.1	0.6	0.2	0.5	0.4	0.2	0.3	0.6
	Other Land	0.5	25.3	46.2	27.8	11.2	9.1	4.5	1.7	0.6
	Water	1.3	71.0	19.1	2.9	0.1	0.1	0.0		
Gambela	Forest	0.7	10.0	7.1	3.2	1.4	0.6	0.2	0.2	0.1
	Other Wooded Land	14.0		181.4	97.7		69.8			
	Other Land	0.1	21.1	44.3	25.9	14.6	9.8	3.9	1.6	0.1
Oromia	Forest	0.8	34.5	16.3	4.6	1.1	0.5	0.2		
	Other Wooded Land	0.3	10.8	13.0	7.4	1.9	0.8	0.3	0.1	0.0
	Other Land		24.1	43.9	14.2					
	Water	4.8	40.6	4.2	0.3	0.0				
SNNPR	Forest	0.1	1.8	0.6	0.3					
	Other Wooded Land		118.0	34.4	2.7	0.3				
	Other Land	6.7	42.0	14.1	3.8	0.7	0.1	0.1		
Somali	Forest	0.3	18.4	5.0	2.2	0.4	0.1			
	Other Wooded Land	4.8	40.6	4.2	0.3	0.0				
	Other Land	0.1	1.8	0.6	0.3					
Tigray	Forest		118.0	34.4	2.7	0.3				
	Other Wooded Land	6.7	42.0	14.1	3.8	0.7	0.1	0.1		
	Other Land	0.3	18.4	5.0	2.2	0.4	0.1			

Appendix A4 Regeneration count and biomass

Table A4.1 Regeneration count by regions and biomes

Region	Biome	Count (N ha ⁻¹)	AGB (t ha ⁻¹)	BGB (t ha ⁻¹)	Biomass (t ha ⁻¹)
Afar	<i>Acacia-Commiphora</i>	761.1	1.0	0.4	1.4
	Other	9 500.0	12.0	4.8	16.8
Amhara	<i>Acacia-Commiphora</i>	2 390.7	3.0	0.8	3.8
	<i>Combretum-Terminalia</i>	1 570.2	2.0	0.8	2.8
	Dry Afromontane	1 155.5	1.5	0.4	1.9
	Moist Afromontane	0.0	0.0	0.0	0.0
Beneshangul Gumu	<i>Combretum-Terminalia</i>	2 599.5	3.3	1.4	4.7
Gambela	<i>Combretum-Terminalia</i>	2 488.2	3.1	1.1	4.3
	Moist Afromontane	4 731.4	6.0	1.2	7.2
Oromia	<i>Acacia-Commiphora</i>	2 829.4	3.6	1.2	4.8
	<i>Combretum-Terminalia</i>	2 002.9	2.5	0.7	3.2
	Dry Afromontane	7 184.2	9.1	2.4	11.5
	Moist Afromontane	2 671.3	3.4	0.9	4.3
SNNPR	<i>Acacia-Commiphora</i>	1 862.4	2.4	0.8	3.1
	<i>Combretum-Terminalia</i>	2 267.0	2.9	0.9	3.8
	Dry Afromontane	2 128.5	2.7	0.7	3.4
	Moist Afromontane	1 991.6	2.5	0.7	3.2
Somali	<i>Acacia-Commiphora</i>	1 984.8	2.5	1.0	3.5
	Other	4 266.7	5.4	2.2	7.5
Tigray	<i>Acacia-Commiphora</i>	482.0	0.6	0.2	0.8
	<i>Combretum-Terminalia</i>	1 410.0	1.8	0.7	2.4
	Dry Afromontane	517.3	0.7	0.2	0.8
	Moist Afromontane	1 000.0	1.3	0.3	1.6

Table A4.2 Regeneration count by regions and FRA classes

Region	FRA class	Count (N ha ⁻¹)	AGB (t ha ⁻¹)	BGB (t ha ⁻¹)	Biomass (t ha ⁻¹)
Afar	Other Wooded Land	1 269.4	1.6	0.7	2.3
Amhara	Forest	1 372.0	1.7	0.7	2.5
	Other Wooded Land	1 528.1	1.9	0.7	2.6
Beneshangul Gumu	Forest	2 853.7	3.6	1.3	4.9
	Other Wooded Land	2 406.9	3.0	1.5	4.5
Gambela	Forest	4 731.4	6.0	1.2	7.2
	Other Wooded Land	2 488.2	3.1	1.1	4.3
Oromia	Forest	4 628.2	5.8	1.6	7.5
	Other Wooded Land	2 757.4	3.5	1.1	4.6
SNNPR	Forest	2 270.6	2.9	0.8	3.6
	Other Wooded Land	2 189.7	2.8	0.9	3.7
Somali	Forest	6 087.9	7.7	3.1	10.8
	Other Wooded Land	1 999.6	2.5	1.0	3.5
Tigray	Forest	2 199.0	2.8	1.0	3.8
	Other Wooded Land	934.7	1.2	0.4	1.6

Appendix A5 Basal area

Table A5.1 Tree basal area by DBH classes. biomes and FRA classes

Biome	FRA Class	10- 14.9 cm	15- 19.9 cm	20- 24.9 cm	25- 29.9 cm	30- 34.9 cm	35- 39.9 cm	40+ cm
Acacia- Commiphora	Forest	0.3	0.3	0.9	0.7	0.7	0.5	3.0
	Other Wooded Land	0.4	0.3	0.3	0.1	0.1	0.1	0.1
	Other Land	0.0	0.0	0.0	0.0	0.0	0.0	0.4
	Water	0.0	0.2	0.2	0.2			0.3
Combretum- Terminalia	Forest	0.2	0.4	1.0	1.2	1.0	0.7	3.9
	Other Wooded Land	0.3	0.4	0.5	0.4	0.3	0.3	0.9
	Other Land	0.1	0.1	0.2	0.2	0.2	0.2	1.1
	Water		0.2	0.1		0.5		
Dry Afromontane	Forest	0.3	0.5	0.8	0.8	0.6	0.6	8.9
	Other Wooded Land	0.1	0.2	0.3	0.2	0.2	0.1	0.3
	Other Land	0.1	0.1	0.1	0.1	0.1	0.1	0.8
	Water	0.1			0.3		0.3	
Moist Afromontane	Forest	0.4	0.7	0.8	0.8	0.7	0.8	16.6
	Other Wooded Land	0.5	0.7	0.9	0.6	0.5	0.3	1.1
	Other Land	0.1	0.2	0.2	0.2	0.3	0.2	5.6
	Water	0.1	1.3	1.6	1.6	3.8	0.9	11.2
Other	Other Wooded Land	0.2	0.0					
	Other Land	0.1		0.1				

Table A5.2 Tree basal area by height classes, biomes and FRA classes

Biome	FRA Class	Basal area (m ² ha ⁻¹)								
		< 5m	5 - 9.9m	10 - 14.9m	15 - 19.9m	20 - 24.9m	25 - 29.9m	30 - 34.9m	35 - 39.9m	40+ m
Acacia-Commiphora	Forest	0.0	0.5	1.7	1.5	0.4	0.6	0.6	0.5	0.6
	Other Wooded Land	0.1	1.0	0.3	0.1	0.0	0.0			
	Other Land	0.0	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.1
	Water		0.7	0.1						
Combretum-Terminalia	Forest	0.0	1.2	2.0	1.5	1.1	1.5	0.7	0.3	0.0
	Other Wooded Land	0.0	1.0	0.9	0.6	0.3	0.1	0.1	0.0	
	Other Land	0.0	0.7	0.7	0.5	0.1	0.1	0.1	0.0	0.0
	Water		0.1	0.4	0.1		0.2			
Dry Afromontane	Forest	0.0	0.8	1.7	2.6	1.4	2.7	2.0	1.3	0.0
	Other Wooded Land	0.0	0.8	0.3	0.2	0.1	0.0	0.0		
	Other Land	0.0	0.4	0.4	0.3	0.2	0.1	0.0	0.0	0.0
	Water		0.2	0.2	0.3					
Moist Afromontane	Forest	0.1	0.9	2.1	3.1	3.6	4.6	3.1	1.7	1.5
	Other Wooded Land	0.0	1.9	1.9	0.4	0.3	0.0			
	Other Land	0.0	0.5	0.7	2.1	0.8	1.3	0.7	0.4	0.3
	Water	1.4		4.6	4.7		9.9			
Other	Other Wooded Land		0.2							
	Other Land		0.1							

Table A5.3 Tree basal area by DBH classes, regions and FRA classes

Region	FRA Class	Basal area (m ² ha ⁻¹)						
		10-14.9 cm	15-19.9 cm	20-24.9 cm	25-29.9 cm	30-34.9 cm	35-39.9 cm	40+ cm
Afar	Other Wooded Land	0.1	0.1	0.1	0.0	0.0	0.0	0.1
	Other Land	0.0	0.0	0.0	0.0	0.0	0.0	
	Water	0.0	0.2	0.2	0.2			0.3
Amhara	Forest	0.4	0.2	0.5	0.6	0.4	0.4	14.2
	Other Wooded Land	0.2	0.3	0.3	0.3	0.2	0.3	0.6
	Other Land	0.1	0.1	0.2	0.2	0.2	0.1	1.0
	Water	0.0	0.1		0.2	0.3	0.2	
Beneshangu I Gumu	Forest	0.3	0.3	1.0	1.2	1.1	1.0	2.8
	Other Wooded Land	0.4	0.8	0.9	0.7	0.6	0.4	1.2
	Other Land	0.0	0.1	0.2	0.2	0.3	0.3	0.9
	Water		0.1	0.4		0.5		
Gambela	Forest	0.1	3.1	0.6	0.5	0.3	0.3	13.5
	Other Wooded Land	0.1	0.2	0.2	0.3	0.2	0.1	0.1
	Other Land	0.0	0.0	0.0	0.0	0.0	0.0	0.8
Oromia	Forest	0.3	0.4	0.9	0.8	0.8	0.7	12.7
	Other Wooded Land	0.5	0.6	0.5	0.3	0.2	0.1	0.4
	Other Land	0.1	0.1	0.2	0.1	0.2	0.1	1.4
	Water	0.2	2.0	2.4	2.5	5.8	1.4	16.8
SNNPR	Forest	0.3	0.5	0.7	0.7	0.7	0.7	10.3
	Other Wooded Land	0.2	0.3	0.4	0.3	0.3	0.2	0.8
	Other Land	0.1	0.2	0.2	0.2	0.2	0.2	4.3
Somali	Forest	0.5	0.5	0.7	0.4	0.2	0.1	
	Other Wooded Land	0.3	0.3	0.2	0.1	0.1	0.0	0.1
	Other Land	0.0	0.0	0.0	0.0		0.0	0.0
Tigray	Forest	0.2	0.6	2.1	1.9	1.1	0.3	0.2
	Other Wooded Land	0.3	0.4	0.3	0.3	0.2	0.2	1.0
	Other Land	0.1	0.1	0.2	0.1	0.1	0.1	0.3

Table A5.4 Tree basal area by regions and biomes

Region	Biome	Basal area (m ² ha ⁻¹)
Afar	<i>Acacia-Commiphora</i>	0.2
	<i>Combretum-Terminalia</i>	0.5
	Other	0.0
Amhara	<i>Acacia-Commiphora</i>	0.2
	<i>Combretum-Terminalia</i>	3.2
	Dry Afromontane	2.0
	Moist Afromontane	0.6
Beneshangul Gumu	<i>Combretum-Terminalia</i>	4.9
Gambela	<i>Combretum-Terminalia</i>	0.7
	Moist Afromontane	16.3
Oromia	<i>Acacia-Commiphora</i>	2.4
	<i>Combretum-Terminalia</i>	3.1
	Dry Afromontane	2.0
	Moist Afromontane	9.6
SNNPR	<i>Acacia-Commiphora</i>	1.6
	<i>Combretum-Terminalia</i>	3.1
	Dry Afromontane	2.6
	Moist Afromontane	18.1
Somali	<i>Acacia-Commiphora</i>	1.0
	Other	0.3
Tigray	<i>Acacia-Commiphora</i>	1.8
	<i>Combretum-Terminalia</i>	2.3
	Dry Afromontane	1.0
	Moist Afromontane	0.1

Table A5.5 Tree basal area by height classes. regions and FRA classes

Region	FRA Class	Basal area (m ² ha ⁻¹)								
		< 5m	5 - 9.9m	10 - 14.9m	15 - 19.9m	20 - 24.9m	25 - 29.9m	30 - 34.9m	35 - 39.9m	40+ m
Afar	Other Wooded Land	0.0	0.2	0.1	0.1					
	Other Land		0.1	0.0						
	Water		0.7	0.1						
Amhara	Forest	0.0	0.7	1.3	2.0	1.4	5.1	4.2	1.8	0.1
	Other Wooded Land	0.0	1.0	0.6	0.4	0.1	0.1	0.0		
	Other Land	0.0	0.5	0.6	0.5	0.1	0.1	0.0	0.0	0.0
	Water		0.2	0.4	0.2					
Beneshangu l Gumu	Forest	0.0	0.9	1.7	1.3	1.0	2.0	0.8	0.1	
	Other Wooded Land	0.0	1.0	1.3	1.2	0.7	0.4	0.3	0.1	
	Other Land	0.0	0.8	0.8	0.3	0.0	0.0			
	Water			0.2	0.3		0.5			
Gambela	Forest		0.2	0.8	1.9	2.8	1.5	1.2	1.4	8.5
	Other Wooded Land		0.1	0.3	0.4	0.2	0.1			
	Other Land	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.4
Oromia	Forest	0.1	0.9	2.2	3.1	2.2	3.3	2.6	1.4	0.9
	Other Wooded Land	0.0	1.5	0.7	0.2	0.0	0.0	0.0		
	Other Land	0.0	0.5	0.5	0.4	0.3	0.2	0.1	0.1	0.1
	Water	2.1		6.9	7.1		14.9			
SNNPR	Forest	0.0	0.6	2.3	2.4	3.6	2.8	1.1	1.0	0.1
	Other Wooded Land	0.0	0.9	0.9	0.4	0.2	0.1	0.0		
	Other Land	0.0	0.3	0.7	1.6	0.6	1.1	0.6	0.3	0.2
Somali	Forest		0.4	1.2	0.8					
	Other Wooded Land	0.1	0.8	0.2	0.0	0.0				
	Other Land	0.0	0.0	0.0	0.0					
Tigray	Forest		4.3	2.0	0.2	0.0				
	Other Wooded Land	0.1	1.0	0.6	0.5	0.3	0.0	0.2		
	Other Land	0.0	0.5	0.3	0.1	0.1	0.0			

Appendix A6 Biodiversity

Table A6.1 Frequency and volume of the 10 most abundant species by FRA classes

FRA Class	Species	N of Trees (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
Forest	<i>Croton macrostachyus</i>	4.9	5.0
	<i>Cupressus lusitanica</i>	4.8	0.8
	<i>Syzygium guineense</i>	4.6	10.1
	<i>Tamarindus indica</i>	4.0	0.6
	<i>Millettia ferruginea</i>	3.9	1.5
	<i>Commiphora erythraea</i>	3.4	0.6
	<i>Juniperus procera</i>	3.0	9.2
	<i>Combretum molle</i>	2.8	2.6
	<i>Pouteria adolfi-friedericii</i>	2.6	15.8
	<i>Boswellia papyrifera</i>	2.6	0.5
Other Wooded Land	<i>Acacia bussei</i>	5.7	0.5
	<i>Commiphora erythraea</i>	4.8	0.4
	<i>Boswellia rivae</i>	3.7	0.2
	<i>Combretum molle</i>	2.4	0.4
	<i>Acacia nilotica</i>	2.0	0.1
	<i>Acacia seyal</i>	1.9	0.2
	<i>Acacia senegal</i>	1.6	0.1
	<i>Acacia tortilis</i>	1.6	0.2
	<i>Terminalia brownii</i>	1.4	0.1
	<i>Commiphora africana</i>	1.2	0.1
Other Land	<i>Eucalyptus camaldulensis</i>	1.7	0.4
	<i>Croton macrostachyus</i>	1.5	0.6
	<i>Eucalyptus globulus</i>	1.5	0.5
	<i>Cordia africana</i>	1.2	1.3
	<i>Combretum molle</i>	0.5	0.1
	<i>Millettia ferruginea</i>	0.4	0.2
	<i>Acacia abyssinica</i>	0.4	0.1
	<i>Ficus sur</i>	0.4	0.5
	<i>Acacia nilotica</i>	0.4	0.0
	<i>Terminalia brownii</i>	0.4	0.1
Water	<i>Acacia senegal</i>	6.9	0.6
	<i>Acacia nilotica</i>	4.3	0.3
	<i>Buddleja sp.</i>	4.1	3.1
	<i>Ximenia americana</i>	1.6	0.5
	<i>Balanites aegyptiaca</i>	1.6	0.1
	<i>Commiphora habessinica</i>	1.1	0.3
	<i>Eucalyptus globulus</i>	1.0	0.3
	<i>Celtis africana</i>	0.7	0.2
	<i>Fagaropsis angolensis</i>	0.7	0.1
	<i>Calotropis procera</i>	0.5	0.0

Table A6.2 Frequency and volume of the 10 most abundant species by biomes

Biome	Species	N of Trees (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
Acacia-Commiphora	<i>Acacia bussei</i>	5.6	0.5
	<i>Commiphora erythraea</i>	5.1	0.5
	<i>Boswellia rivae</i>	3.6	0.2
	<i>Acacia nilotica</i>	1.8	0.1
	<i>Acacia tortilis</i>	1.6	0.2
	<i>Acacia senegal</i>	1.5	0.1
	<i>Combretum molle</i>	1.4	0.1
	<i>Terminalia brownii</i>	1.3	0.1
	<i>Commiphora africana</i>	1.2	0.1
	<i>Acacia seyal</i>	1.1	0.1
Combretum-Terminalia	<i>Combretum molle</i>	3.1	1.0
	<i>Terminalia laxiflora</i>	2.0	0.6
	<i>Acacia seyal</i>	1.9	0.2
	<i>Boswellia papyrifera</i>	1.5	0.3
	<i>Anogeissus leiocarpus</i>	1.1	0.5
	<i>Eucalyptus camaldulensis</i>	1.0	0.2
	<i>Ziziphus spina-christi</i>	0.9	0.3
	<i>Ficus sur</i>	0.9	0.8
	<i>Acacia lahai</i>	0.9	0.2
	<i>Terminalia brownii</i>	0.8	0.2
Dry Afromontane	<i>Eucalyptus globulus</i>	3.7	1.3
	<i>Eucalyptus camaldulensis</i>	3.1	0.6
	<i>Croton macrostachyus</i>	2.2	0.7
	<i>Cupressus lusitanica</i>	1.8	0.3
	<i>Juniperus procera</i>	1.7	2.9
	<i>Acacia abyssinica</i>	1.5	0.3
	<i>Cordia africana</i>	1.1	0.3
	<i>Acacia nilotica</i>	0.7	0.0
	<i>Acacia albida</i>	0.7	0.1
	<i>Syzygium guineense</i>	0.6	0.6
Moist Afromontane	<i>Croton macrostachyus</i>	5.6	4.5
	<i>Millettia ferruginea</i>	4.3	1.4
	<i>Syzygium guineense</i>	2.6	5.7
	<i>Cordia africana</i>	2.5	6.7
	<i>Maesa lanceolata</i>	2.3	0.8
	<i>Terminalia laxiflora</i>	1.9	0.2
	<i>Pouteria adolfi-friedericii</i>	1.9	13.5
	<i>Combretum molle</i>	1.5	1.0
	<i>Ficus sur</i>	1.5	4.0
	<i>Cyathea manniana</i>	1.5	0.1
Other	<i>Prosopis juliflora</i>	0.1	0.0

Table A6.3 Frequency and volume of the 10 most abundant species by regions and biomes

Region	Biome	Species	N of Trees (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
Afar	Acacia-Commiphora	<i>Acacia nilotica</i>	1.5	0.1
		<i>Buddleja polystachya</i>	1.0	0.0
		<i>Acacia etbaica</i>	0.6	0.0
		<i>Casuarina cunninghamiana</i>	0.6	0.0
		<i>Acacia</i> sp.	0.5	0.0
		<i>Acacia asak</i>	0.5	0.2
		<i>Acacia seyal</i>	0.4	0.1
		<i>Acacia senegal</i>	0.3	0.0
		<i>Commiphora</i> sp.	0.3	0.0
		<i>Dobera glabra</i>	0.1	0.0
	Combretum-Terminalia	<i>Pithecellobium dulce</i>	6.0	1.5
	Other	<i>Prosopis juliflora</i>	0.5	0.0
Amhara	Acacia-Commiphora	<i>Acacia tortilis</i>	3.9	0.3
		<i>Ziziphus spina-christi</i>	0.9	0.3
		<i>Euphorbia tirucalli</i>	0.8	0.0
		<i>Acacia asak</i>	0.6	0.0
		<i>Combretum molle</i>	0.5	0.1
		<i>Acacia bussei</i>	0.3	0.0
		<i>Schinus molle</i>	0.2	0.0
		<i>Balanites aegyptiaca</i>	0.2	0.0
		<i>Ficus sur</i>	0.1	0.0
		<i>Acacia nilotica</i>	0.1	0.0
	Combretum-Terminalia	<i>Acacia seyal</i>	4.5	0.3
		<i>Combretum molle</i>	3.3	0.9
		<i>Anogeissus leiocarpus</i>	3.2	1.6
		<i>Boswellia papyrifera</i>	2.3	0.4
		<i>Celtis toka</i>	2.1	0.9
		<i>Dalbergia melanoxylon</i>	2.0	0.2
		<i>Combretum aculeatum</i>	1.6	0.5
		<i>Albizia lophantha</i>	1.4	0.2
		<i>Eucalyptus camaldulensis</i>	1.2	0.1
		<i>Combretum collinum</i>	1.1	0.2
	Dry Afromontane	<i>Eucalyptus globulus</i>	5.1	2.0
		<i>Eucalyptus camaldulensis</i>	3.4	0.5
		<i>Croton macrostachyus</i>	3.0	0.8
		<i>Acacia abyssinica</i>	1.7	0.2
		<i>Cordia africana</i>	1.1	0.2
		<i>Juniperus procera</i>	0.9	5.9
		<i>Albizia gummifera</i>	0.6	0.8
		<i>Erythrina abyssinica</i>	0.4	0.1
		<i>Grewia ferruginea</i>	0.3	0.0
	Moist Afromontane	<i>Olea africana</i>	0.3	0.2
		<i>Eucalyptus globulus</i>	8.8	6.4
		<i>Dombeya torrida</i>	0.3	0.0
		<i>Juniperus procera</i>	0.3	0.3
		<i>Acacia abyssinica</i>	0.1	0.1
		<i>Maytenus senegalensis</i>	0.1	0.1
		<i>Hypericum roeperianum</i>	0.1	0.0

Table A6.3 Frequency and volume of the 10 most abundant species by regions and biomes

Region	Biome	Species	N of Trees (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
Beneshangu I Gumu	Combretum-Terminalia	<i>Combretum molle</i>	7.1	3.5
		<i>Acacia lahai</i>	3.6	1.0
		<i>Ziziphus spina-christi</i>	3.4	1.3
		<i>Terminalia laxiflora</i>	3.4	1.2
		<i>Strychnos spinosa</i>	2.4	0.6
		<i>Trilepisium madagascariense</i>	2.3	0.4
		<i>Entada abyssinica</i>	2.1	0.8
		<i>Ficus sur</i>	2.0	1.7
		<i>Combretum collinum</i>	1.9	0.1
		<i>Acacia seyal</i>	1.5	0.6
Gambela	Moist Afromontane	<i>Cordia africana</i>	2.4	6.8
		<i>Baphia abyssinica</i>	1.2	0.8
		<i>Ilex mitis</i>	0.5	2.8
		<i>Galiniera coffeoides</i>	0.3	0.2
		<i>Syzygium guineense</i>	0.2	1.6
		<i>Macaranga lophostigma</i>	0.2	0.6
		<i>Allophylus abyssinicus</i>	0.1	0.7
		<i>Ekebergia capensis</i>	0.1	0.2
		<i>Ocotea kenyensis</i>	0.1	0.0
Oromia	Acacia-Commiphora	<i>Acacia bussei</i>	9.1	0.8
		<i>Commiphora erythraea</i>	6.3	0.8
		<i>Combretum molle</i>	4.5	0.4
		<i>Commiphora africana</i>	4.3	0.2
		<i>Terminalia brownii</i>	3.9	0.4
		<i>Acacia nilotica</i>	3.4	0.2
		<i>Commiphora baluensis</i>	2.6	0.2
		<i>Filicium decipiens</i>	2.3	0.3
		<i>Kirkia burgeri</i>	2.0	0.2
		<i>Olea africana</i>	1.7	0.3
	Combretum-Terminalia	<i>Terminalia laxiflora</i>	6.3	1.5
		<i>Dombeya torrida</i>	3.4	0.5
		<i>Eucalyptus camaldulensis</i>	3.0	1.2
		<i>Cordia africana</i>	2.2	0.7
		<i>Acacia seyal</i>	2.1	0.3
		<i>Croton macrostachyus</i>	2.0	0.6
		<i>Stereospermum kunthianum</i>	1.9	0.4
		<i>Combretum molle</i>	1.7	0.2
		<i>Acacia nilotica</i>	1.6	0.2
		<i>Entada abyssinica</i>	1.3	0.2
	Dry Afromontane	<i>Cupressus lusitanica</i>	3.7	0.6
		<i>Eucalyptus globulus</i>	2.8	0.7
		<i>Eucalyptus camaldulensis</i>	2.7	0.7
		<i>Croton macrostachyus</i>	1.7	0.5
		<i>Juniperus procera</i>	1.5	1.0
		<i>Cordia africana</i>	1.3	0.5
		<i>Maesa lanceolata</i>	0.9	0.3
		<i>Olea africana</i>	0.8	0.3
		<i>Erythrina brucei</i>	0.8	0.1
		<i>Acacia abyssinica</i>	0.6	0.2
	Moist Afromontane	<i>Croton macrostachyus</i>	7.1	6.1
		<i>Syzygium guineense</i>	4.0	8.8
		<i>Terminalia laxiflora</i>	3.1	0.4
		<i>Millettia ferruginea</i>	2.6	1.0
		<i>Pouteria adolfi-friedericii</i>	2.2	20.0
		<i>Grewia bicolor</i>	2.0	0.2
		<i>Allophylus abyssinicus</i>	2.0	1.5
		<i>Olea welwitschii</i>	1.9	0.8
		<i>Eucalyptus camaldulensis</i>	1.9	0.6
		<i>Cordia africana</i>	1.9	1.2

Table A6.3 Frequency and volume of the 10 most abundant species by regions and biomes

Region	Biome	Species	N of Trees (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
SNNPR	Acacia-Commiphora	<i>Acacia tortilis</i>	6.2	1.1
		<i>Rytigynia neglecta</i>	2.3	1.6
		<i>Croton macrostachyus</i>	1.6	0.6
		<i>Cordia africana</i>	1.2	0.3
		<i>Olea capensis</i>	1.1	1.0
		<i>Grevillea robusta</i>	1.0	0.2
		<i>Podocarpus falcatus</i>	0.8	1.1
		<i>Celtis africana</i>	0.8	0.9
		<i>Eucalyptus camaldulensis</i>	0.7	0.1
		<i>Casuarina equisetifolia</i>	0.6	0.1
	Combretum-Terminalia	<i>Terminalia brownii</i>	3.7	0.9
		<i>Combretum molle</i>	2.2	0.4
		<i>Terminalia laxiflora</i>	1.8	0.5
		<i>Cordia africana</i>	1.6	0.8
		<i>Acacia senegal</i>	1.1	0.1
		<i>Eucalyptus camaldulensis</i>	1.1	0.1
		<i>Baphia abyssinica</i>	1.1	0.5
		<i>Eucalyptus grandis</i>	1.1	0.9
		<i>Acacia tortilis</i>	1.1	0.2
		<i>Ficus sur</i>	1.0	0.8
	Dry Afromontane	<i>Eucalyptus camaldulensis</i>	5.4	1.2
		<i>Syzygium guineense</i>	4.8	4.8
		<i>Eucalyptus globulus</i>	3.4	1.9
		<i>Juniperus procera</i>	3.2	1.3
		<i>Croton macrostachyus</i>	2.5	1.3
		<i>Cupressus lusitanica</i>	2.2	0.3
		<i>Eucalyptus grandis</i>	1.7	2.4
		<i>Cordia africana</i>	1.5	0.4
		<i>Maesa lanceolata</i>	1.0	0.2
		<i>Acacia seyal</i>	0.8	0.2
	Moist Afromontane	<i>Millettia ferruginea</i>	12.4	3.8
		<i>Cyathea manniana</i>	6.9	0.6
		<i>Cordia africana</i>	6.1	26.9
		<i>Croton macrostachyus</i>	5.9	3.4
		<i>Maesa lanceolata</i>	5.8	2.7
		<i>Combretum molle</i>	4.0	3.9
		<i>Galiniera coffeoides</i>	3.9	0.2
		<i>Celtis africana</i>	2.3	7.6
		<i>Pouteria adolfi-friedericii</i>	2.2	5.5
		<i>Ficus sur</i>	1.8	2.2
Somali	Acacia-Commiphora	<i>Commiphora erythraea</i>	6.3	0.5
		<i>Boswellia rivae</i>	5.9	0.3
		<i>Acacia bussei</i>	5.8	0.5
		<i>Acacia senegal</i>	2.0	0.1
		<i>Acacia tortilis</i>	1.9	0.2
		<i>Acacia mellifera</i>	1.6	0.1
		<i>Ximenia americana</i>	1.4	0.1
		<i>Acacia seyal</i>	1.4	0.1
		<i>Acacia nilotica</i>	1.2	0.1
		<i>Terminalia polycarpa</i>	1.1	0.1

Table A6.3 Frequency and volume of the 10 most abundant species by regions and biomes

Region	Biome	Species	N of Trees (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
Tigray	<i>Acacia-Commiphora</i>	<i>Euphorbia candelabrum</i>	83.5	3.1
	<i>Combretum-Terminalia</i>	<i>Boswellia papyrifera</i>	6.2	1.3
		<i>Teclea nobilis</i>	3.4	0.3
		<i>Combretum molle</i>	2.7	0.8
		<i>Ziziphus spina-christi</i>	2.3	0.2
		<i>Acacia seyal</i>	2.3	0.2
		<i>Anogeissus leiocarpus</i>	2.0	0.4
		<i>Grewia ferruginea</i>	2.0	0.3
		<i>Acacia abyssinica</i>	1.8	0.1
		<i>Commiphora habessinica</i>	1.6	0.2
		<i>Lannea schimperi</i>	1.5	0.1
	Dry Afromontane	<i>Acacia albida</i>	6.3	0.6
		<i>Acacia abyssinica</i>	6.2	0.6
		<i>Acacia nilotica</i>	6.1	0.3
		<i>Juniperus procera</i>	5.1	0.4
		<i>Eucalyptus globulus</i>	2.0	0.2
		<i>Eucalyptus camaldulensis</i>	1.3	0.2
		<i>Eucalyptus grandis</i>	1.1	0.1
		<i>Croton macrostachyus</i>	1.0	0.1
		<i>Euclea racemosa</i>	0.7	0.1
		<i>Pittosporum viridiflorum</i>	0.7	0.1
		<i>Olea africana</i>	0.7	0.1
	Moist Afromontane	<i>Juniperus procera</i>	2.2	0.2

Table A6.4 Frequency and volume of the 10 most abundant species by regions (Forest only. Ranked by frequency)

Region	Species	N of Trees (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
Amhara	<i>Eucalyptus camaldulensis</i>	23.6	5.3
	<i>Eucalyptus globulus</i>	11.6	16.5
	<i>Juniperus procera</i>	9.9	116.7
	<i>Dalbergia melanoxylon</i>	8.2	0.9
	<i>Pittosporum viridiflorum</i>	3.7	1.5
	<i>Croton macrostachyus</i>	3.7	1.6
	<i>Ekebergia capensis</i>	3.7	0.2
	<i>Albizia gummifera</i>	3.3	8.4
	<i>Maesa lanceolata</i>	2.8	0.5
	<i>Anogeissus leiocarpus</i>	2.6	1.4
Beneshangul Gumu	<i>Combretum molle</i>	9.8	8.0
	<i>Acacia lahai</i>	6.8	2.3
	<i>Ziziphus spina-christi</i>	6.6	2.9
	<i>Entada abyssinica</i>	2.3	1.4
	<i>Fagaropsis angolensis</i>	1.8	2.6
	<i>Croton macrostachyus</i>	1.8	0.2
	<i>Acacia sieberiana</i>	1.6	0.9
	<i>Anogeissus leiocarpus</i>	1.4	0.7
	<i>Acacia bussei</i>	1.3	0.1
	<i>Stereospermum kunthianum</i>	1.2	0.7

Table A6.4 Frequency and volume of the 10 most abundant species by region (Forest only, ranked by frequency)

Region	Species	N of Trees (N ha ⁻¹)	Volume (m ³ ha ⁻¹)
Gambela	<i>Baphia abyssinica</i>	1.5	1.0
	<i>Ilex mitis</i>	0.7	3.7
	<i>Cordia africana</i>	0.6	2.1
	<i>Galiniera coffeoides</i>	0.4	0.2
	<i>Syzygium guineense</i>	0.3	2.2
	<i>Allophylus abyssinicus</i>	0.1	1.0
	<i>Ekebergia capensis</i>	0.1	0.2
	<i>Macaranga lophostigma</i>	0.1	0.0
	<i>Ocotea kenyensis</i>	0.1	0.4
Oromia	<i>Cupressus lusitanica</i>	11.1	1.9
	<i>Croton macrostachyus</i>	8.4	9.3
	<i>Commiphora erythraea</i>	7.9	1.5
	<i>Syzygium guineense</i>	7.7	20.1
	<i>Millettia ferruginea</i>	4.3	1.7
	<i>Pouteria adolfi-friedericii</i>	4.0	33.4
	<i>Terminalia laxiflora</i>	3.7	0.7
	<i>Allophylus abyssinicus</i>	3.6	2.7
	<i>Maesa lanceolata</i>	2.9	1.2
	<i>Ficus sur</i>	2.7	9.6
SNNPR	<i>Millettia ferruginea</i>	10.2	3.9
	<i>Cyathea manniana</i>	9.9	0.9
	<i>Syzygium guineense</i>	6.6	7.5
	<i>Galiniera coffeoides</i>	5.5	0.2
	<i>Pouteria adolfi-friedericii</i>	4.1	7.8
	<i>Maesa lanceolata</i>	4.0	3.0
	<i>Croton macrostachyus</i>	3.8	4.4
	<i>Juniperus procera</i>	3.7	1.7
	<i>Eucalyptus grandis</i>	3.1	2.9
	<i>Terminalia brownii</i>	2.9	0.5
Somali	<i>Tamarindus indica</i>	81.0	12.7
Tigray	<i>Boswellia papyrifera</i>	52.1	8.7
	<i>Grewia ferruginea</i>	23.6	3.5
	<i>Juniperus procera</i>	16.6	1.9
	<i>Commiphora habessinica</i>	15.8	1.7
	<i>Stereospermum kunthianum</i>	12.4	1.8
	<i>Lannea schimperi</i>	5.7	0.5
	<i>Eucalyptus globulus</i>	3.2	0.8
	<i>Olea africana</i>	3.0	0.3
	<i>Acacia abyssinica</i>	3.0	0.1
	<i>Sterculia africana</i>	2.4	0.7

Appendix A7 Volume

Table A7.1 Tree volume by regions and FRA classes

Region	FRA Class	Volume (m ³ ha ⁻¹)
Afar	Other Wooded Land	1.6
	Other Land	0.3
	Water	2.5
Amhara	Forest	181.1
	Other Wooded Land	10.8
	Other Land	10.8
	Water	4.1
Beneshangul Gumuz	Forest	66.3
	Other Wooded Land	35.3
	Other Land	8.6
	Water	8.5
Gambela	Forest	262.5
	Other Wooded Land	7.3
	Other Land	12.4
Oromia	Forest	166.5
	Other Wooded Land	10.4
	Other Land	15.2
	Water	248.2
SNNPR	Forest	127.9
	Other Wooded Land	12.9
	Other Land	48.4
	Water	
Somali	Forest	13.2
	Other Wooded Land	3.4
	Other Land	0.4
Tigray	Forest	24.2
	Other Wooded Land	15.2
	Other Land	4.7

Table A7.2 Tree volume by height classes, regions and FRA classes

Region	FRA Class	Volume (m ³ ha ⁻¹)								
		< 5m	5 - 9.9m	10 - 14.9m	15 - 19.9m	20 - 24.9m	25 - 29.9m	30 - 34.9m	35 - 39.9m	40+ m
Afar	Other Wooded Land	0.0	0.7	0.3	0.5					
	Other Land		0.3	0.0						
	Water		2.1	0.4						
Amhara	Forest	0.0	2.5	6.8	14.4	13.2	58.7	56.3	27.9	1.4
	Other Wooded Land	0.0	3.1	3.1	2.6	1.3	0.7	0.0		
	Other Land	0.0	1.5	3.1	3.7	1.2	0.7	0.4	0.1	0.0
	Water		0.6	2.3	1.1					
Beneshangul Gumuz	Forest	0.0	2.9	8.8	9.1	9.6	22.4	11.4	2.1	
	Other Wooded Land	0.0	3.2	6.6	8.8	6.4	5.1	4.0	1.2	
	Other Land	0.0	2.4	4.1	1.7	0.2	0.2			
	Water			0.9	2.5		5.2			
Gambela	Forest		0.7	3.8	14.1	25.3	17.2	15.9	22.2	163.2
	Other Wooded Land		0.4	1.6	3.0	1.6	0.7			
	Other Land	0.0	0.0	0.2	0.1	0.8	0.8	1.5	1.5	7.4
Oromia	Forest	0.1	2.9	11.4	22.1	19.7	37.4	34.8	21.2	16.9
	Other Wooded Land	0.1	4.5	3.5	1.6	0.3	0.4	0.1		
	Other Land	0.0	1.4	2.3	3.0	2.4	2.2	1.2	1.3	1.3
	Water	1.8		37.2	51.2		158.1			
SNNPR	Forest	0.0	1.8	12.4	16.9	34.0	31.0	14.9	15.4	1.3
	Other Wooded Land	0.0	2.8	4.4	2.4	1.6	1.0	0.7		
	Other Land	0.0	1.1	3.6	11.9	5.2	12.0	7.8	4.0	2.9
Somali	Forest		1.3	6.4	5.5					
	Other Wooded Land	0.1	2.2	0.9	0.2	0.0				
	Other Land	0.0	0.1	0.2	0.1					
Tigray	Forest		13.3	9.3	1.4	0.2				
	Other Wooded Land	0.2	2.9	3.0	4.0	2.6	0.1	2.5		
	Other Land	0.0	1.5	1.4	1.0	0.7	0.2			

Table A7.3 Tree volume by height classes, regions and biomes

Region	Biome	Volume (m ³ ha ⁻¹)								
		< 5m	5 - 9.9m	10 - 14.9m	15 - 19.9m	20 - 24.9m	25 - 29.9m	30 - 34.9m	35 - 39.9m	40+ m
Afar	<i>Acacia-Commiphora</i>	0.0	0.4	0.1	0.2					
	<i>Combretum-Terminalia</i>		1.5							
	Other		0.0							
Amhara	<i>Acacia-Commiphora</i>	0.0	0.4	0.3	0.2					
	<i>Combretum-Terminalia</i>	0.0	3.3	4.8	6.4	1.6	1.1	0.0		
	Dry Afromontane	0.0	1.1	2.6	2.2	1.7	3.3	3.1	1.4	0.1
	Moist Afromontane		0.2	0.3	0.7	0.5	0.9	2.7	1.3	0.4
Beneshangul Gumu	<i>Combretum-Terminalia</i>	0.0	2.9	6.4	6.8	5.4	8.6	4.9	1.1	
Gambela	<i>Combretum-Terminalia</i>		0.2	1.0	1.9	1.0	0.4			
	Moist Afromontane	0.0	0.6	3.4	10.9	21.4	15.4	16.6	21.2	144.8
Oromia	<i>Acacia-Commiphora</i>	0.0	3.5	2.8	1.6	0.7	1.0	0.8	0.8	1.7
	<i>Combretum-Terminalia</i>	0.0	2.6	5.4	3.9	2.0	2.2	0.8	1.7	0.1
	Dry Afromontane	0.0	1.3	2.2	3.7	2.5	1.9	0.8	0.2	0.0
	Moist Afromontane	0.1	3.0	6.6	11.7	11.3	20.3	18.4	11.2	8.0
SNNPR	<i>Acacia-Commiphora</i>	0.0	0.7	2.2	1.9	2.8	2.3	0.3	0.7	2.4
	<i>Combretum-Terminalia</i>	0.0	2.3	4.6	4.3	2.7	3.1	2.9	0.9	0.2
	Dry Afromontane	0.0	0.8	2.9	3.1	3.4	5.2	2.2	5.0	
	Moist Afromontane	0.0	1.8	10.2	36.6	29.7	45.6	25.0	15.2	7.8
Somali	<i>Acacia-Commiphora</i>	0.1	2.0	0.9	0.2	0.0				
	Other		0.6							
Tigray	<i>Acacia-Commiphora</i>	1.8	1.3	0.0						
	<i>Combretum-Terminalia</i>	0.0	3.0	3.1	2.7	1.8	0.2	1.1		
	Dry Afromontane	0.0	1.8	1.1	0.5	0.2	0.0			
	Moist Afromontane		0.2							

Table A7.4 Tree volume by height classes, regions and biomes

Region	Biome	Volume (m ³ ha ⁻¹)								
		< 5m	5 - 9.9m	10 - 14.9m	15 - 19.9m	20 - 24.9m	25 - 29.9m	30 - 34.9m	35 - 39.9m	40+ m
Afar	<i>Acacia-Commiphora</i>	0.0	0.4	0.1	0.2					
	<i>Combretum-Terminalia</i>		1.5							
	Other		0.0							
Amhara	<i>Acacia-Commiphora</i>	0.0	0.4	0.3	0.2					
	<i>Combretum-Terminalia</i>	0.0	3.3	4.8	6.4	1.6	1.1	0.0		
	Dry Afromontane	0.0	1.1	2.6	2.2	1.7	3.3	3.1	1.4	0.1
	Moist Afromontane		0.2	0.3	0.7	0.5	0.9	2.7	1.3	0.4
Beneshangul Gumu	<i>Combretum-Terminalia</i>	0.0	2.9	6.4	6.8	5.4	8.6	4.9	1.1	
Gambela	<i>Combretum-Terminalia</i>		0.2	1.0	1.9	1.0	0.4			
	Moist Afromontane	0.0	0.6	3.4	10.9	21.4	15.4	16.6	21.2	144.8
Oromia	<i>Acacia-Commiphora</i>	0.0	3.5	2.8	1.6	0.7	1.0	0.8	0.8	1.7
	<i>Combretum-Terminalia</i>	0.0	2.6	5.4	3.9	2.0	2.2	0.8	1.7	0.1
	Dry Afromontane	0.0	1.3	2.2	3.7	2.5	1.9	0.8	0.2	0.0
	Moist Afromontane	0.1	3.0	6.6	11.7	11.3	20.3	18.4	11.2	8.0
SNNPR	<i>Acacia-Commiphora</i>	0.0	0.7	2.2	1.9	2.8	2.3	0.3	0.7	2.4
	<i>Combretum-Terminalia</i>	0.0	2.3	4.6	4.3	2.7	3.1	2.9	0.9	0.2
	Dry Afromontane	0.0	0.8	2.9	3.1	3.4	5.2	2.2	5.0	
	Moist Afromontane	0.0	1.8	10.2	36.6	29.7	45.6	25.0	15.2	7.8
Somali	<i>Acacia-Commiphora</i>	0.1	2.0	0.9	0.2	0.0				
	Other		0.6							
Tigray	<i>Acacia-Commiphora</i>	1.8	1.3	0.0						
	<i>Combretum-Terminalia</i>	0.0	3.0	3.1	2.7	1.8	0.2	1.1		
	Dry Afromontane	0.0	1.8	1.1	0.5	0.2	0.0			
	Moist Afromontane		0.2							

Appendix A8 Biomass and carbon

Table A8.1 Tree biomass and carbon by biome and FRA classes

FRA Class	Major LUCC	LUCC	AG biomass (t ha ⁻¹)	BG biomass (t ha ⁻¹)	Biomass (t ha ⁻¹)	AG carbon (t ha ⁻¹)	BG carbon (t ha ⁻¹)	Carbon (t ha ⁻¹)
Forest	Natural regenerated forest	Evergreen forest	149.6	43.6	193.3	74.8	21.8	96.6
		Deciduous forest	83.4	24.5	107.9	41.7	12.3	54.0
		Semi-deciduous forest	28.3	11.7	39.9	14.1	5.8	20.0
		Bamboo forest	51.8	24.8	76.6	25.9	12.4	38.3
		High woodland	33.8	9.1	43.0	16.9	4.6	21.5
	Plantation	Broadleaved planted forest	67.3	18.2	85.5	33.7	9.1	42.7
		Coniferous planted forest	96.7	26.1	122.9	48.4	13.1	61.4
		Forest Plantation of Bamboo	28.4	7.7	36.1	14.2	3.8	18.0
Other Wooded Land	Other wooded land	Woodland	7.4	2.7	10.1	3.7	1.4	5.1
		Wooded grassland	9.1	2.6	11.8	4.6	1.3	5.9
		Wooded wetland	7.5	2.2	9.7	3.8	1.1	4.8
Other Land	Other land - Natural	Natural Forest Coffee	550.7*	148.7 *	699.3*	275.3*	74.3*	349.7 *
		Barren Land	0.2	0.1	0.2	0.1	0.0	0.1
		Natural Grassland	5.8	1.8	7.5	2.9	0.9	3.8
		Marsh	3.8	1.0	4.8	1.9	0.5	2.4
	Other land - Cultivated	Coffee plantation	169.2*	45.7*	214.9*	84.6*	22.8*	107.4*
		Annual Crop	8.6	2.7	11.3	4.3	1.3	5.6
		Perennial crop	78.4	24.3	102.7	39.2	12.2	51.4
		Mixed annual and perennial crop	29.7	8.0	37.7	14.8	4.0	18.9
		Fallow	6.4	2.2	8.5	3.2	1.1	4.3
		Wood lot of Bamboo	9.2	2.5	11.7	4.6	1.2	5.8
		Wood lot	32.6	8.8	41.3	16.3	4.4	20.7
	Other land - Build up	Built up area	17.1	4.8	21.9	8.5	2.4	10.9
	Other land - Quarry/ Mining	Quarry/Mining site	7.3	2.0	9.3	3.7	1.0	4.6
Water	Inland water	Perennial River	16.8	4.5	21.3	8.4	2.3	10.7
	Inland water	Intermittent River (seasonal)	3.6	1.3	4.9	1.8	0.7	2.5

*Results not sufficiently representative.

Table A8.2 Tree biomass and carbon by biome and LUCC

Biome	FRA Class	AG biomass (t ha ⁻¹)	BG biomass (t ha ⁻¹)	Biomass (t ha ⁻¹)	AG carbon (t ha ⁻¹)	BG carbon (t ha ⁻¹)	Carbon (t ha ⁻¹)
<i>Acacia-Commiphora</i>	Forest	56.8	20.0	76.8	28.4	10.0	38.4
	Other Wooded Land	5.0	1.8	6.8	2.5	0.9	3.4
	Other Land	4.5	1.6	6.1	2.3	0.8	3.0
	Water	2.5	1.0	3.5	1.3	0.5	1.8
<i>Combretum-Terminalia</i>	Forest	65.5	26.9	92.4	32.7	13.4	46.2
	Other Wooded Land	17.9	6.7	24.5	8.9	3.3	12.3
	Other Land	12.8	4.3	17.0	6.4	2.1	8.5
	Water	5.1	1.4	6.5	2.6	0.7	3.3
Dry Afromontane	Forest	114.4	30.9	145.3	57.2	15.5	72.7
	Other Wooded Land	6.5	1.8	8.3	3.2	0.9	4.1
	Other Land	7.8	2.1	9.9	3.9	1.1	4.9
	Water	3.7	1.0	4.7	1.8	0.5	2.3
Moist Afromontane	Forest	202.9	53.9	256.9	101.5	27.0	128.4
	Other Wooded Land	21.5	5.8	27.3	10.7	2.9	13.6
	Other Land	57.8	15.6	73.4	28.9	7.8	36.7
	Water	162.6	43.9	206.5	81.3	21.9	103.2
Other	Other Wooded Land	0.5	0.2	0.7	0.3	0.1	0.4
	Other Land	0.3	0.1	0.5	0.2	0.1	0.2

Table A8.3 Tree biomass and carbon by region and biome

Region	Biome	AG biomass (t ha ⁻¹)	BG biomass (t ha ⁻¹)	Biomass (t ha ⁻¹)	AG carbon (t ha ⁻¹)	BG carbon (t ha ⁻¹)	Carbon (t ha ⁻¹)
Afar	<i>Acacia-Commiphora</i>	0.7	0.3	1.0	0.4	0.1	0.5
	<i>Combretum-Terminalia</i>	1.5	0.8	2.3	0.8	0.4	1.1
	Other	0.0	0.0	0.0	0.0	0.0	0.0
Amhara	<i>Acacia-Commiphora</i>	0.9	0.2	1.1	0.4	0.1	0.5
	<i>Combretum-Terminalia</i>	17.0	6.5	23.5	8.5	3.3	11.8
	Dry Afromontane	14.6	4.0	18.6	7.3	2.0	9.3
	Moist Afromontane	6.8	1.8	8.6	3.4	0.9	4.3
Beneshangul Gumu	<i>Combretum-Terminalia</i>	36.1	17.3	53.3	18.0	8.6	26.7
Gambela	<i>Combretum-Terminalia</i>	4.6	1.3	5.9	2.3	0.6	3.0
	Moist Afromontane	215.5	46.4	261.9	107.7	23.2	130.9
Oromia	<i>Acacia-Commiphora</i>	12.8	4.4	17.2	6.4	2.2	8.6
	<i>Combretum-Terminalia</i>	18.5	5.4	23.8	9.2	2.7	11.9
	Dry Afromontane	12.5	3.4	15.9	6.2	1.7	7.9
	Moist Afromontane	85.6	23.2	108.9	42.8	11.6	54.4
SNNPR	<i>Acacia-Commiphora</i>	12.8	3.6	16.5	6.4	1.8	8.2
	<i>Combretum-Terminalia</i>	20.6	5.4	26.0	10.3	2.7	13.0
	Dry Afromontane	21.7	5.9	27.6	10.9	2.9	13.8
	Moist Afromontane	159.0	42.9	202.0	79.5	21.5	101.0
Somali	<i>Acacia-Commiphora</i>	3.3	1.3	4.6	1.7	0.7	2.3
	Other	0.7	0.3	1.0	0.3	0.1	0.5
Tigray	<i>Acacia-Commiphora</i>	3.3	0.9	4.2	1.7	0.4	2.1
	<i>Combretum-Terminalia</i>	11.9	4.6	16.5	6.0	2.3	8.2
	Dry Afromontane	3.6	1.0	4.6	1.8	0.5	2.3
	Moist Afromontane	0.2	0.1	0.2	0.1	0.0	0.1

Table A8.4 Tree biomass and carbon by region and FRA class

Region	FRA Class	AG biomass (t ha ⁻¹)	BG biomass (t ha ⁻¹)	Biomass (t ha ⁻¹)	AG carbon (t ha ⁻¹)	BG carbon (t ha ⁻¹)	Carbon (t ha ⁻¹)
Afar	Other Wooded Land	1.6	0.6	2.2	0.8	0.3	1.1
	Other Land	0.3	0.1	0.4	0.1	0.1	0.2
	Water	2.6	1.0	3.6	1.3	0.5	1.8
Amhara	Forest	170.2	47.8	218.1	85.1	23.9	109.0
	Other Wooded Land	10.9	4.2	15.2	5.5	2.1	7.6
	Other Land	10.5	3.4	13.9	5.3	1.7	7.0
	Water	4.1	1.1	5.2	2.1	0.6	2.6
Beneshangul Gumu	Forest	65.8	33.1	98.9	32.9	16.5	49.4
	Other Wooded Land	35.5	16.6	52.0	17.7	8.3	26.0
	Other Land	8.6	3.2	11.9	4.3	1.6	5.9
	Water	8.6	2.3	10.9	4.3	1.2	5.5
Gambela	Forest	240.5	49.2	289.7	120.3	24.6	144.9
	Other Wooded Land	7.4	2.1	9.4	3.7	1.0	4.7
	Other Land	11.6	3.1	14.7	5.8	1.6	7.4
Oromia	Forest	157.3	43.8	201.1	78.6	21.9	100.5
	Other Wooded Land	10.6	3.3	13.9	5.3	1.7	7.0
	Other Land	14.7	4.3	19.0	7.3	2.2	9.5
	Water	244.2	65.9	310.2	122.1	33.0	155.1
SNNPR	Forest	122.1	33.0	155.0	61.0	16.5	77.5
	Other Wooded Land	13.0	3.3	16.3	6.5	1.6	8.1
	Other Land	44.7	12.1	56.9	22.4	6.1	28.4
Somali	Forest	13.5	5.4	19.0	6.8	2.7	9.5
	Other Wooded Land	3.5	1.4	4.9	1.8	0.7	2.5
	Other Land	0.4	0.2	0.6	0.2	0.1	0.3
Tigray	Forest	24.9	9.5	34.4	12.5	4.8	17.2
	Other Wooded Land	14.9	5.5	20.4	7.5	2.8	10.2
	Other Land	4.8	1.7	6.5	2.4	0.9	3.3

Appendix A9 Confidence Intervals. Standards Errors and Variance

Table A9.1 Variance. CI and SE per FRA class and LUCC: Number of trees

FRA class	Major LUCC	LUCC	N of Trees (N ha ⁻¹)	Count -Variance	Count - Standard Error	Count CI (95%)	Count - CI95 relative (%)
Forest	Natural regenerate d forest	Evergreen forest	124.9	404.8	20.1	40.0	32%
		Deciduous forest	114.7	536.9	23.2	48.5	42%
		Semi-deciduous forest	147.9	7530.0	86.8	223.1	151%
		Bamboo forest	93.3	797.5	28.2	63.9	68%
		High woodland	54.7	433.6	20.8	264.6	484%
	Plantation	Broadleaved planted forest*	116.6	439.5	21.0	44.7	38%
		Coniferous planted forest*	453.0	48658.6	220.6	612.4	135%
		Forest Plantation of Bamboo	31.6	10.1	3.2	40.5	128%
Other Wooded Land	Other wooded land	Woodland	61.7	26.5	5.1	10.1	16%
		Wooded grassland	57.1	92.2	9.6	19.4	34%
		Wooded wetland	53.5	693.5	26.3	64.4	120%
Other Land	Other land - Natural	Natural Forest Coffee*	101.6	100551.4	317.1	1009.1	993%
		Barren Land	1.8	0.2	0.4	0.9	47%
		Natural Grassland	18.3	3.4	1.8	3.6	20%
		Marsh	41.1	653.8	25.6	81.4	198%
	Other land - Cultivated	Coffee plantation*	125.1	8010.9	89.5	183.3	147%
		Annual Crop	20.9	2.1	1.5	2.9	14%
		Perennial crop	45.7	1216.9	34.9	71.3	156%
		Mixed annual and perennial crop	38.6	129.0	11.4	24.4	63%
		Fallow	16.5	25.1	5.0	10.1	61%
		Wood lot of Bamboo	6.0				
		Wood lot	142.9	3235.5	56.9	157.9	111%
	Other land - Build up	Built up area	49.4	90.8	9.5	19.4	39%
	Other land - Quarry/Mining	Quarry/Mining site	35.1				
Water	Inland water	Perennial River	26.9	280.9	16.8	36.9	137%
	Inland water	Intermittent River (seasonal)	24.6	260.1	16.1	35.9	146%

*Results not sufficiently representative

Table A9.2 Variance. CI and SE per Biome and FRA class: Number of trees

Biome	FRA class	N of Trees (N ha ⁻¹)	Count -Variance	Count - Standard Error	Count CI (95%)	Count - CI95 relative (%)
<i>Acacia-Commiphora</i>	Forest	97.3	2120.2	46.0	118.4	122%
	Other Wooded Land	58.9	32.3	5.7	11.2	19%
	Other Land	6.3	5.4	2.3	4.6	74%
	Water	21.1	188.0	13.7	35.2	167%
<i>Combretum-Terminalia</i>	Forest	117.4	276.5	16.6	34.0	29%
	Other Wooded Land	74.2	96.2	9.8	19.5	26%
	Other Land	32.3	11.8	3.4	6.8	21%
	Water	15.6	43.3	6.6	16.1	103%
Dry Afromontane	Forest	116.2	1840.8	42.9	87.8	76%
	Other Wooded Land	37.5	45.8	6.8	13.7	36%
	Other Land	20.1	3.0	1.7	3.4	17%
	Water	13.0	25.4	5.0	12.3	95%
Moist Afromontane	Forest	145.1	488.7	22.1	44.2	30%
	Other Wooded Land	120.3	1373.4	37.1	78.2	65%
	Other Land	36.3	878.9	29.6	59.2	163%
	Water	241.5	11397.0	106.8	339.7	141%
Other	Other Wooded Land	18.5	249.2	15.8	67.9	367%
	Other Land	6.9	38.8	6.2	26.8	391%

Table A9.3 Variance. CI and SE by Biome and FRA class: Volume

Biome	FRA class	Volume (m3 ha ⁻¹)	Volume Variance	Volume SE	Volume CI (95%)	Volume CI95 relative (%)
<i>Acacia-Commiphora</i>	Forest	58.2	658.2	25.7	65.9	113%
	Other Wooded Land	4.8	0.4	0.6	1.2	25%
	Other Land	4.7	5.0	2.2	4.5	95%
	Water	2.4	0.1	0.4	0.9	38%
<i>Combretum-Terminalia</i>	Forest	66.8	113.9	10.7	21.8	33%
	Other Wooded Land	17.8	7.3	2.7	5.4	30%
	Other Land	13.1	3.7	1.9	3.8	29%
	Water	5.1	12.7	3.6	8.7	172%
Dry Afromontane	Forest	120.0	1836.4	42.9	87.6	73%
	Other Wooded Land	6.4	1.8	1.3	2.7	42%
	Other Land	8.0	1.4	1.2	2.3	29%
	Water	3.6	7.3	2.7	6.6	182%
Moist Afromontane	Forest	216.1	528.9	23.0	45.9	21%
	Other Wooded Land	21.2	10.9	3.3	7.0	33%
	Other Land	62.3	854.2	29.2	58.4	94%
	Water	165.2	11571.3	107.6	342.3	207%
Other	Other Wooded Land	0.5	0.0	0.1	0.6	117%
	Other Land	0.3	0.2	0.4	1.7	533%

Table A9.4 Variance. CI and SE by Biome and FRA class: AGB

Biome	FRA class	AG biomass (t ha ⁻¹)	AGB Variance	AGB SE	AGB CI (95%)	AGB CI95 relative (%)
<i>Acacia-Commiphora</i>	Forest	56.8	658.2	25.7	65.9	116%
	Other Wooded Land	5.0	0.4	0.6	1.2	24%
	Other Land	4.5	5.0	2.2	4.5	99%
	Water	2.5	0.1	0.4	0.9	37%
<i>Combretum-Terminalia</i>	Forest	65.5	113.9	10.7	21.8	33%
	Other Wooded Land	17.9	7.3	2.7	5.4	30%
	Other Land	12.8	3.7	1.9	3.8	30%
	Water	5.1	12.7	3.6	8.7	170%
Dry Afromontane	Forest	114.4	1836.4	42.9	87.6	77%
	Other Wooded Land	6.5	1.8	1.3	2.7	42%
	Other Land	7.8	1.4	1.2	2.3	30%
	Water	3.7	7.3	2.7	6.6	179%
Moist Afromontane	Forest	202.9	528.9	23.0	45.9	23%
	Other Wooded Land	21.5	10.9	3.3	7.0	32%
	Other Land	57.8	854.2	29.2	58.4	101%
	Water	162.6	11571.3	107.6	342.3	211%
Other	Other Wooded Land	0.5	0.0	0.1	0.6	108%
	Other Land	0.3	0.2	0.4	1.7	503%

Table A9.5 Variance. CI and SE per Region Biome and FRA class: Number of trees

Region	Biome	FRA class	N of Trees (N ha ⁻¹)	Count -Variance	Count - Standard Error	Count CI (95%)	Count - CI95 relative (%)
Afar	<i>Acacia-Commiphora</i>	Other Wooded Land	14.7	18.0	4.2	9.2	63%
		Other Land	1.6	0.2	0.5	1.0	62%
		Water	21.9	219.7	14.8	47.2	216%
	<i>Combretum-Terminalia</i>	Other Land	6.0	20.1	4.5	57.0	951%
	Other	Other Wooded Land	0.5				
Amhara	<i>Acacia-Commiphora</i>	Other Wooded Land	3.6	4.9	2.2	7.1	197%
		Other Land	17.6	214.8	14.7	63.1	359%
		Forest	59.7	567.5	23.8	75.8	127%
	<i>Combretum-Terminalia</i>	Other Wooded Land	70.1	206.7	14.4	30.0	43%
		Other Land	44.8	38.7	6.2	12.7	28%
		Water	17.1	123.8	11.1	35.4	207%
	Dry Afromontane	Forest	116.0	23185.4	152.3	351.1	303%
		Other Wooded Land	29.8	66.3	8.1	17.3	58%
		Other Land	19.2	7.7	2.8	5.6	29%
		Water	14.7	37.8	6.1	15.8	107%
	Moist Afromontane	Forest	83.3	1787.4	42.3	537.2	645%
		Other Land	8.3	23.7	4.9	11.9	144%
Beneshangul Gumu	<i>Combretum-Terminalia</i>	Forest	127.9	595.4	24.4	53.2	42%
		Other Wooded Land	125.4	768.7	27.7	58.8	47%
		Other Land	25.0	46.8	6.8	15.1	60%
		Water	23.5	207.5	14.4	62.0	263%
Gambela	<i>Combretum-Terminalia</i>	Other Wooded Land	29.8	89.3	9.5	23.1	78%
	Moist Afromontane	Forest	182.4	4524.3	67.3	172.9	95%
		Other Land	34.9				
Oromia	<i>Acacia-Commiphora</i>	Forest	102.6				
		Other Wooded Land	93.8	205.8	14.3	28.9	31%
		Other Land	15.4	81.1	9.0	18.5	120%
	<i>Combretum-Terminalia</i>	Forest	115.2	2860.1	53.5	137.5	119%
		Other Wooded Land	142.7	2946.5	54.3	125.2	88%
		Other Land	31.0	29.3	5.4	11.3	36%
	Dry Afromontane	Forest	119.3	1897.5	43.6	93.4	78%
		Other Wooded Land	26.2	56.5	7.5	15.9	60%
		Other Land	19.5	6.8	2.6	5.2	27%
	Moist Afromontane	Forest	137.0	912.6	30.2	60.8	44%
		Other Wooded Land	122.3	1850.6	43.0	93.7	77%
		Other Land	36.0	44.5	6.7	13.5	37%
		Water	362.8	12288.4	110.9	477.0	131%

Table A9.5 Variance. CI and SE per Region Biome and FRA class: Number of trees

Region	Biome	FRA class	N of Trees (N ha ⁻¹)	Count -Variance	Count - Standard Error	Count CI (95%)	Count - CI95 relative (%)
SNNPR	<i>Acacia-Commiphora</i>	Forest	154.0	185.5	13.6	173.1	112%
		Other Wooded Land	60.0	677.0	26.0	72.2	120%
		Other Land	11.8	27.6	5.3	13.5	114%
	<i>Combretum-Terminalia</i>	Forest	75.2	517.5	22.7	51.5	68%
		Other Wooded Land	56.0	276.9	16.6	34.7	62%
		Other Land	36.6	87.0	9.3	19.8	54%
	Dry Afromontane	Forest	121.1	1129.1	33.6	93.3	77%
		Other Wooded Land	38.1	269.6	16.4	70.6	185%
		Other Land	26.6	34.3	5.9	13.1	49%
	Moist Afromontane	Forest	164.3	753.2	27.4	60.4	37%
		Other Wooded Land	113.3	1415.2	37.6	104.4	92%
		Other Land	64.7	14201.1	119.2	254.0	393%
Somali	<i>Acacia-Commiphora</i>	Forest	82.3				
		Other Wooded Land	50.7	43.0	6.6	13.1	26%
		Other Land	2.5	0.5	0.7	1.6	62%
	Other	Other Wooded Land	23.4				
		Other Land	25.0				
Tigray	<i>Acacia-Commiphora</i>	Other Wooded Land	131.2				
		Other Land	15.7				
		Forest	196.1				
	<i>Combretum-Terminalia</i>	Other Wooded Land	57.4	206.8	14.4	31.1	54%
		Other Land	31.8	45.4	6.7	14.2	45%
		Forest	87.7	2973.1	54.5	173.5	198%
	Dry Afromontane	Other Wooded Land	78.7	1498.4	38.7	94.7	120%
		Other Land	20.4	34.6	5.9	13.1	64%
	Moist Afromontane	Other Land	2.4				

Table A9.6 Variance. CI and SE per Region Biome and FRA class: Volume

Region	Biome	FRA class	Volume (m ³ ha ⁻¹)	Volume Variance	Volume SE	Volume CI (95%)	Volume CI95 relative (%)
Afar	<i>Acacia-Commiphora</i>	Other Wooded Land	1.7	0.7	0.8	1.8	109%
		Other Land	0.1	0.0	0.1	0.1	126%
		Water	2.5	0.1	0.3	1.0	40%
	<i>Combretum-Terminalia</i>	Other Land	1.5				
	Other	Other Wooded Land	0.0				
Amhara	<i>Acacia-Commiphora</i>	Other Wooded Land	0.3	0.1	0.2	0.8	254%
		Other Land	2.0	0.0	0.1	0.6	32%
	<i>Combretum-Terminalia</i>	Forest	24.1	14.8	3.8	12.2	51%
		Other Wooded Land	15.7	8.2	2.9	6.0	38%
		Other Land	18.0	9.2	3.0	6.2	35%
		Water	5.0	26.9	5.2	16.5	333%
	Dry Afromontane	Forest	261.8	21891.1	148.0	341.2	130%
		Other Wooded Land	4.1	1.6	1.3	2.7	66%
		Other Land	7.4	4.3	2.1	4.2	56%
		Water	4.1	9.3	3.0	7.8	190%
	Moist Afromontane	Forest	40.8				
Beneshangul Gumu	Moist Afromontane	Other Land	6.4	23.0	4.8	11.7	184%
		Forest	66.3	240.0	15.5	33.8	51%
		Other Wooded Land	35.3	49.5	7.0	14.9	42%
		Other Land	8.6	10.6	3.3	7.2	83%
		Water	8.5	55.6	7.5	32.1	375%
Gambela	<i>Combretum-Terminalia</i>	Other Wooded Land	7.3	5.3	2.3	5.7	78%
	Moist Afromontane	Forest	262.5	3292.8	57.4	147.5	56%
		Other Land	151.7	674.5	26.0	111.7	74%
Oromia	<i>Acacia-Commiphora</i>	Forest	83.0	2014.8	44.9	142.9	172%
		Other Wooded Land	9.1	3.8	1.9	3.9	43%
		Other Land	16.2	81.1	9.0	18.5	114%
	<i>Combretum-Terminalia</i>	Forest	47.3	108.5	10.4	26.8	57%
		Other Wooded Land	24.8	18.6	4.3	10.0	40%
		Other Land	15.6	14.3	3.8	7.9	50%
	Dry Afromontane	Forest	70.9	848.3	29.1	62.5	88%
		Other Wooded Land	9.1	12.2	3.5	7.4	80%
		Other Land	9.2	3.3	1.8	3.7	40%
	Moist Afromontane	Forest	231.6	892.5	29.9	60.1	26%
		Other Wooded Land	17.4	5.7	2.4	5.2	30%
		Other Land	28.9	36.0	6.0	12.1	42%
		Water	248.2	11089.2	105.3	453.1	183%

Table A9.6 Variance. CI and SE per Region Biome and FRA class: Volume

Region	Biome	FRA class	Volume (m ³ ha ⁻¹)	Volume Variance	Volume SE	Volume CI (95%)	Volume CI95 relative (%)
SNNPR	<i>Acacia-Commiphora</i>	Forest	142.6				
		Other Wooded Land	14.9	55.5	7.5	20.7	138%
		Other Land	7.0	15.8	4.0	10.2	146%
	<i>Combretum-Terminalia</i>	Forest	98.8	576.4	24.0	54.3	55%
		Other Wooded Land	11.5	16.5	4.1	8.5	74%
		Other Land	18.3	55.4	7.4	15.8	86%
	Dry Afromontane	Forest	108.9	1261.2	35.5	98.6	91%
		Other Wooded Land	15.3	22.3	4.7	20.3	133%
		Other Land	10.5	7.2	2.7	6.0	57%
	Moist Afromontane	Forest	163.5	641.8	25.3	55.8	34%
		Other Wooded Land	47.9	108.9	10.4	29.0	61%
		Other Land	189.3	11257.7	106.1	226.2	119%
Somali	<i>Acacia-Commiphora</i>	Forest	13.2				
		Other Wooded Land	3.5	0.1	0.4	0.7	21%
		Other Land	0.4	0.1	0.2	0.5	119%
	Other	Other Wooded Land	0.6				
		Other Land	1.1				
Tigray	<i>Acacia-Commiphora</i>	Other Wooded Land	4.8				
		Other Land	0.6				
	<i>Combretum-Terminalia</i>	Forest	31.9				
		Other Wooded Land	19.2	54.8	7.4	16.0	83%
		Other Land	6.8	2.9	1.7	3.6	53%
	Dry Afromontane	Forest	10.8	16.1	4.0	12.8	118%
		Other Wooded Land	6.6	2.6	1.6	3.9	60%
		Other Land	2.2	0.4	0.7	1.5	69%
	Moist Afromontane	Other Land	0.2				

Table A9.7 Variance. CI and SE per Region Biome and FRA class: AGB

Region	Biome	FRA	AG biomass (t ha ⁻¹)	AGB Variance	AGB SE	AGB CI (95%)	AGB CI95 relative (%)
Afar	Acacia-Commiphora	Other Wooded Land	1.7	0.7	0.8	1.8	107%
		Other Land	0.1	0.0	0.1	0.1	119%
		Water	2.6	0.1	0.3	1.0	38%
	Combretum-Terminalia	Other Land	1.5				
	Other	Other Wooded Land	0.0				
Amhara	Acacia-Commiphora	Other Wooded Land	0.3	0.1	0.2	0.8	243%
		Other Land	2.0	0.0	0.1	0.6	31%
	Combretum-Terminalia	Forest	24.0	14.8	3.8	12.2	51%
		Other Wooded Land	15.8	8.2	2.9	6.0	38%
		Other Land	17.6	9.2	3.0	6.2	35%
		Water	5.0	26.9	5.2	16.5	328%
	Dry Afromontane	Forest	245.3	21891.1	148.0	341.2	139%
		Other Wooded Land	4.2	1.6	1.3	2.7	64%
		Other Land	7.2	4.3	2.1	4.2	58%
		Water	4.2	9.3	3.0	7.8	188%
	Moist Afromontane	Forest	41.1				
		Other Land	6.2	23.0	4.8	11.7	188%
Beneshangu I Gumu	Combretum-Terminalia	Forest	65.8	240.0	15.5	33.8	51%
		Other Wooded Land	35.5	49.5	7.0	14.9	42%
		Other Land	8.6	10.6	3.3	7.2	83%
		Water	8.6	55.6	7.5	32.1	372%
Gambela	Combretum-Terminalia	Other Wooded Land	7.4	5.3	2.3	5.7	77%
	Moist Afromontane	Forest	240.5	3292.8	57.4	147.5	61%
		Other Land	141.9	674.5	26.0	111.7	79%
Oromia	Acacia-Commiphora	Forest	80.3	2014.8	44.9	142.9	178%
		Other Wooded Land	9.3	3.8	1.9	3.9	42%
		Other Land	15.4	81.1	9.0	18.5	120%
	Combretum-Terminalia	Forest	46.8	108.5	10.4	26.8	57%
		Other Wooded Land	25.0	18.6	4.3	10.0	40%
		Other Land	15.2	14.3	3.8	7.9	52%
	Dry Afromontane	Forest	69.4	848.3	29.1	62.5	90%
		Other Wooded Land	9.0	12.2	3.5	7.4	82%
		Other Land	8.9	3.3	1.8	3.7	41%
	Moist Afromontane	Forest	217.4	892.5	29.9	60.1	28%
		Other Wooded Land	17.8	5.7	2.4	5.2	29%
		Other Land	27.8	36.0	6.0	12.1	44%
		Water	244.2	11089.2	105.3	453.1	186%

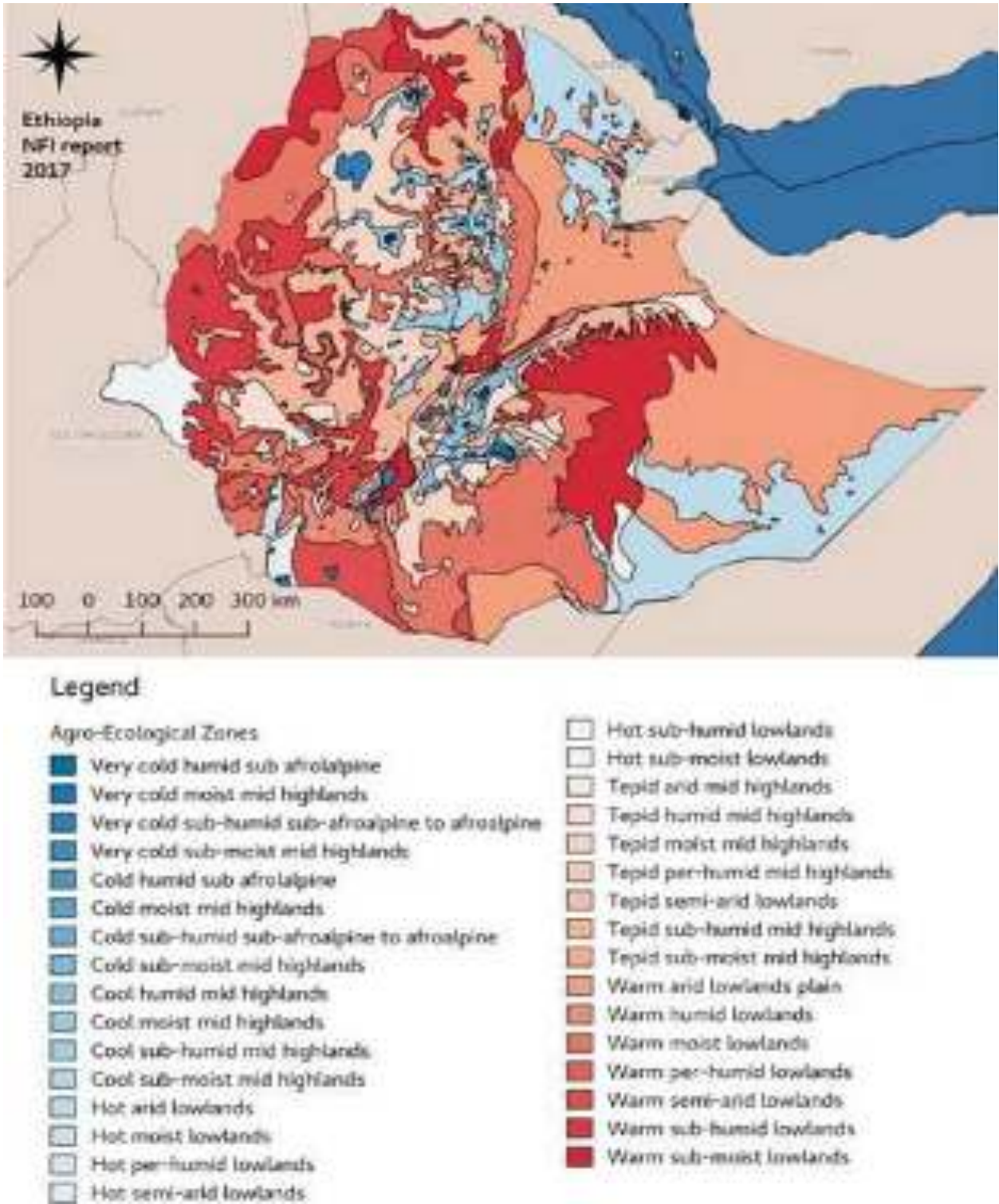
Table A9.7 Variance. CI and SE per Region Biome and FRA class: AGB

Region	Biome	FRA	AG biomass (t ha ⁻¹)	AGB Variance	AGB SE	AGB CI (95%)	AGB CI95 relative (%)
SNNPR	Acacia-Commiphora	Forest	140.4				
		Other Wooded Land	14.8	55.5	7.5	20.7	140%
		Other Land	6.7	15.8	4.0	10.2	152%
	Combretum-Terminalia	Forest	94.0	576.4	24.0	54.3	58%
		Other Wooded Land	11.6	16.5	4.1	8.5	73%
		Other Land	17.5	55.4	7.4	15.8	90%
	Dry Afromontane	Forest	105.0	1261.2	35.5	98.6	94%
		Other Wooded Land	15.3	22.3	4.7	20.3	133%
		Other Land	10.2	7.2	2.7	6.0	59%
	Moist Afromontane	Forest	155.6	641.8	25.3	55.8	36%
		Other Wooded Land	47.3	108.9	10.4	29.0	61%
		Other Land	173.0	11257.7	106.1	226.2	131%
Somali	Acacia-Commiphora	Forest	13.5				
		Other Wooded Land	3.6	0.1	0.4	0.7	20%
		Other Land	0.4	0.1	0.2	0.5	117%
	Other	Other Wooded Land	0.6				
		Other Land	1.2				
Tigray	Acacia-Commiphora	Other Wooded Land	5.2				
		Other Land	0.7				
	Combretum-Terminalia	Forest	32.8				
		Other Wooded Land	18.7	54.8	7.4	16.0	86%
		Other Land	6.9	2.9	1.7	3.6	53%
	Dry Afromontane	Forest	11.1	16.1	4.0	12.8	115%
		Other Wooded Land	6.9	2.6	1.6	3.9	57%
		Other Land	2.2	0.4	0.7	1.5	67%
	Moist Afromontane	Other Land	0.2				

Appendix A10 Agro-ecological zones

In 2000 the Ministry of Agriculture (MOA) developed the Agro-Ecological Zone (AEZ) map with the aim to introduce a classification comprehensive of all the relevant characters and not discipline oriented. The AEZ classification¹ is based on the basic ecological elements of climate, physiography, soils, vegetation, farming systems, etc. The major characters included in the AEZs are eighteen, delineated and named by terms describing the broad moisture and elevation conditions of areas. The final AEZ map classifies Ethiopia into 33 agro-ecological sub-zones identified as homogeneous in terms of climate, physiography, soils, vegetation, land use, farming system and animal production. The AEZ has been combined with the Digital Elevation Model (DEM) to develop the Strata (4.2.3 Agro- ecological zones) used as pre-stratification layer for the NFA sampling.

Table A10.1 Agro ecological zones of Ethiopia



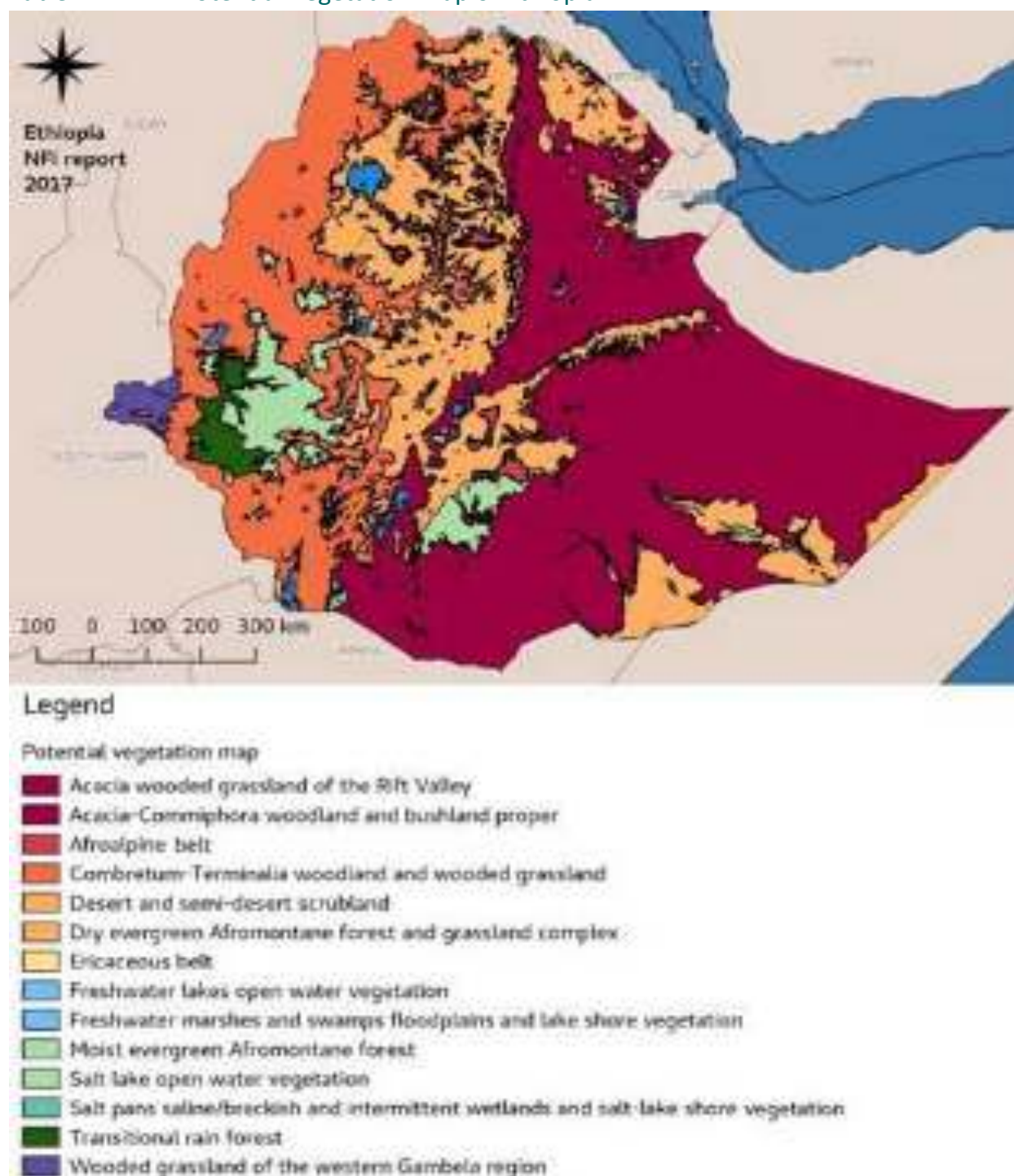
Appendix A11 Potential vegetation map

Published in 2010 the Atlas of potential Vegetation of Ethiopia¹, a map in scale 1:2000.000, aims to classify the Ethiopian flora in twelve potential vegetation types and subtypes.

1. Desert and semi-desert scrubland.
2. Acacia-Commiphora woodland and bushland (with the subtypes (2a) Acacia-Commiphora woodland and bushland proper and (2b) Acacia wooded grassland of the Rift Valley).
3. Wooded grassland of the western Gambela region.
4. Combretum-Terminalia woodland and wooded grassland.
5. Dry evergreen Afromontane forest and grassland complex (with the subtypes (5a) Undifferentiated Afromontane forest, (5b) Dry single-dominant Afromontane forest of the Ethiopian highlands, (5c) Afromontane woodland, wooded grassland and grassland, (5d) Transition between Afromontane vegetation and Acacia-Commiphora bushland on the Eastern escarpment).
6. Moist evergreen Afromontane forest (with the subtypes (6a) Primary or mature secondary moist evergreen Afromontane forest, and (6b) Edges of moist evergreen Afromontane forest, bushland, woodland and wooded grassland.
7. Transitional rainforest.
8. Ericaceous belt.
9. Afroalpine belt.
10. Riverine vegetation.
11. Fresh-water lakes, etc. (with the subtypes (na) Fresh-water lake vegetation (open water) and (nb) Freshwater marshes and swamps, floodplains and lake shore vegetation).
12. Salt lakes, etc. (with the subtypes (12a) Salt lake vegetation (open-water) and (12b) Saltpans, saline/brackish and intermittent wetlands and salt-lake shore vegetation).

The Atlas has been realized using SRTM DEM of Ethiopia, field surveys, rainfall datasets, the Global Lakes and Wetlands Database (GLWD) data and by revising the Flora of Ethiopia and Eritrea². The Atlas of Potential Vegetation in Ethiopia as been used in the definitions of the Biomes (4.2.3. Potential vegetation of Ethiopia).

Table A11.1 Potential Vegetation Map of Ethiopia



Appendix A12 Wall-to-wall biomass map of Ethiopia

General Introduction

This report describes the creation of a wall-to-wall biomass map done for Ethiopia within the National Forest Inventory project. The map is referenced for the year 2016. The main idea is to use the field inventory data and use it for the calibration of a multi-sensor satellite data stack, consisting of optical and radar imagery, in order to get spatially explicit values of biomass. While those estimates are affected by high uncertainties on a pixel level, and therefore are critical to use for carbon stock estimates, the map can give important information on the locations where the actual biomass deviates considerably from the expected one. The spatial information can then be used for planning of carbon stock enhancements.

Data preparation

NFI data

For the combination of the field and satellite data, the spatial scale and potential location errors of both datasets has to be taken into account. The sampling unit design consists of 4 subplots with a respective size of about 2 ha. In order to account for location errors of the satellite data and the field inventory plots, the 4 subplots have been aggregated by considering their mean biomass value per Sampling Unit (SU). The location of the whole SU has therefore been generalized by calculating the center point of the available subplots. Subsequently, a buffer zone of 300 m radius is created around this point and the mean value of the subplot biomass measurements is attributed to that area.

Applied for each zone separately. This assures that the model represents the relation between the satellite data

Optical data

A cloud-free wall-to-wall mosaic of optical Landsat-8 data has been generated by using the System for Earth Observation Data Access, Processing and Analysis for Land Monitoring (SEPAL) main interface. This allows to create a best-pixel composite algorithm, that according to some basic parameter settings, detects a set of scenes for a given year and area. One of the most important parameters is the target day. This datum decides on the importance of the selected scenes and will give precedence to the scenes that are closest

to that day. For the given task the 1st of November 2016 has been chosen. This day falls within the beginning of the dry period and should ideally pick up the vegetation signal of still photosynthetically active woody vegetation, while the herbaceous layer already dried out. In this way the reflectance is most related to aboveground woody biomass. The minimum number of scenes per path/row combination had been set to 10, assuring cloud-free observations for every pixel. The weighting factor between cloud-freeness and target day importance was kept on 50%. Apart from the shadow tolerance, which was set to the maximum, all other parameter settings have been kept to the default values. Snow masking, BRDF correction as well as the median composite option were activated for the final retrieval of the mosaic. The reason for activating the median option were mainly agricultural fields that dependent on the pixel selection resulted in non-homogeneous artifacts in the final mosaic.

The final mosaic was subsequently downloaded for all of Ethiopia, including all available bands (i.e. blue, green, red, NIR, SWIR 1 and SWIR 2) on a geometric resolution of 30 m. Further band ratios have been calculated, i.e. the NDVI, MSAVI, NDWI and the NBR.

ALOS Kyoto & Carbon mosaic

Long-wavelength L-band data (wavelength = 23 cm) demonstrated its sensitivity to structural forest attributes such as biomass. While there is no free and open access to raw data, the Japanese Aerospace Exploration Agency (JAXA) provides already preprocessed data tiles for ALOS PALSAR data. These datasets are generated on a yearly basis and exhibit a spatial resolution of 25 m. The Open SAR Toolkit (OST) on the SEPAL platform eases access, applies a Refined Lee Speckle filter and adds a ratio band for RGB visualization. The 2016 mosaic, based on data from ALOS-2 PALSAR-2, was used for the elaboration of the biomass map.

Sentinel-1 timescan mosaic

C-band radar data (wavelength = 5 cm) is actually less suited for biomass mapping itself since the backscatter is rather a function of the canopy structure than the structure of the tree. In addition, the signal is more sensitive to changes in soil moisture and the roughness of the surface. However, the dense temporal sampling of the Sentinel-1 data allows to aggregate the temporal dynamics by calculating basic descriptive statistics, such as mean, minimum, maximum and standard deviation, in time. This is referred to as timescan method and enhances the signal with respect to single imagery. For the mapping, 7 wall- to-wall mosaics between September 2016 and May 2017 and a spatial resolution of 30 m, were created. This included about 310 full Sentinel-1 scenes. The resultant time-series data of the full country has then been used to calculate the respective timescan layers for both the VV and the VH polarization bands, respectively. All processing has been done with OST on the SEPAL platform, which provides a fully-automated workflow for the data download, processing and mosaicking.

SRTM

The Shuttle Radar Topography Mission (SRTM) provides elevation data for most parts of the earth. Recent studies on biomass mapping integrate this data into machine learning algorithms for better estimates. For the biomass map of Ethiopia, the height and slope information were derived on a 90 m spatial resolution.

Multi-Sensor Integration

All the above-mentioned satellite data has been merged to a single image stack consisting of 24 layers. A visual inspection did not result in any data offsets, so that a separate co-registration of the different layers was not necessary. The extent and pixel spacing of the Landsat data was used as a reference for the other datasets. Therefore, all layers were resampled to a 30 m resolution.

Classification

For the integration of the NFI data with the satellite data, a data-driven approach based on machine learning was chosen. This approach is integrated into the OST and was consequently run on SEPAL. In detail, the random forest regression algorithm (RF) was used to create a model between the averaged biomass values for each of the generalized SUs, and the average value of each of the satellite layers underlying the generalized SUs. While this includes spatial averaging, it is assumed to account for localization errors and spatial inhomogeneity. RF showed to be highly efficient for non-linear relations between a set of multiple predictors (i.e. the satellite data) and the target parameter (i.e. biomass), while at the same time easy to parameterize.

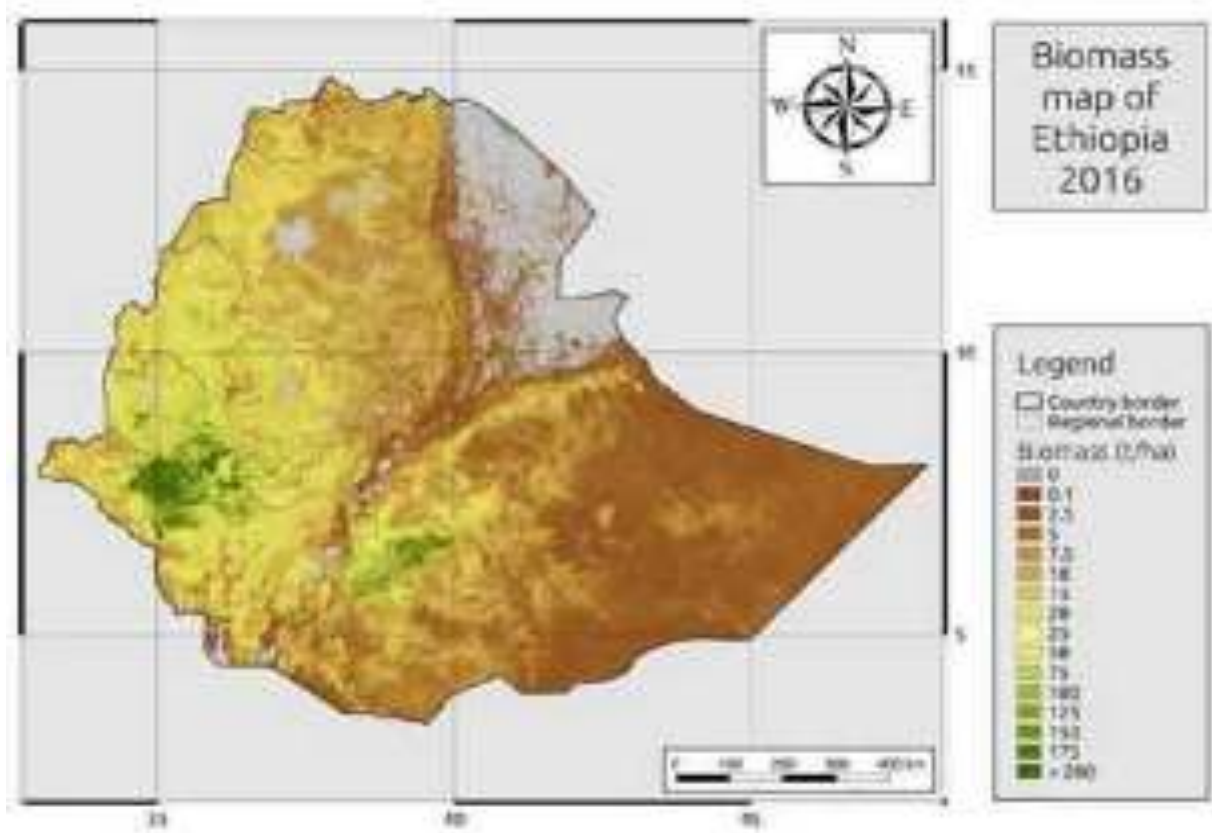
During the model elaboration, different parameter sets for the classifier are tested on the fly. The best model was chosen based on the Out-Of-Bag estimate from the RF classifier itself, and a Leave 5-out cross-validation procedure. In addition, the importance of features has been used to create a subset of input features (i.e. the satellite layers) for an additional parameter testing. All those steps are applied to avoid overfitting of the final model, which RF is prone to when using correlated feature input. Once the best model has been selected, it is applied on the unseen satellite data in order to derive the final biomass estimates. While the model has been elaborated on a reduced resolution of the satellite data (underlying the 300 m radius SU), it was applied on the full resolution of 30 m, which represents the geometric resolution of the final product.

In order to attribute for the structural differences of forests between the different biome zones of Ethiopia, the country has been split into 4 zones beforehand (Figure 1). The above-mentioned model elaboration was then applied for each zone separately. This assures that the model represents the relation between the satellite data and the biomass data according to the specific forest biome, which is assumed to be homogeneous. One issue is that within the ALOS and Sentinel-1 data missing values appeared due to strong topography. Due to the side-looking geometry of SAR instruments, in some cases no information can be retrieved at steep slopes. In these cases, the same classification approach has been followed for those

data layers with valid values. While the classification for this subsetting data stacks has been done for the whole are of each biome, only the parts not covered by one or the other datasets have been filled in the final product.

Another issue is that some land cover classes exhibit similar SAR backscatter as woody vegetation. As a result, high woody biomass values appear at places where no vegetation is present, for example in urban areas. Therefore, a mask has been derived from the continuous tree and shrub cover product from the ESA Copernicus program (<https://proba-v-mep.esa.int/success-stories/land-cover-map-over-africa-based-proba-v-100-m-data>). All places with less than either 5 % tree or shrub cover have been masked out and the map had been resampled on the same extent and resolution of the multi-sensor stack.

Table A12.1 Biomass map of Ethiopia



The map reflects the distribution of aboveground woody biomass (AWB) in Ethiopia on a 30 m resolution (Figure 2). The estimates are not limited to actual forest areas, which makes it possible to assess woody resources also outside forested areas. Although visual interpretation results in a good general representation of biomass on a local scale, statistical uncertainties on a pixel level are high. This is expected for this kind of mapping approach, since the remote sensing data cannot capture all the variability that determines the amount of AWB within a Sampling Unit. This is also depicted in the model evaluations based on the Out-Of-Bag (OOB) score coming from the random forest regressor and the leave k-out cross validation (k=5), where the r^2 ranges between 0.24 and 0.44, respective to the biome (Table 1 & 2). The RMSE varies considerably between the biomes and is lowest

for the low-biomass Acacia Commisphère biome, while highest in the Moist Afromountain biome, where biomass reaches values of over 300 t/ha. Low variations of biomass on a local scale should therefore be treated with care. However, when aggregating the samples on the biome scale the bias between measured and predicted values is very low, which means that on larger scales the uncertainty of the estimates decreases considerably.

